Efficient Estimation of Nonlinear DSGE Models

Eva F. Janssens[†] and Sean McCrary[‡]

October, 2026

Most Recent Version

Abstract

This paper introduces a computationally efficient method for full-information estimation of nonlinear Dynamic Stochastic General Equilibrium (DSGE) models. Rather than treating model solution and filtering as separate computationally intensive steps, we integrate the solution into the filtering procedure. The method computes a sequence of local linear solutions at the forecasted state, yielding a time-varying linear state-space system for likelihood evaluation via the Kalman filter. Across three canonical benchmark models, the method accurately captures strong nonlinearities, accommodates occasionally binding constraints and large state spaces, and delivers computational gains of several orders of magnitude relative to global solution methods with nonlinear filtering. In an empirical application to an estimated New Keynesian model with labor market frictions, we find state-dependent monetary policy transmission consistent with evidence from local projections. The framework extends to models with regime switching and heterogeneous agents.

Keywords: Bayesian inference, nonlinear filtering, computational methods

JEL classification codes: C63, C68, C32, E32, E52, J64

* Acknowledgements: We are grateful for Hilde Bjornland's discussion at Midwest Econometrics, and for the helpful comments of Ryan Michaels, Frank Schorfheide, Mikkel Plagborg-Møller, Marco del Negro, Ulrich Müller, Adam McCloskey, Jean-Jacques Forneron, Christian Matthes, and Kaspar Wüthrich, seminar, brown-bag, and conferences participants at UNC, OSU, Notre Dame, Richmond Fed, NY Fed, SED, SEA, SEM, IAAE, Midwest Macro, Midwest Econometrics, NAWMES, Women in Econometrics, CEF, Rome Conference in Macroeconometrics and Time Series, and T2M. Part of this paper was earlier presented under the name "Unemployment and the State-Dependent Effects of Monetary Policy". Any errors are our own.

Contact information:
† Eva F. Janssens: University of Michigan, Department of Economics, USA. evafj@umich.edu
‡ Sean McCrary: Ohio State University, Department of Economics, USA. mccrary.65@osu.edu

This repository contains example code for several of the applications in this paper.

1 Introduction

Nonlinearities play a fundamental role in macroeconomic dynamics, and a growing empirical literature underscores their importance.[1] While macroeconomic models often feature such nonlinearities, whether they affect model predictions depends on the choice of solution and estimation method. Global solution methods combined with nonlinear filters are the most accurate way to solve and estimate such nonlinear models, but their computational cost quickly becomes prohibitive: estimation can take days, weeks, or months.[2]

The methodological contribution of this paper is a computationally efficient and broadly applicable approach to full-information estimation of nonlinear Dynamic Stochastic General Equilibrium (DSGE) models. Traditional estimation methods treat model solution and likelihood evaluation (using a filter) as two separate steps. Instead, our method characterizes the solution *within* the filtering procedure. The approach builds on the observation that even highly nonlinear models exhibit local behavior that can be well approximated by linear functions, but nonlinearity implies that these functions vary across the state space. The main idea is that, for estimation, an accurate characterization of nonlinear model dynamics can be obtained by repeatedly solving for such local approximations in the regions of the state space where the data indicates the economy is most likely to be.

Concretely, the method constructs a dynamic sequence of local linear solutions at the best forecast of the current state, and embeds the resulting policy rules in a time-varying linear state-space system. Each local solution is fast to compute and accurate in a neighborhood of the current state. The dynamic implementation of local linear methods preserves nonlinearities such as state dependence and, despite the policy functions being linear, yields solutions that are not certainty equivalent. The linear state-space form enables likelihood evaluation via the time-varying Kalman filter. Theoretically, we show that the proposed procedure can be close in terms of accuracy to the filter one would obtain if the extended Kalman filter were applied to the true, but unknown, global policy and transition functions. The key advantage of our approach is that this accuracy is attained without characterizing the global solution. The method therefore mitigates the two main computational bottlenecks in nonlinear DSGE estimation, i.e., global solutions and nonlinear filtering, while retaining accuracy.

[1] For example, Stock and Watson (2020) and Benigno and Eggertsson (2024) examine nonlinearities in the Phillips curve; Tenreyro and Thwaites (2016) and Barnichon et al. (2022) analyze state and sign dependence in the transmission of economic shocks, respectively.

[2] For example, in Gust et al. (2017), a single solution and likelihood evaluation takes six minutes in serial. For 100,000 likelihood evaluations, this would imply over a year of computation time.

We benchmark the proposed estimation method using three canonical models. First, a labor search model of the Diamond-Mortensen-Pissarides (Diamond, 1982; Mortensen and Pissarides, 1994) type, which admits an analytical global solution, allows for the direct evaluation of the theoretical approximation to the extended Kalman filter applied to the true global policy functions. The local method closely reproduces this oracle benchmark and is computationally inexpensive. Second, the three equation New Keynesian model of Galí (2015), with and without the zero lower bound, shows how the framework accommodates occasionally binding constraints. Third, a multi-country real business cycle model shows that the method remains computationally tractable as the dimension of the state space increases.

Empirical Application We apply the proposed estimation framework to study state dependence in monetary policy transmission within a New Keynesian model with labor market frictions. In search models, the flow surplus, i.e., the profitability of job creation, determines how productivity shocks propagate through the economy (Ljungqvist and Sargent, 2017). We show this mechanism extends to monetary policy shocks. In a linearized model, the flow surplus is fixed, but in the nonlinear model, it evolves endogenously, giving rise to state-dependent transmission. The method captures this state dependence by solving the model locally along the filtered state path, allowing elasticities to vary.

Model-implied state-dependent impulse response functions show that monetary policy shocks affect unemployment differently depending on the state of the economy, but finds less evidence for state-dependence in inflation. This generates time variation in the unemployment-inflation trade-off (Phillips Multiplier, Barnichon and Mesters, 2021), which ranges from approximately -2 in recessions to -5 in expansions. These results align closely with evidence from reduced-form state-dependent local projections. More broadly, these results demonstrate that nonlinear features found in empirical work (e.g., Tenreyro and Thwaites, 2016; Benigno and Eggertsson, 2024) can arise endogenously in standard macroeconomic models when solved and estimated nonlinearly.

Extensions We show how the method can be extended to accommodate regime switching, including, for example, models with rare disasters (Barro, 2009). Additionally, we extend the method to accommodate heterogeneous agent models. Unlike existing full-information approaches (Winberry, 2018; Liu and Plagborg-Møller, 2023; Auclert et al., 2021), which rely on perturbation and impose certainty equivalence, our approach accommodates aggregate uncertainty in individual decision rules while preserving a linear state-space structure. We illustrate this using a simple heterogeneous household model in which job loss and job finding probabilities are determined by aggregate productivity. The method captures time-varying

precautionary saving motives and reveals state dependence and asymmetry in aggregate outcomes.

Related literature. Our methodological contribution is a new framework for full-information estimation of nonlinear DSGE models. A key benchmark is Fernández-Villaverde and Rubio-Ramírez (2007), who introduce the particle filter to estimate nonlinear DSGE models. While broadly applicable, it is computationally intensive, suffers from the curse of dimensionality, and can be unstable. Herbst and Schorfheide (2019) propose the tempered particle filter, which is more stable but at higher computational cost. Farmer (2021) proposes the discretization filter, which is fast, but is limited to small state spaces. Winschel and Krätzig (2010) use sparse grids solution methods and parallelization to evaluate the likelihood, but the computational burden remains high. All of these methods apply a filter to a separately solved model. In contrast, our approach solves the model dynamically, avoids simulation-based filters and scales well in large dimensions. Relatedly, Childers et al. (2022) avoid filtering altogether but are limited to low-order perturbation solutions.

In terms of solution method, we build on Levintal (2018) and Mennuni et al. (2025). Levintal (2018) formalizes Taylor Projection, originally used in Krusell et al. (2002), which computes a locally optimal policy rule at a given point in the state space. Fernández-Villaverde and Levintal (2024) apply this approach to a heterogeneous agent model. We apply Taylor Projections (Levintal, 2018) and Dynamic Perturbation (Mennuni et al., 2025) dynamically within our filtering framework, updating the local policy rule at each forecasted state. None of these papers considered full-information estimation, which is the focus of our paper.

Cagliarini and Kulish (2013) and Kulish and Pagan (2017) study linear rational expectations models with predictable structural change, including likelihood-based estimation. Guerrieri and Iacoviello (2015) use piecewise linear perturbations for occasionally binding constraints. Our approach differs, because the structural model is fixed, and within a given constraint regime the local policies continue to vary with the state.

Our empirical application integrates New Keynesian and search-and-matching models, similar to Christiano et al. (2016), Thomas (2011), and Galí et al. (2012). The model is closest to Krause et al. (2008), but a key difference is that they use a linear perturbation, which fixes elasticities at steady-state values, whereas our method allows these elasticities to vary. This generates state-dependent monetary policy shock transmission, our main contribution relative to this literature. Our results also relate to recent work on the Phillips curve, e.g., Benigno and Eggertsson (2024) shows that it steepens in tight labor markets by imposing an explicit kink. We show this cyclicality arises endogenously from a nonlinear solution.

Outline The rest of this paper is structured as follows. Section 2 presents the method and theory. Section 3 demonstrates the performance of this method in the Diamond-Mortensen-Pissarides model, three-equation New-Keynesian model, and multi-country real business cycle model. Section 4 presents the empirical application. Section 5 discusses extensions with regime switching and heterogeneous households. Section 6 concludes.

2 Local Linear Solutions and the Time-Varying Kalman Filter

This section presents our main methodological contribution: a novel full-information estimation framework for nonlinear DSGE models and its theoretical properties.

2.1 A General Dynamic Stochastic General Equilibrium model

A standard DSGE model can be represented as follows (Schmitt-Grohé and Uribe, 2004):

$$\mathbb{E}_t \left[f(y_{t+1}, y_t, x_{t+1}, x_t; \theta) \right] = 0, \tag{1}$$

The vector θ collects the model parameters, y_t represents the control variables, and the state variables x_t can be partitioned into endogenous variables $x_{1,t}$ and exogenous variables $x_{2,t}$. Often, the exogenous variables are assumed to follow a stationary VAR(1):

$$x_{2,t+1} = Ax_{2,t} + B\eta_{t+1}, \tag{2}$$

where $\eta_t \sim N(0, Q)$ is a vector of Gaussian exogenous shocks. This restriction is not essential, for example, Section 5 considers an extension with regime switching. Endogenous variables $x_{1,t+1}$ depend on x_t through an unknown nonlinear mapping. Hence, a solution to (1) is characterized by two implicit and unknown functions,

$$x_{1,t+1} = h(x_t; \theta), \qquad y_t = g(x_t; \theta), \tag{3}$$

where g is the policy function. Equation (3), together with the law of motion for $x_{2,t}$ in (2), implies the state transition equation

$$x_{t+1} = \begin{pmatrix} h(x_t; \theta) \\ Ax_{2,t} \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ B \end{pmatrix}}_{R} \eta_{t+1}. \tag{4}$$

Local linear methods. While there are many different methods to recover approximations for g and h, in this paper we focus on the class of local linear solution methods. Assume that we can approximate the functions g and h, centered at some x, with linear functions

parameterized by $G_0(x)$, $G_x(x)$, $H_0(x)$, $H_x(x)$ as:

$$\tilde{g}(x_t;\theta) = G_0(x) + G_x(x)\,x_t, \qquad \tilde{h}(x_t;\theta) = H_0(x) + H_x(x)\,x_t. \tag{5}$$

Note that coefficients G_0, G_x, H_0, H_x are implicit functions of the point of approximation x and any structural parameters θ. Although the solutions in Equations (5) are linear, their coefficients depend on the expansion point x. They are therefore more flexible than standard linear perturbations around the nonstochastic steady state (a single point). Intuitively, resolving the model at different states x captures the nonlinearities of g and h by changing the intercepts, $G_0(x)$ and $H_0(x)$, and slopes, $G_x(x)$ and $H_x(x)$, of the local linear policy and transition functions.

The estimation procedure we propose below is compatible with any local linear solution of the form in Equations (5). In the implementation, we focus on Taylor Projection (TP) of Levintal (2018) and Dynamic Perturbation (DP) of Mennuni et al. (2025), both of which produce state-dependent local linear decision rules that are not certainty equivalent because they account for curvature and uncertainty through expectations over future shocks. TP constructs the local solution directly at the current approximation point, while DP constructs the continuation policy along an auxiliary equilibrium path, making it particularly useful for regime dependence and occasionally binding constraints such as the zero lower bound. Further details on the implementation, computational properties, accuracy, and speed of various local solution methods, including TP and DP, are provided in McCrary (2026).

2.2 Likelihood Evaluation

Estimating the parameters θ of a DSGE model requires specifying an observation equation that maps the model's states and controls into observed variables through function f^{obs}:

$$y_t^{\text{obs}} = f^{\text{obs}}(y_t, x_t;\theta) + \varepsilon_t^{\text{ME}} = f^{\text{obs}}(g(x_t;\theta), x_t;\theta) + \varepsilon_t^{\text{ME}} = Z(x_t;\theta) + \varepsilon_t^{\text{ME}}, \tag{6}$$

where $\varepsilon_t^{\text{ME}} \overset{i.i.d.}{\sim} N(0, H)$ is measurement error. We assume that the initial state, structural shocks, and measurement errors are mutually independent. Equation (6) shows any observation mapping that depends on states and controls implicitly depends on only the state x_t. To estimate parameters using full-information methods, we must evaluate the model's likelihood function:

$$p(Y_{1:T}^{\text{obs}}) = \prod_{t=1}^{T} p(y_t^{\text{obs}} | Y_{1:t-1}^{\text{obs}}), \qquad \text{where} \quad Y_{1:T}^{\text{obs}} = (y_1^{\text{obs}}, y_2^{\text{obs}}, ..., y_T^{\text{obs}}).$$

However, evaluation is complicated by the dependence of y_t^{obs} on the latent state x_t. Computing the sequence $p\left(y_t^{\text{obs}} \mid Y_{1:t-1}^{\text{obs}}\right)$ recursively requires a filter, that is, an algorithm that integrates out the latent state.

Following Herbst and Schorfheide (2016), a general filter can be written as follows. Initialize $p(x_0) = p\left(x_0|Y_{1:0}^{\text{obs}}\right)$, here, as a Gaussian density with mean $\hat{x}_0$ (e.g., equal to the steady state of the model) and an initial covariance matrix P_0. Next, for $t = 1, ..., T$, we iterate between forecasting t given $t-1$, using:

$$p\left(x_t \mid Y_{1:t-1}^{\text{obs}}\right) = \int \underbrace{p\left(x_t \mid x_{t-1}, Y_{1:t-1}^{\text{obs}}\right)}_{\text{state-transition kernel}} p\left(x_{t-1} \mid Y_{1:t-1}^{\text{obs}}\right) dx_{t-1} \quad \text{(Transition Eq.)} \tag{7}$$

$$p\left(y_t^{\text{obs}} \mid Y_{1:t-1}^{\text{obs}}\right) = \int \underbrace{p\left(y_t^{\text{obs}} \mid x_t, Y_{1:t-1}^{\text{obs}}\right)}_{\text{observation kernel}} p\left(x_t \mid Y_{1:t-1}^{\text{obs}}\right) dx_t \quad \text{(Observation Eq.)} \tag{8}$$

and updating with Bayes rule:

$$p\left(x_t \mid Y_{1:t}^{\text{obs}}\right) = p\left(x_t \mid y_t^{\text{obs}}, Y_{1:t-1}^{\text{obs}}\right) = \frac{p\left(y_t^{\text{obs}} \mid x_t, Y_{1:t-1}^{\text{obs}}\right) p\left(x_t \mid Y_{1:t-1}^{\text{obs}}\right)}{p\left(y_t^{\text{obs}} \mid Y_{1:t-1}^{\text{obs}}\right)}. \tag{9}$$

Traditionally, the problem of solving the model, that is, obtaining functional forms for the policy function (3) and the state-transition equation (4), is treated separately from the filtering problem in Equations (7)–(9). However, the choice of the solution method, specifically, the functional forms of g and h, affects the complexity of the filtering problem.

When the functions $x_{1,t+1} = h(x_t)$ and $y_t = g(x_t)$ in Equations (3)–(4), together with the observation equation (6), are approximated by linear functions, the state-transition and observation kernels are Gaussian. In this case, Equations (7)–(8) reduce to convolutions of normal densities, and the filtering problem simplifies to the Kalman filter.[3] When the policy function (3) and/or state-transition function (4) are nonlinear, likelihood evaluation requires nonlinear filters, which can be costly for larger models.

The insight that our method builds on is that although general nonlinear solutions complicate filtering, the Kalman filter remains valid when the state-transition and observation equations take the form (Hamilton, 1994; Durbin and Koopman, 2012):

$$x_{t+1} = c\left(Y_{1:t-1}^{\text{obs}}\right) + T\left(Y_{1:t-1}^{\text{obs}}\right) x_t + R\eta_{t+1}, \tag{10}$$

$$y_t^{\text{obs}} = d\left(Y_{1:t-1}^{\text{obs}}\right) + Z\left(Y_{1:t-1}^{\text{obs}}\right) x_t + \varepsilon_t^{\text{ME}}, \tag{11}$$

[3]This corresponds to the standard approach in which a DSGE model is solved using a first-order perturbation and filtered via the Kalman filter (for a textbook treatment, see Herbst and Schorfheide, 2016).

where $c(\cdot)$, $T(\cdot)$, $d(\cdot)$, and $Z(\cdot)$ are functions of the observed data history $Y_{1:t-1}^{\text{obs}}$, and R as in (4). Conditional on data through period $t-1$, the system is linear and Gaussian, allowing its likelihood to be evaluated with the Kalman filter. The question now is when a state-space system of this form arises and what its properties are.

Suppose that, given the data history $Y_{1:t-1}^{\text{obs}}$, a state forecast $\hat{x}_{t|t-1}$ is obtained. We take this forecast as given for now. Assume we compute a local linear solution of the form (5) at expansion point $x = \hat{x}_{t|t-1}$. Then, the transition equation becomes:

$$x_{t+1} = \underbrace{\begin{pmatrix} H_0\left(\hat{x}_{t|t-1}\right) \\ 0 \end{pmatrix}}_{c_t} + \underbrace{\begin{pmatrix} H_x\left(\hat{x}_{t|t-1}\right) \\ A \end{pmatrix}}_{T_t} x_t + \underbrace{\begin{pmatrix} 0 \\ B \end{pmatrix}}_{R} \eta_{t+1}, \tag{12}$$

where $c_t \equiv c(\hat{x}_{t|t-1})$ and $T_t \equiv T(\hat{x}_{t|t-1})$. Furthermore, if the observation equation mapping $Z(x_t)$ is linear, or also approximated using a local linearization around $\hat{x}_{t|t-1}$, we obtain

$$y_t^{\text{obs}} = d_t + Z_t x_t + \varepsilon_t^{\text{ME}}. \tag{13}$$

The coefficients $d_t \equiv d(\hat{x}_{t|t-1})$ and $Z_t \equiv Z(\hat{x}_{t|t-1})$ may include elements of G_0, G_x as well as other model parameters, depending on the observables. Since $\hat{x}_{t|t-1}$ is a function of $Y_{1:t-1}^{\text{obs}}$, Equations (12)–(13) are consistent with Equations (10)–(11) and using the Kalman filter. In each applications below, we show how to obtain the state-space mappings in (12)–(13).

Time-Varying Kalman Filter Taking stock, the time-varying linear state-space system in Equations (12)–(13) give rise to the filtering procedure described in Algorithm 1. In words, Algorithm 1 computes a local linear approximation of the policy function g and the law of motion h at the forecasted state $\hat{x}_{t|t-1}$, which represents the best estimate of the current state x_t based on data through period $t-1$. This approximation is then mapped into the linear state-space form given by Equations (12) and (13), and the Kalman filter is used to generate a forecast for the next state, $\hat{x}_{t+1|t}$. The procedure is then repeated: a new local linear solution is computed at $\hat{x}_{t+1|t}$, mapped into state-space form, and used to forecast $\hat{x}_{t+2|t+1}$. Below, we refer to Taylor Projection combined with the time-varying Kalman filter as TP-TVKF, and for Dynamic Perturbation DP-TVKF.

The corresponding smoothing procedure is presented in Algorithm 2 in Online Appendix A. The Kalman smoother relies on the same sequence of system matrices computed in Algorithm 1 and therefore does not require re-solving the model.

Algorithm 1 Time-Varying Kalman Filter

Initialize: Set $t = 1$, initialize $\widehat{x}_{1|0}$ at the steady state, and initialize $P_{1|0}$.

while $t \leq T$ **do**

Solution Step: Solve the model locally at $\widehat{x}_{t|t-1}$, obtaining the state-space system in Equations (12)–(13).

Update Step:

Compute the innovation covariance:

$$F_t = Z_t P_{t|t-1} Z_t' + H.$$

Compute the filtered-state gain and one-step ahead prediction gain:

$$G_t = P_{t|t-1} Z_t' F_t^{-1}, \qquad K_t = T_t G_t = T_t P_{t|t-1} Z_t' F_t^{-1}.$$

Compute the innovation:

$$v_t = y_t^{\text{obs}} - Z_t \widehat{x}_{t|t-1} - d_t.$$

Update the state estimate and covariance:

$$\widehat{x}_{t|t} = \widehat{x}_{t|t-1} + G_t v_t, \qquad P_{t|t} = P_{t|t-1} - G_t F_t G_t'.$$

Prediction Step:

Compute the predicted state estimate:

$$\widehat{x}_{t+1|t} = T_t \widehat{x}_{t|t} + c_t.$$

Compute the predicted covariance:

$$P_{t+1|t} = T_t P_{t|t} T_t' + RQR'.$$

Store $\widehat{x}_{t|t-1}$, $P_{t|t-1}$, T_t, Z_t, K_t, v_t, and F_t for the smoothing recursion.

Advance Time: Set $t \leftarrow t + 1$ and repeat.

end while

2.3 Relation to the Extended Kalman Filter

The previous subsection outlined a filtering algorithm based on local linear solution methods. The objective of this section is to assess whether we can say anything about its accuracy. In particular, this section establishes that the extended Kalman filter (EKF) applied to the true global model solution is the natural (oracle) benchmark for our approach. The EKF is a standard method for nonlinear state space models; see, e.g., Durbin and Koopman (2012).

Suppose that the true equilibrium policy $g^*(x_t)$ and endogenous transition $h^*(x_t)$ are known. The nonlinear state space system can then be written as

$$y_t^{\text{obs}} = Z(x_t) + \varepsilon_t^{\text{ME}}, \tag{14}$$

$$x_{t+1} = \begin{pmatrix} h^*(x_t) \\ Ax_{2,t} \end{pmatrix} + \begin{pmatrix} 0 \\ B \end{pmatrix} \eta_{t+1}, \tag{15}$$

where $Z(x_t) = f^{\text{obs}}(g^*(x_t), x_t)$ in the notation of (6). The EKF constructs a time-varying linear state space system by taking first order Taylor expansions of g^* and h^* around the forecasted state $x = \hat{x}_{t|t-1}$:

$$g^*(x_t) \approx g^*(x) + D_x g^*(x)(x_t - x), \tag{16}$$

$$h^*(x_t) \approx h^*(x) + D_x h^*(x)(x_t - x). \tag{17}$$

Thus, the oracle EKF relies on knowing the levels and slopes of the global solution at x.[4]

The Ideal Local Linear Solution. In practice, local linear methods such as TP and DP will not exactly reproduce the expansions in Equations (16) and (17). We therefore first describe an idealized local solution method to clarify the source of this discrepancy. We begin with the observation that the same first order objects, namely levels and slopes, can be recovered from the equilibrium system itself once a candidate continuation policy $g(x_{t+1}) = y_{t+1}$ for future controls is specified. To see this, fix a current state $x_t = (x_{1,t}', x_{2,t}')'$. Conditional on continuation policy g, the period t equilibrium system (1) implicitly defines the current control $y_t = \hat{g}(x_t; g)$ and the endogenous transition $x_{1,t+1} = \hat{h}(x_t; g)$. The next period state is

$$x_{t+1} = \begin{pmatrix} \hat{h}(x_t; g) \\ Ax_{2,t} \end{pmatrix} + \begin{pmatrix} 0 \\ B \end{pmatrix} \eta_{t+1}, \tag{18}$$

where $(\hat{g}, \hat{h})$ solve the system of nonlinear equations defined by

$$\mathbb{E}_t[f(g(x_{t+1}), \hat{g}(x_t; g), x_{t+1}, x_t)] = 0, \qquad x_{t+1} \text{ given by (18)}. \tag{19}$$

Fix a candidate continuation policy g and an expansion point x. Suppose that the system in Equation (19), with x_{t+1} given by Equation (18), admits a unique solution $\left(\widehat{g}(x; g), \widehat{h}(x; g)\right)$. Assume that the expected residual in Equation (19) is continuously differentiable jointly in the current state x_t and the unknowns $(y_t, x_{1,t+1})$ in a neighborhood of this solution. Assume also that its Jacobian with respect to $(y_t, x_{1,t+1})$ is nonsingular at the solution. Whenever

[4]The EKF likewise linearizes the observation mapping around $x = \hat{x}_{t|t-1}$, when not already linear.

these conditions hold, the continuation policy g locally induces the mapping

$$g \longmapsto \Big(\widehat{g}(\cdot; g), \widehat{h}(\cdot; g)\Big).$$

Furthermore, the implicit function theorem then implies that the induced mappings $\widehat{g}(\cdot; g)$ and $\widehat{h}(\cdot; g)$ are continuously differentiable in a neighborhood of x. Hence, given some continuation policy g, the local linear solution at the point x is the first-order expansion of the induced equilibrium mapping:

$$\widetilde{g}(x_t; x, g) = \widehat{g}(x; g) + D_x\widehat{g}(x; g)(x_t - x), \tag{20}$$

$$\widetilde{h}(x_t; x, g) = \widehat{h}(x; g) + D_x\widehat{h}(x; g)(x_t - x). \tag{21}$$

If $g = g^*$, then $\Big(g^*(x), h^*(x)\Big)$ solves Equation (19). Uniqueness implies that the induced solution coincides with the true solution, as well as its derivative w.r.t. x:

$$\widehat{g}(x; g^*) = g^*(x), \qquad \widehat{h}(x; g^*) = h^*(x), \tag{22}$$

$$D_x\widehat{g}(x; g^*) = D_x g^*(x), \qquad D_x\widehat{h}(x; g^*) = D_x h^*(x). \tag{23}$$

Hence, when $g = g^*$, (22)–(23) imply that the levels and slopes entering the local solutions in Equations (20)–(21) coincide with the Taylor coefficients used by the oracle EKF. The EKF applied to the true global solution is therefore the oracle case of these local linear methods.

Approximation to the Oracle EKF. Since the true continuation policy g^* is unknown, a local solution method must approximate the continuation policy entering (19). TP restricts the continuation policy to the same local rule used at the current state, whereas DP constructs the continuation policy along an auxiliary path. In both cases, the accuracy of the local solution depends on the quality of this continuation-policy approximation.

We now characterize how different such local methods, combined with the TVKF, are from the oracle EKF. Departures from the oracle arise through two channels: (i) errors in the state-space coefficients despite using the same expansion point due to approximating g^*, in other words, how different $\hat{g}(x; g)$, $D_x\hat{g}(x; g)$, $\hat{h}(x; g)$, and $D_x\hat{h}(x; g)$ are when using an approximate continuation policy rather than g^* and how this maps into the state space form, and (ii) the resulting difference of the expansion points generated by the two filters. We derive pathwise bounds on how these discrepancies propagate through the filtering recursion and translate them into bounds on filtered states, likelihoods, and parameter estimates.

Fix a parameter value θ and a realized observation path $\{y_t^{\text{obs}}\}_{t\geq 1}$. Conditional on these, the oracle and local filtering recursions are deterministic. Let $T \geq 1$ denote the sample size. Let $\|\cdot\|$ denote the Euclidean norm for vectors and the Frobenius norm for matrices. For collections of vectors and matrices, we use the product norm, defined as the square root of the sum of the squared component norms. $\|A\|_{\text{op}} := \sup_{\|z\|=1}\|Az\|$ is the operator norm.

For $j \in \{O, m\}$, where O denotes the oracle EKF and m denotes the local solution method,

$$\Xi^j(x;\theta) := \Big(c^j(x;\theta), T^j(x;\theta), d^j(x;\theta), Z^j(x;\theta)\Big)$$

collects the transition and observation coefficients implied by method j at expansion point x. Denote by $\hat{x}^j_{t|t-1}(\theta)$ the forecasted state by method j at time t given a $t-1$ history. Then, we can define the same-expansion-point error of method m relative to the Oracle by

$$r_{t,m}(\theta) := \Xi^m\left(\hat{x}^m_{t|t-1}(\theta);\theta\right) - \Xi^O\left(\hat{x}^m_{t|t-1}(\theta);\theta\right), \qquad e_{t,m}(\theta) := \|r_{t,m}(\theta)\|. \tag{24}$$

The error $e_{t,m}$ isolates the approximation error of the local solution from additional differences in the coefficients that would be induced by using different expansion points. Below, we express all deviations in terms of $e_{t,m}(\theta)$, but underlying this error are deviations in the policy functions and laws of motion g and h, which map into the state space matrices in Ξ.

To make this connection explicit, at the common expansion point $\widehat{x}^m_{t|t-1}(\theta)$, the transition coefficients c^j and T^j are determined by the level and slope of the induced law of motion h^j, while the observation coefficients d^j and Z^j are determined by the level and slope of the induced policy function g^j. If the mapping from these levels and slopes into the state-space coefficients is continuously differentiable between the local and oracle values, the mean-value theorem implies that there exists a finite constant $L_{\Xi,t,m}(\theta)$ such that

$$\begin{aligned} e_{t,m}(\theta) \leq L_{\Xi,t,m}(\theta)\Bigg[&\left\|\widehat{g}(\widehat{x}^m_{t|t-1}(\theta); g_{t,m}) - g^*(\widehat{x}^m_{t|t-1}(\theta))\right\|^2 + \left\|D_x\widehat{g}(\widehat{x}^m_{t|t-1}(\theta); g_{t,m}) - D_x g^*(\widehat{x}^m_{t|t-1}(\theta))\right\|_F^2 \\ &+ \left\|\widehat{h}(\widehat{x}^m_{t|t-1}(\theta); g_{t,m}) - h^*(\widehat{x}^m_{t|t-1}(\theta))\right\|^2 + \left\|D_x\widehat{h}(\widehat{x}^m_{t|t-1}(\theta); g_{t,m}) - D_x h^*(\widehat{x}^m_{t|t-1}(\theta))\right\|_F^2\Bigg]^{1/2}. \end{aligned}$$

Here $L_{\Xi,t,m}(\theta)$ measures the local sensitivity of the state-space coefficients (c, T, d, Z) to changes in the levels and slopes of the policy function and law of motion.

Collect the filtered and forecasted means and covariances after filtering observations $y^{\text{obs}}_{1:t}$ in

$$S^j_t(\theta) := \Big(\hat{x}^j_{t|t}(\theta), P^j_{t|t}(\theta), \hat{x}^j_{t+1|t}(\theta), P^j_{t+1|t}(\theta)\Big).$$

Let Π_x denote the selection matrix that selects the forecasted state from S, so that $\Pi_x S^j_{t-1}(\theta) = \hat{x}^j_{t|t-1}(\theta)$. For a fixed parameter value θ and observation y^{obs}_t, let $\Psi_{\theta,t}$ denote the Kalman update-and-prediction map taking as inputs the previous filtered and forecasted states and the state-space matrices. We can then write

$$S^j_t(\theta) = \Psi_{\theta,t}\left(S^j_{t-1}(\theta), \Xi^j\left(\Pi_x S^j_{t-1}(\theta); \theta\right)\right), \tag{25}$$

with $S^j_0(\theta)$ all filter moments supplied to the first recursive step. Define the oracle propagation map $\Phi^O_{t:s,\theta}(S)$ by starting from state S at date s and applying the oracle EKF recursion through date t, recomputing the oracle state-space coefficients at the forecasted state generated at each step:

$$\Psi^O_{\theta,t}(S) := \Psi_{\theta,t}\left(S, \Xi^O(\Pi_x S; \theta)\right) \quad \text{and, for } t \geq s, \quad \Phi^O_{t:s,\theta} := \Psi^O_{\theta,t} \circ \cdots \circ \Psi^O_{\theta,s+1}, \quad \Phi^O_{s:s,\theta} = I.$$

At date t, define the counterfactual filter state

$$\widetilde{S}^{O|m}_t(\theta) := \Psi_{\theta,t}\left(S^m_{t-1}(\theta), \Xi^O\left(\hat{x}^m_{t|t-1}(\theta); \theta\right)\right), \tag{26}$$

i.e., performing a Kalman update using the filtered and forecasted means and covariances implied by the local linear method m, but the transition and observation coefficients implied by the Oracle. Furthermore, define the corresponding state-disturbance segment

$$\mathcal{L}^S_{t,m}(\theta) := \left[\widetilde{S}^{O|m}_t(\theta), S^m_t(\theta)\right],$$

where $[a, b] := \{(1-u)a + ub : u \in [0, 1]\}$ denotes the line segment joining a and b. The states $S^m_t(\theta)$ and $\widetilde{S}^{O|m}_t(\theta)$ have the same incoming filter state and expansion point and differ only in the state space system matrices used at date t. For each t, define the same-expansion-point coefficient segment

$$\mathcal{L}^{\Xi}_{t,m}(\theta) := \left[\Xi^O\left(\hat{x}^m_{t|t-1}(\theta); \theta\right), \Xi^m\left(\hat{x}^m_{t|t-1}(\theta); \theta\right)\right].$$

Assumption 1. *For every $T \geq 1$ and every $1 \leq s \leq t \leq T$, the oracle propagation map $\Phi^O_{t:s,\theta}$ is well defined on $\mathcal{L}^S_{s,m}(\theta)$. There exist constants $C_F \geq 1$ and $\rho \in (0, 1)$, independent of s, t, and T, such that*

$$\left\|\Phi^O_{t:s,\theta}(S) - \Phi^O_{t:s,\theta}(\widetilde{S})\right\| \leq C_F \rho^{t-s} \left\|S - \widetilde{S}\right\| \tag{27}$$

for every $S, \widetilde{S} \in \mathcal{L}^S_{s,m}(\theta)$.

Assumption 1 is a local exponential-forgetting condition on the full oracle EKF recursion. It requires two oracle filter paths initialized at different filter states, but propagated by the same subsequent oracle maps $\Phi^O_{t:s,\theta}$ and observations, to converge geometrically. This condition is consistent with classical filter stability results (Bougerol, 1993; Le Gland and Mevel, 2000) and EKF stability analyses such as Bonnabel and Slotine (2015).

Assumption 2. *For every $T \geq 1$ and each $t = 1, \ldots, T$, there exists an open set $\mathcal{U}^{\Xi}_{t,m}(\theta)$ containing $\mathcal{L}^{\Xi}_{t,m}(\theta)$ such that the map $\Xi \longmapsto \Psi_{\theta,t}\left(S^m_{t-1}(\theta), \Xi\right)$ is continuously differentiable on $\mathcal{U}^{\Xi}_{t,m}(\theta)$. The corresponding Kalman update is admissible throughout this set; in particular, every innovation covariance matrix entering the update is positive definite.*

Assumption 2 imposes local smoothness along the coefficient segments connecting the oracle and the local method's coefficients at the same expansion point. Under Assumption 2, define

$$L_{K,T} := \max_{1 \leq t \leq T} \sup_{\Xi \in \mathcal{L}^{\Xi}_{t,m}(\theta)} \left\| D_{\Xi} \Psi_{\theta,t}\left(S^m_{t-1}(\theta), \Xi\right) \right\|_{\mathrm{op}}. \tag{28}$$

Compactness of the segments and continuity of the derivatives imply that $L_{K,T} < \infty$.

Theorem 1. *Suppose Assumptions 1 and 2 hold. For any $T \geq 1$, let $d_{t,m}(\theta) := \left\| S^m_t(\theta) - S^O_t(\theta) \right\|$ and $C_{\Psi,T} := C_F L_{K,T}$. Suppose that the oracle and local filters have a common initialization, $S^m_0(\theta) = S^O_0(\theta)$. Then, for every $t = 1, \ldots, T$,*

$$d_{t,m}(\theta) \leq C_{\Psi,T} \sum_{r=1}^{t} \rho^{t-r} e_{r,m}(\theta). \tag{29}$$

Consequently,

$$\max_{0 \leq t \leq T} d_{t,m}(\theta) \leq \frac{C_{\Psi,T}}{1-\rho} \max_{1 \leq t \leq T} e_{t,m}(\theta), \quad \text{and} \quad \frac{1}{T}\sum_{t=1}^{T} d_{t,m}(\theta) \leq \frac{C_{\Psi,T}}{1-\rho} \frac{1}{T} \sum_{t=1}^{T} e_{t,m}(\theta). \tag{30}$$

Proof in Appendix B.1.

Theorem 1 bounds the filter discrepancy $d_{t,m}(\theta)$ between the local method combined with the TVKF and the oracle EKF in terms of the same-expansion-point state-space coefficient errors $e_{t,m}(\theta)$. The proof decomposes this discrepancy into the difference between the state space matrices of the oracle EKF and the local method at the same expansion point, and propagates these differences forward under the full oracle EKF recursion. Assumption 1 ensures that the effect of each perturbation decays geometrically as it is propagated forward, including the resulting feedback through subsequent expansion points and coefficients.

Remark 1. *Theorem 1 also applies to a filter that computes its state-space coefficients once at a fixed expansion point $x^\circ(\theta)$, typically the steady state, and uses $\Xi^{\text{fix}}(x^\circ(\theta);\theta)$ in every period. This nests the common approach of using log-linearization to solve the model combined with the standard Kalman filter. By the triangle inequality,*

$$\begin{aligned} e_{t,\text{fix}}(\theta) &= \left\|\Xi^{\text{fix}}(x^\circ(\theta);\theta) - \Xi^O\left(\widehat{x}^{\text{fix}}_{t|t-1}(\theta);\theta\right)\right\| \\ &\leq \left\|\Xi^{\text{fix}}(x^\circ(\theta);\theta) - \Xi^O(x^\circ(\theta);\theta)\right\| + \left\|\Xi^O(x^\circ(\theta);\theta) - \Xi^O\left(\widehat{x}^{\text{fix}}_{t|t-1}(\theta);\theta\right)\right\|. \end{aligned}$$

The first term is the difference in state-space matrices between the fixed method and Oracle EKF at the fixed expansion point, while the second term measures variation in the oracle coefficients as the filtered state moves away from that point. Theorem 1 therefore implies

$$d_{t,\text{fix}}(\theta) \leq C_{\Psi,T} \sum_{r=1}^{t} \rho^{t-r} \left[\left\|\Xi^{\text{fix}}(x^\circ(\theta);\theta) - \Xi^O(x^\circ(\theta);\theta)\right\| + \left\|\Xi^O(x^\circ(\theta);\theta) - \Xi^O\left(\widehat{x}^{\text{fix}}_{r|r-1}(\theta);\theta\right)\right\| \right].$$

If the oracle coefficient map is Lipschitz on the relevant segments, then

$$\left\|\Xi^O(x^\circ(\theta);\theta) - \Xi^O\left(\widehat{x}^{\text{fix}}_{t|t-1}(\theta);\theta\right)\right\| \leq L_{O,\text{fix},T} \left\|\widehat{x}^{\text{fix}}_{t|t-1}(\theta) - x^\circ(\theta)\right\|.$$

Thus, the error bound for a fixed linearization depends on its accuracy at the fixed expansion point and how far the filtered state moves away from that point. Note that these results provide separate error bounds for the fixed and time-varying linearizations, but not necessarily a ranking between the methods themselves.

Next, we study how the filtering approximation affects the log likelihood and the corresponding maximum likelihood estimates. Let $\mathcal{V} \subseteq \Theta$ be a compact parameter set. For each t and $\theta \in \mathcal{V}$, define the expansion-point segment

$$\mathcal{L}^x_{t,m}(\theta) := \left[\widehat{x}^O_{t|t-1}(\theta), \widehat{x}^m_{t|t-1}(\theta)\right].$$

Assumption 3. *For every $\theta \in \mathcal{V}$, there exists an open set $\mathcal{U}^x_m(\theta)$ containing $\bigcup_{t=1}^T \mathcal{L}^x_{t,m}(\theta)$ such that the map $x \longmapsto \Xi^O(x;\theta)$ is continuously differentiable on $\mathcal{U}^x_m(\theta)$. Moreover, the derivatives are uniformly bounded on the relevant segments:*

$$L_{O,T} := \max_{1\leq t\leq T} \sup_{\theta\in\mathcal{V}} \sup_{x\in\mathcal{L}^x_{t,m}(\theta)} \left\|D_x \Xi^O(x;\theta)\right\|_{\text{op}} < \infty. \tag{31}$$

For $j \in \{O, m\}$, define the average log likelihood

$$Q_{T,j}(\theta) := \frac{1}{T}\sum_{t=1}^{T} \ell_t^j(\theta), \quad \text{where} \quad \ell_t^j(\theta) = -\frac{1}{2}\left[\log\det F_t^j(\theta) + v_t^j(\theta)' F_t^j(\theta)^{-1} v_t^j(\theta)\right],$$

up to a common additive constant. Here, $v_t^j(\theta)$ and $F_t^j(\theta)$ denote the innovation and innovation covariance generated by method j, as defined in Algorithm 1. Write

$$\zeta_t^j(\theta) := \left(S_{t-1}^j(\theta), \Xi^j\left(\widehat{x}_{t|t-1}^j(\theta); \theta\right)\right),$$

so that $v_t^j(\theta) = v_t\left(\zeta_t^j(\theta); \theta\right)$, and $F_t^j(\theta) = F_t\left(\zeta_t^j(\theta); \theta\right)$. For each t, define

$$\mathcal{G}_{t,m} := \left\{(\zeta, \theta) : \theta \in \mathcal{V},\ \zeta \in \left[\zeta_t^O(\theta), \zeta_t^m(\theta)\right]\right\}.$$

Assumption 4. *For each $t = 1, \ldots, T$ and $j \in \{O, m\}$, the map $\theta \mapsto \zeta_t^j(\theta)$ is continuous on $\mathcal{V}$. The maps $(\zeta, \theta) \longmapsto v_t(\zeta; \theta)$ and $(\zeta, \theta) \longmapsto F_t(\zeta; \theta)$ are continuous on an open neighborhood of $\mathcal{G}_{t,m}$ and are differentiable with respect to ζ there. The maps $(\zeta, \theta) \longmapsto D_\zeta v_t(\zeta; \theta)$ and $(\zeta, \theta) \longmapsto D_\zeta F_t(\zeta; \theta)$ are jointly continuous on that neighborhood. Moreover,*

$$\underline{f} := \min_{1 \le t \le T} \inf_{(\zeta,\theta) \in \mathcal{G}_{t,m}} \lambda_{\min}\left(F_t(\zeta; \theta)\right) > 0. \tag{32}$$

Assumption 3 controls changes in the oracle state-space coefficients induced by differences between the oracle and local expansion points. Assumption 4 imposes local smoothness and nonsingularity of the Gaussian innovation likelihood along the segments joining the oracle and local likelihood arguments. Compactness of $\mathcal{V}$, continuity of the filter paths, and finiteness of T ensure that the relevant likelihood quantities and derivatives are uniformly bounded on these segments.

Assumption 5. *The parameter set $\mathcal{V}$ is compact and convex. The oracle criterion $Q_{T,O}(\theta)$ is twice continuously differentiable on an open neighborhood of $\mathcal{V}$, and there exists $\mu > 0$ such that*

$$-\nabla_\theta^2 Q_{T,O}(\theta) \succeq \mu I, \qquad \theta \in \mathcal{V}. \tag{33}$$

The criterion $Q_{T,m}(\theta)$ of the local method is continuous on $\mathcal{V}$.

Assumption 5 imposes a global quadratic curvature condition on the realized finite-sample oracle criterion. Alternatively, the following assumption requires quadratic curvature only in a neighborhood of a unique oracle maximizer.

Assumption 6. *The parameter set $\mathcal{V}$ is compact and convex. The oracle criterion $Q_{T,O}$ is continuous on $\mathcal{V}$ and has a unique maximizer $\widehat{\theta}_{T,O}$. There exists $r > 0$ such that $\mathcal{A}_r := \left\{\theta \in \mathcal{V} : \left\|\theta - \widehat{\theta}_{T,O}\right\| \geq r\right\} \neq \varnothing$. Define $\mathcal{B}_r := \left\{\theta \in \mathcal{V} : \left\|\theta - \widehat{\theta}_{T,O}\right\| \leq r\right\}$. The oracle criterion is twice continuously differentiable on an open neighborhood of $\mathcal{B}_r$, and*

$$\mu := \inf_{\theta \in \mathcal{B}_r} \lambda_{\min}\left(-\nabla_\theta^2 Q_{T,O}(\theta)\right) > 0.$$

The criterion $Q_{T,m}$ for the local method is continuous on $\mathcal{V}$.

Under Assumption 6, define

$$\varepsilon_0 := \frac{1}{2}\left[Q_{T,O}(\widehat{\theta}_{T,O}) - \max_{\theta \in \mathcal{A}_r} Q_{T,O}(\theta)\right]. \tag{34}$$

Because $\mathcal{A}_r$ is nonempty and compact, the displayed maximum exists. Continuity of $Q_{T,O}$, uniqueness of $\widehat{\theta}_{T,O}$, and $r > 0$ imply that $\varepsilon_0 > 0$.

Assumption 6 replaces global curvature with local quadratic curvature around a unique oracle maximizer. Compactness and continuity then imply separation of this maximizer from parameter values outside the local neighborhood. A condition like in Assumption 5 or 6 is necessary: if the criterion is locally flat, a small perturbation can lead to a large change in its maximizer.

Define the uniform average same-expansion-point error by

$$\mathcal{E}_{T,m} := \sup_{\theta \in \mathcal{V}} \left\{\frac{1}{T}\sum_{t=1}^{T} e_{t,m}(\theta)\right\}. \tag{35}$$

Theorem 2. *Suppose Assumption 1 holds uniformly over $\theta \in \mathcal{V}$ with common constants C_F and ρ, which are independent of T. For a given T, suppose Assumptions 2 and 3 hold uniformly over $\theta \in \mathcal{V}$ with finite constants $L_{K,T}$ and $L_{O,T}$, and suppose the two filters use common initial conditions for every $\theta \in \mathcal{V}$. If Assumption 4 also holds, then, for method m, there exists a finite constant $C_{\ell,m,T}$, independent of $\theta \in \mathcal{V}$, such that*

$$\sup_{\theta \in \mathcal{V}} |Q_{T,m}(\theta) - Q_{T,O}(\theta)| \leq C_{\ell,m,T}\mathcal{E}_{T,m}. \tag{36}$$

Suppose, in addition, that either of the following conditions holds: (i) Assumption 5 holds; or (ii) Assumption 6 holds and $C_{\ell,m,T}\mathcal{E}_{T,m} < \varepsilon_0$, where ε_0 is defined in (34)*. Then, for*

$\widehat{\theta}_{T,j} \in \arg\max_{\theta\in\mathcal{V}} Q_{T,j}(\theta),\ j \in \{O, m\}$, *we have*

$$\left\|\widehat{\theta}_{T,m} - \widehat{\theta}_{T,O}\right\| \leq 2\left[\frac{C_{\ell,m,T}}{\mu}\mathcal{E}_{T,m}\right]^{1/2}, \tag{37}$$

where μ denotes the curvature constant in Assumption 5 or 6, respectively.

Proof. See Appendix B.2.

Theorem 2 extends the filtering bounds to the average log likelihood and the corresponding maximum likelihood estimates. Regularity of the oracle coefficient map controls differences induced by the oracle and local expansion points, while smoothness of the innovation likelihood converts differences in the filter and coefficient paths into uniform differences between the oracle and local criteria. The global identification Assumption 5, or the local identification Assumption 6 together with the separation threshold, then converts likelihood closeness into closeness of their maximizers. The result is pathwise with bounds uniform over $\theta \in \mathcal{V}$.

Thus, our method delivers a computationally tractable approximation to the oracle EKF without needing to know or compute the global nonlinear solution: we approximate the unknown continuation policy, solve the equilibrium system locally, and then apply the time-varying Kalman filter. Of course, the oracle EKF itself remains a first-order filtering approximation to the full nonlinear state-space model. The quantitative applications below help assess the remaining difference relative to global-solution nonlinear filtering.

3 Three Canonical Models

In this section, we compare the performance of our proposed estimation framework to existing methods using simulations of the Diamond-Mortensen-Pissarides (DMP) model, and further demonstrate the method by revisiting the textbook three-equation New Keynesian (NK) model, with and without the zero-lower bound, as well as the multi-country real business cycle model in Juillard and Villemot (2011). The first model is known to be poorly approximated by standard perturbation approaches (Petrosky-Nadeau and Zhang, 2017) and has an analytical global model solution, which allows us to test the theory developed in the previous section. The second model demonstrates how the method handles occasionally-binding constraints. The third model demonstrates the scalability of the method to large state spaces.

3.1 Labor Search Model

We briefly describe the model environment. The only non-standard assumptions are those made with respect to the timing of separations and hires, and the simplification of the wage

which allows for an analytic global solution. Following the literature on labor search in NK models (Christiano et al., 2016), we assume workers who lose their jobs at the start of the period can search within the period for a new job.[5]

Firms post vacancies V_t to maximize the present discounted value of profit:

$$\mathbb{E}_0 \left[\sum_{t=1}^{\infty} \beta^t \Big\{ Z_t N_t - W_t N_t - c V_t \Big\} \right] \quad \text{subject to} \quad N_t = (1-\delta) N_{t-1} + Q_t V_t,$$

taking wages W_t as given. Here, the flow payoff is output $Z_t N_t$, net of wage expenditures $W_t N_t$ and vacancy posting costs cV_t, δ is the job destruction rate, and Q_t the vacancy filling rate. We close the model by assuming the wage is given by $W_t = \eta Z_t + (1-\eta)\nu$, where η is the worker's bargaining weight, and ν is the outside option of the worker[6]. The model consists of one policy Θ_t, market tightness, and one state, Z_t, labor productivity. The equilibrium is characterized by two equations; a job creation condition and law of motion for productivity:

$$\frac{c}{Q_t} = (1-\eta)\left(Z_t - \nu\right) + \beta(1-\delta)\mathbb{E}_t \left[\frac{c}{Q_{t+1}} \right], \quad \text{and} \quad \ln Z_{t+1} = \rho \ln Z_t + \sigma \varepsilon_{t+1}.$$

Let lower case letters denote log deviations from steady state, e.g. $x_t = \ln(X_t / X_{ss})$ for a generic x_t, We assume the matching function is Cobb-Douglas which implies $q_t = -\alpha\theta_t$ where α is the matching function elasticity. Let $\gamma = \beta(1-\delta)$, rewriting the system in log deviations from steady state, the exact equilibrium system becomes:

$$e^{\alpha\theta_t} = \frac{1-\gamma}{1-\nu}\left[e^{z_t} - \nu\right] + \gamma \mathbb{E}_t \left[e^{\alpha\theta_{t+1}} \right], \quad \text{and} \quad z_{t+1} = \rho z_t + \sigma \varepsilon_{t+1}.$$

Model Solutions. An attractive feature of this model is that its global policy function and derivatives are available analytically. This allows us to construct the actual oracle EKF and compare its performance with TP+TVKF and DP+TVKF. We also report results on the log-linear solution with the Kalman filter, various higher-order perturbations with particle filters, and the exact global solution with a particle filter. See McCrary (2026) for further details on each solution method. Note the perturbation solution to this model is stable and a high-enough perturbation order can approximate the global solution well.

[5] This assumption is typically justified as a way of making low-frequency (quarterly) NK models conform with labor markets where most unemployment spells last only a few weeks. Ahn and Hamilton (2022) find two-thirds of workers exit unemployment within the first quarter of an unemployment spell.

[6] As noted in McCrary (2026), this is not the Nash bargaining solution, but is a convenient assumption that allows for a closed form global solution.

Solution Accuracy. Our theoretical results above imply that the distance between the local linear filters and the oracle EKF based on the global solution can be bounded in terms of the error in the policy rule $e_{t,m}(\theta)$. Since in this application, the global policy function is available analytically (i.e., the Oracle is known), we can evaluate this error directly. Table 1 reports stationary mean absolute errors and supremum errors for the log market-tightness policy, together with the corresponding errors in its slope with respect to TFP. The slope errors measure the distance between the slopes of the local linear methods and the oracle EKF. The results show that the local linear methods perform well. Taylor Projection substantially improves on the log-linear approximation and, overall, also performs better than a second-order perturbation solution. Dynamic Perturbation performs especially well; the accuracy of DP is better than a third-order and slightly worse than a sixth-order perturbation solution, overall close to global accuracy.

Table 1: Policy approximation errors relative to the global solution.

Method	Stationary mean abs. error	Supremum abs. error
Panel A: Log market-tightness policy		
Log-linear	0.0150	0.1869
Perturbation 2nd	0.00195	0.0484
Perturbation 3rd	0.00033	0.0139
Perturbation 6th	3.51×10^{-6}	0.0004
Taylor Projection	0.00297	0.0130
Dynamic Perturbation	0.00268	0.0063
Panel B: Local policy slope		
Log-linear	1.0712	6.289
Perturbation 2nd	0.1646	2.348
Perturbation 3rd	0.0320	0.875
Perturbation 6th	0.0005	0.045
Taylor Projection	0.0875	0.799
Dynamic Perturbation	0.0008	0.001

Note: Errors are measured relative to the analytic global solution. Panel A reports errors in the log market-tightness policy, $|\theta^m(z) - \theta^*(z)|$. Panel B reports errors in the local policy slope, $|\theta_z^m(z) - \theta_z^*(z)|$. Stationary mean absolute errors use stationary Gaussian weights over the state grid, while supremum absolute errors are computed over $z/\sigma_z \in [-3, 3]$. Calibration: $\beta = 0.96^{1/52}$, $\alpha = 0.50$, $\eta = 0.50$, $\nu = 0.84$; $f_{ss} = 0.50$, $u_{ss} = 0.10$; $\rho = 0.9895$, $\sigma = 0.0034$.

State-Space Representation. To make explicit how a local linear solution of the form $\theta_t = G_0(z) + G_z(z)(z_t - z)$ maps into the state space model, assume we observe market tightness and log wages, with measurement errors. The local linear solution methods imply a linear mapping of log market tightness $\hat{\theta}_t$ to the state, and wages $w_t = \log(W_t/W_{ss})$ can be linearized using a first order expansion around the current state forecast $\hat{z}_{t|t-1}$: $w(z_t) \approx$

$w\left(\hat{z}_{t|t-1}\right) + \frac{\partial w(z)}{\partial z}\Big|_{z=\hat{z}_{t|t-1}} (z_t - \hat{z}_{t|t-1})$. This implies the following state-space representation:

$$\begin{pmatrix} \theta_t^{\text{obs}} \\ w_t^{\text{obs}} \end{pmatrix} = \begin{pmatrix} G_0\left(\hat{z}_{t|t-1}\right) - G_z\left(\hat{z}_{t|t-1}\right)\hat{z}_{t|t-1} \\ \log\left(\frac{\eta e^{\hat{z}_{t|t-1}} + (1-\eta)\nu}{\eta + (1-\eta)\nu}\right) - \frac{\hat{z}_{t|t-1}\eta e^{\hat{z}_{t|t-1}}}{\eta e^{\hat{z}_{t|t-1}} + (1-\eta)\nu} \end{pmatrix} + \begin{bmatrix} G_z\left(\hat{z}_{t|t-1}\right) \\ \frac{\eta e^{\hat{z}_{t|t-1}}}{\eta e^{\hat{z}_{t|t-1}} + (1-\eta)\nu} \end{bmatrix} z_t + \varepsilon_t^{ME}$$

$$z_{t+1} = \rho_z z_t + \eta_{t+1}, \quad \eta_t \sim (0, \sigma_z^2).$$

We can then use Algorithm 1 to filter the underlying states and evaluate the likelihood.

Monte Carlo We simulate 1,000 datasets of length $T = 200$ from the DMP model using the global solution, adding measurement error to observable market tightness and wages with a standard deviation equal to 20% of its true standard deviation. We compare eight solution–filtering approaches: (i) log-linear perturbation with the Kalman Filter; (ii)–(iv) second-, third-, and sixth-order perturbations with the Particle Filter; (v) the global solution with the Particle Filter; (vi) TP+TVKF; (vii) DP+TVKF; and (viii) the global solution with the EKF. Structural parameters are fixed at their true values. Table 2 reports normalized MSEs for the filtered states and computation times.

Table 2: Comparison of solution and filtering methods for the DMP model at $T = 200$. MSE is scaled by the unconditional variance of the corresponding variable.

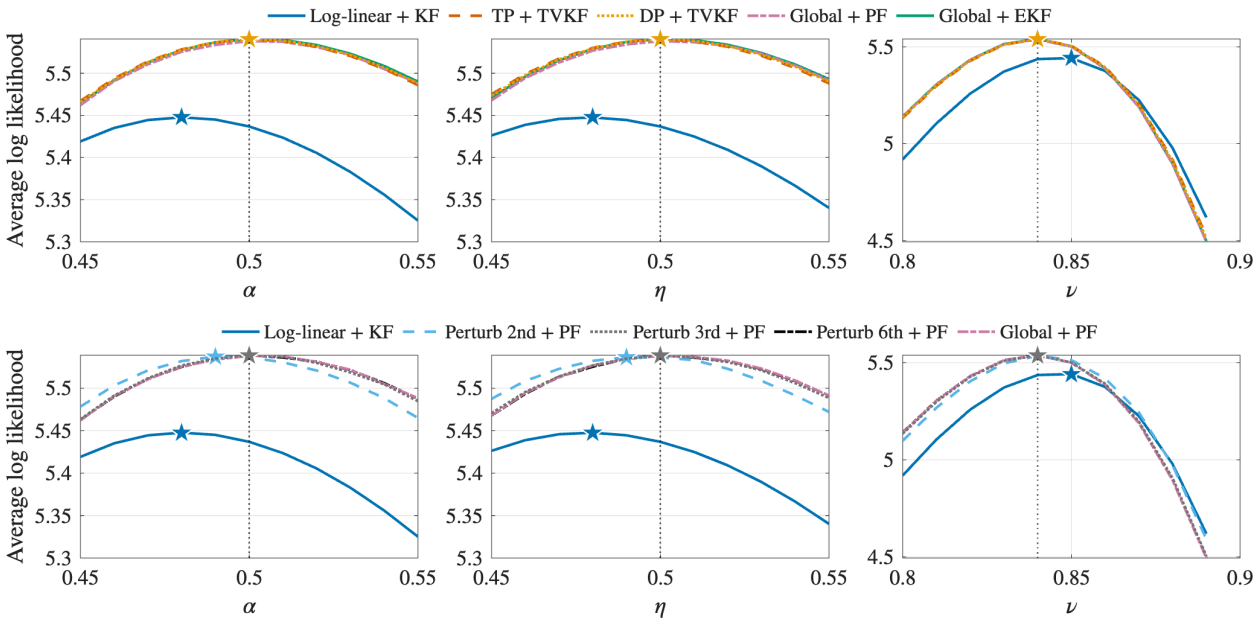
Method	Particles	$100 \times \text{MSE}_\theta$	$100 \times \text{MSE}_w$	$100 \times \text{MSE}_z$	Time (s)
Global + EKF	–	1.249	1.259	1.257	0.0019
DP + TV-KF	–	1.253	1.263	1.261	0.0069
TP + TV-KF	–	1.254	1.264	1.262	0.0220
Log-linear + KF	–	1.564	1.538	1.453	0.0006
	200	1.283	1.290	1.289	0.0044
2nd-order pert. + PF	2,000	1.258	1.267	1.266	0.0517
	20,000	1.255	1.265	1.264	0.1947
	200	1.274	1.283	1.282	0.0065
3rd-order pert. + PF	2,000	1.251	1.260	1.259	0.0753
	20,000	1.248	1.258	1.256	0.2610
	200	1.274	1.283	1.282	0.0260
6th-order pert. + PF	2,000	1.250	1.260	1.258	0.2231
	20,000	1.247	1.257	1.256	0.6982
	200	1.273	1.282	1.281	0.0048
Global + PF	2,000	1.250	1.260	1.258	0.0401
	20,000	1.247	1.257	1.256	0.2171

Note: Calibration: $\beta = 0.96^{1/52}$, $\alpha = 0.50$, $\eta = 0.50$, $\nu = 0.84$; $f_{ss} = 0.50$, $u_{ss} = 0.10$; $\rho = 0.9895$, $\sigma = 0.0034$.

The log-linear approximation with the Kalman Filter is the fastest method, but is also noticeably less accurate. DP+TVKF, TP+TVKF, and Global+EKF deliver nearly identical filtering accuracy. These results are consistent with the results of Theorem 1 that local time-varying approximations can closely track the oracle EKF. With only 200 particles, the Particle Filter is less accurate than the local time-varying Kalman approaches. Higher-order perturbations combined with sufficiently many particles can essentially reproduce the accuracy of the global solution, but become increasingly costly as the perturbation order and number of particles increase. Global+EKF and Global+PF are relatively inexpensive in this application because the model admits an analytical global solution; the computational advantage of the local time-varying approaches matters more in larger models for which characterizing the global solution is more costly and the curse of dimensionality hurts the particle filter performance.

Comparing Likelihoods. To compare likelihood functions, data of length $T = 5000$ are simulated from the global solution, adding measurement error to observable market tightness and wages. Figure 1 shows two times three panels of likelihood functions where all parameters are kept fixed at their true value, except for one parameter that is varied on a grid. We consider matching elasticity α, outside option ν and bargaining power η.

Figure 1: Comparing likelihoods for match elasticity α, bargaining power η and outside option ν.



Note: Calibration: $\beta = 0.96^{1/52}$, $\alpha = 0.50$, $\eta = 0.50$, $\nu = 0.84$; $f_{ss} = 0.50$, $u_{ss} = 0.10$; $\rho = 0.9895$, $\sigma = 0.0034$. The figure compares likelihoods computed using the time-varying Kalman filter with Dynamic Perturbation and Taylor Projection, the Particle Filter with second-, third-, and sixth-order perturbation and with the global model solution, the Extended Kalman Filter with the global model solution, and the log-linearized model with the Kalman Filter. Stars indicate the maximizer of each conditional likelihood. Results are based on a simulation of length $T = 5000$ from the global model solution. All remaining parameters are held fixed at their true values.

Figure 1 shows that the Log-linear+KF approach leads to biased estimates. In contrast, TP+TV-KF, Global+PF, and Global+EKF yield almost indistinguishable likelihoods centered around the true parameter values, consistent with Theorem 2. The second row in Figure 1 shows the perturbation approaches alongside the global solution, confirming the results of Table 2 that a third order perturbation is approximately as accurate as the global solution for this particular model. Second-order perturbations still lead to biased estimates, albeit with smaller bias than the log-linearization.

3.2 Three-Equation New Keynesian Model with and without Zero Lower Bound

Next, consider the canonical three-equation New Keynesian model (Galí, 2015), with and without a zero lower bound on the interest rate. Given the model is well known, we leave most model details to Online Appendix C. A minimal state-space representation of the model consists of the state variables: productivity z_t, discount factor shocks d_t and monetary policy shocks m_t, all following an AR(1), collected in $x_t = (z_t, d_t, m_t)$, and policy functions for inflation and employment $\{\Pi(x_t), N(x_t)\}$. These policies are determined by two equilibrium conditions: the Euler equation (IS curve) and the optimal price-setting condition (New Keynesian Phillips Curve). A Taylor Rule determines the nominal interest rate R_t.

Given the minimal state representation, we solve for two policies $\pi_t = \log(\Pi_t/\Pi)$ and $n_t = \log(N_t/N)$, that is inflation and employment in log deviations from their steady state values. The local approximation is computed as:

$$
\begin{aligned}
g^{\pi}(z_t, d_t, m_t) &= \alpha_0 + \alpha_1 z_t + \alpha_2 d_t + \alpha_3 m_t \\
g^{n}(z_t, d_t, m_t) &= \beta_0 + \beta_1 z_t + \beta_2 d_t + \beta_3 m_t,
\end{aligned}
$$

where the coefficients $\alpha_0, \alpha_1, \alpha_2, \alpha_3, \beta_0, \beta_1, \beta_2, \beta_3$ will differ at each point of the state-space, but we have suppressed the dependence on x for notational convenience. With a ZLB, the policies become regime dependent with $j \in \{C, U\}$ denoting constrained or unconstrained:

$$
\begin{aligned}
g^{\pi}_j(z_t, d_t, m_t) &= \alpha_{0,j} + \alpha_{1,j} z_t + \alpha_{2,j} d_t + \alpha_{3,j} m_t \\
g^{n}_j(z_t, d_t, m_t) &= \beta_{0,j} + \beta_{1,j} z_t + \beta_{2,j} d_t + \beta_{3,j} m_t,
\end{aligned}
$$

State-Space Representation First focus on the model without ZLB. The states of the model $x_t = (z_t, d_t, m_t)$ are exogenous, so the state-transition equation is not time-varying, with system matrix $T = \text{diag}(\rho_z, \rho_d, \rho_m)$, and $Q = \text{diag}(\sigma_z^2, \sigma_d^2, \sigma_m^2)$ determined by the AR(1) parameters. Assume we observe detrended log output y_t^{obs}, log inflation π_t^{obs}, and log interest

rates r_t^{obs}. The observation equation, conditional on the solution at $x = \hat{x}_{t|t-1}$, is:

$$\begin{pmatrix} y_t^{\text{obs}} \\ \pi_t^{\text{obs}} \\ r_t^{\text{obs}} \end{pmatrix} = \begin{pmatrix} \beta_0(x) \\ \pi_{ss} + \alpha_0(x) \\ r_{ss} + \psi_\pi \alpha_0(x) + \psi_y \beta_0(x) \end{pmatrix} + Z(x) \begin{pmatrix} z_t \\ d_t \\ m_t \end{pmatrix} + \varepsilon_t^{ME}, \quad \text{where}$$

$$Z(x) = \begin{pmatrix} \beta_1(x) + 1 & \beta_2(x) & \beta_3(x) \\ \alpha_1(x) & \alpha_2(x) & \alpha_3(x) \\ \psi_\pi \alpha_1(x) + \psi_y(\beta_1(x) + 1) & \psi_\pi \alpha_2(x) + \psi_y \beta_2(x) & \psi_\pi \alpha_3(x) + \psi_y \beta_3(x) + 1 \end{pmatrix},$$

where $\pi_{ss} = \log(\Pi)$ and $r_{ss} = \log(R)$. The bottom row in $Z(x)$ reflects how interest rates depend on both output and inflation through the monetary policy rule $r_t = \psi_\pi \pi_t + \psi_y y_t + m_t$. This state-space representation allows us to solve and filter the model as in Algorithm 1.

To filter the model with the ZLB, we assume that it is known to the econometrician when the ZLB episodes take place. The observed ZLB indicator is used to select the relevant local policy rule and is not treated as an additional observable. In periods where the ZLB does not bind, one uses the unconstrained (U) policies and the state space system above. When the ZLB binds, one uses the constrained (C) policies, the intercept of the observation equation for r_t^{obs} becomes 0, and the row of $Z(x)$ corresponding to r_t^{obs} becomes a row of zeros. The resulting sequence of state space systems is consistent with Theorems 1–2, provided their stated assumptions hold and the regime-specific policies themselves are smooth.

Table 3: Computation time (seconds) for a single likelihood evaluation for $T = 250$.

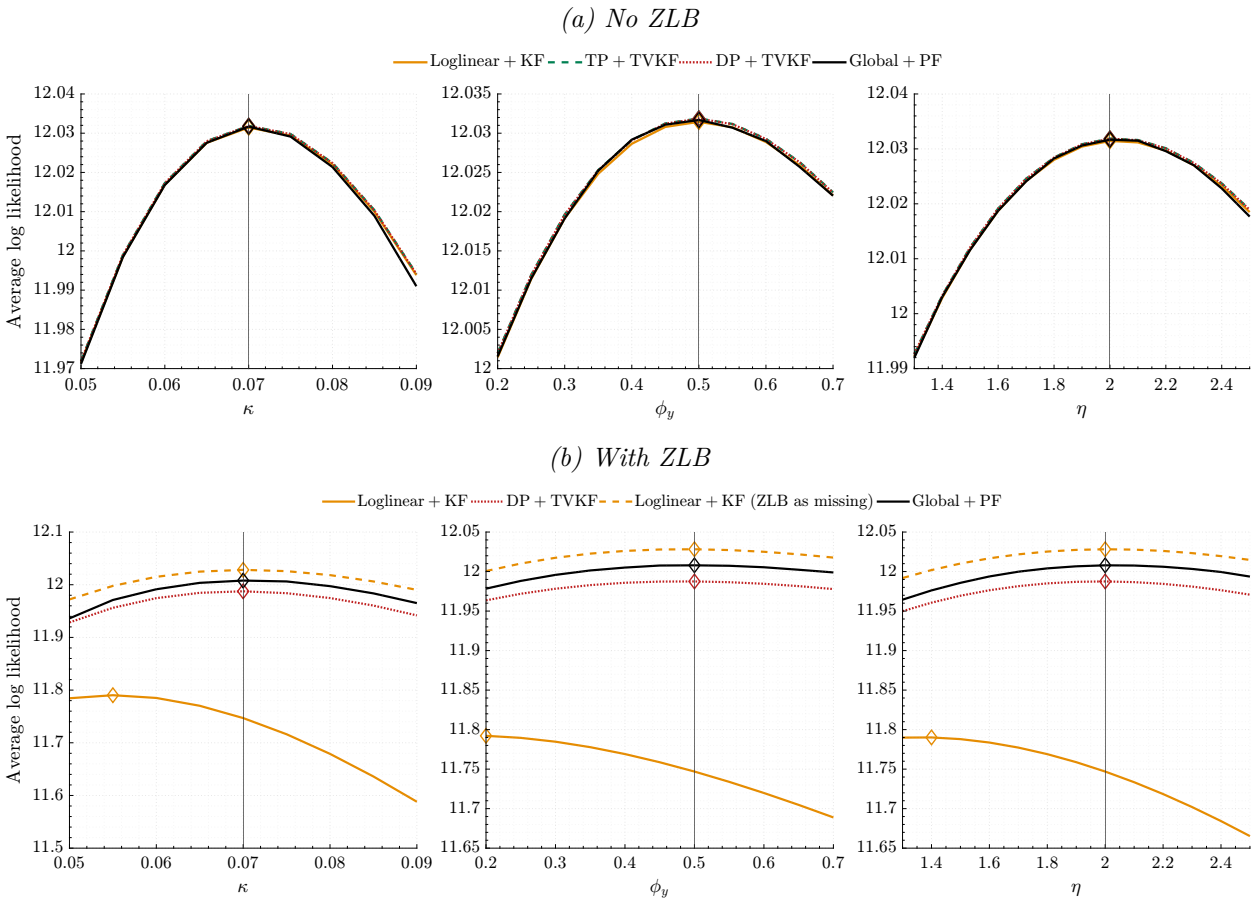
Method	No ZLB	With ZLB
Loglinear + KF	0.002	0.002
DP + TVKF	0.052	0.091
TP + TVKF	0.120	—
Global + PF (100,000)	8.89	9.03
Global + PF (1,000,000)	57.9	59.1

Runtime We summarize the computation times[7] in Table 3. One thing to note is to get accurate estimates with the particle filter, one needs at least 100,000 particles, despite only having three states, making the particle filter computationally costly. This is because the number of particles grows exponentially with the number of states. In addition, we provide computation times for 1,000,000 particles to demonstrate how the computational cost of

[7]We use Matlab 2025b on a standard laptop without parallelization. Computations are done on an Apple Macbook Pro with an Apple M3 Max processor and 64GB RAM.

the particle filter scales. When using DP+TVKF with ZLB, the computation time roughly doubles relative to no ZLB, because DP needs to compute a policy for each regime.

Figure 2: Comparing likelihoods for the Phillips curve slope κ, weight on output in the Taylor rule ϕ_y, and inverse Frisch elasticity η in the Three Equation New Keynesian Model



Note: Calibration: $\tau = 1$, $\beta = 0.96^{1/4}$, $R^{ss} = \beta^{-1}$, $\eta = 2$, $\kappa = 0.07$, $\phi_\pi = 2$, *and* $\phi_y = 0.5$*;* $\rho_z = 0.85$, $\sigma_z = 0.005$, $\rho_d = 0.92$, $\sigma_d = 0.012$, $\rho_m = 0.7$, *and* $\sigma_m = 0.007$. *The measurement-error standard deviations are* $(\sigma_{\varepsilon_n}, \sigma_{\varepsilon_\pi}, \sigma_{\varepsilon_r}) = (0.002, 0.006, 0.001)$. *Figure compares likelihoods computed based on the time-varying Kalman filter with Dynamic Perturbation and Taylor Projection, the Particle Filter (100,000 particles) + global model solution, and the linearized model with Kalman filter. Based on a simulation of length* $T = 10,000$ *from the global model solution. All other parameters are kept constant at their true values.*

Comparing Likelihoods. Figure 2 compares the likelihoods implied by the different solution methods and filters, for both the model with and without the ZLB. To do so, we simulate a series of length $T = 10{,}000$ from the global model solution. The top row of Figure 2 shows that, for the three-equation New Keynesian model without a ZLB, using a log-linearization introduces no bias, which is unsurprising given that the functional forms make the model essentially log-linear. For the model with the ZLB, however, the log-linear model solution, which ignores the ZLB, generates substantial bias. Interestingly, using the log-linear solution and treating the ZLB dates as missing eliminates this bias, although the figure masks that this approach discards approximately 10% of the sample because it reports the average log

likelihood by observations actually used. By contrast, dynamic perturbation combined with the time-varying Kalman filter uses all the data and closely matches the likelihood produced by the global method with the particle filter (100,000 particles).

3.3 Multi-Country RBC Model

To assess how likelihood evaluation scales with model size, we use the multi-country RBC economy developed for the computational comparison in the special issue of the *Journal of Economic Dynamics and Control*; see Den Haan et al. (2011), Juillard and Villemot (2011), and Kollmann et al. (2011). The model is well known, so we summarize it only to fix notation. There are N countries, complete asset markets, and one freely traded good. With Negishi weights $\tau_j > 0$, the social planner chooses $\{C_t^j, L_t^j, I_t^j, K_{t+1}^j\}_{j,t}$ to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \sum_{j=1}^{N} \tau_j \left[\frac{(C_t^j)^{1-\gamma_j} - 1}{1-\gamma_j} - \psi_j \frac{(L_t^j)^{1+1/\eta_j}}{1+1/\eta_j} \right],$$

where γ_j is relative risk aversion, η_j is the Frisch elasticity, and $\psi_j > 0$ scales labor disutility. The consumption term is interpreted as $\log C_t^j$ when $\gamma_j = 1$. Country j produces and accumulates capital according to

$$Y_t^j = A_t^j (K_t^j)^{\alpha} (L_t^j)^{1-\alpha}, \quad \text{and} \quad K_{t+1}^j = (1-\delta) K_t^j + I_t^j, \tag{38}$$

and incurs the adjustment cost $\chi_t^j = (\phi/2) K_t^j (I_t^j / K_t^j - \delta)^2$. The planner is subject to the world resource constraint

$$\sum_{j=1}^{N} \left(C_t^j + I_t^j + \chi_t^j \right) = \sum_{j=1}^{N} Y_t^j. \tag{39}$$

Writing $a_t^j = \log A_t^j$, productivity follows $a_{t+1} = \rho_a \odot a_t + L_a \varepsilon_{t+1}$, where $\varepsilon_{t+1} \sim \mathcal{N}(0, I_N)$ and L_a captures the potential covariance of innovations across countries. Capital and productivity in every country are state variables, so the state vector has dimension $2N$. This linear increase in dimension makes the model a useful benchmark to assess scalability.

We compare two approaches to likelihood evaluation. Loglinear+KF applies the Kalman filter to the fixed state-space representation implied by the steady-state loglinear solution. DP+TVKF instead recomputes a first-order Dynamic Perturbation solution at each forecasted state and applies the filter using the resulting time-varying state-space matrices. Further details of the local solution methods are provided in McCrary (2026); Appendix D describes the calibration, filtering and timing protocol.

Table 4 reports computation times. For each N, we simulate a sample of length $T = 200$ and average computation time over 100 evaluations of the same sample. DP+TVKF scales

Table 4: Likelihood-evaluation time (serial) in the multi-country RBC model

	Avg. time per evaluation (secs)		Proj. time for 100,000 evaluations (hrs)	
N	DP+TVKF	Loglinear+KF	DP+TVKF	Loglinear+KF
2	0.29	0.002	8.05	0.06
4	0.30	0.004	8.22	0.11
8	0.59	0.020	16.51	0.56
16	0.69	0.040	19.11	1.10
32	1.00	0.112	27.85	3.11
50	1.70	0.486	47.32	13.49

Notes: Each entry is the average of 100 evaluations of the same simulated sample of length $T = 200$.

favorably with model size: increasing N from 2 to 50 raises the state dimension 25-fold, from 4 to 100, but raises average evaluation time only 5.9-fold, from 0.29 to 1.70 seconds. Consequently, 100,000 serial evaluations of the 100-state model require approximately 47 hours. The loglinear benchmark is faster, but its relative advantage falls from a factor of 138 at $N = 2$ to 3.5 at $N = 50$. We conclude that Bayesian estimation of a nonlinear model with 100 state variables is feasible using DP+TVKF, while characterizing the global model solution and applying the particle filter for such a large state space would go beyond the current computational frontier.

4 A New Keynesian Model with Labor Search

Here, we present a New Keynesian (NK) model with Diamond Mortensen Pissarides (DMP) labor market frictions, leading to state-dependent monetary policy transmission.

4.1 Model

Overview. The economy consists of households that consume and supply labor, intermediate-good producers that set prices and hire in a frictional labor market, and final-good producers that aggregate intermediate goods while taking prices as given. Wages are determined by Nash bargaining between the representative household and intermediate goods firms, and the government sets the nominal interest rate according to a Taylor rule.

Timing. We follow the same timing assumption as the DMP model in Section 3. Each period consists of four stages: (i) separations, (ii) search and matching, (iii) bargaining, and (iv) production and consumption/saving.

Households. Households are composed of a continuum of members $i \in [0, 1]$ who derive utility from consumption $c_t(i)$ and supply one unit of labor $n_t(i) \in \{0, 1\}$ if employed. Mem-

bers perfectly insure each other against income fluctuations. Defining aggregate consumption and employment as $C_t := \int c_t(i)\,di$ and $N_t := \int n_t(i)\,di$, the household maximizes utility:

$$\begin{aligned} & \mathbb{E}_0\left[\sum_{t=0}^{\infty}\beta^t D_t \frac{C_t^{1-\tau}-1}{1-\tau}\right], \\ \text{s.t.} \quad & P_tC_t + B_t = P_tW_tN_t + P_t(1-N_t)b + R_{t-1}B_{t-1} + T_t, \\ & N_t = (1-\delta_t)N_{t-1} + q_tV_t. \end{aligned}$$

Here, β is the discount factor, D_t is a preference shock, $1/\tau$ is the intertemporal elasticity of substitution, P_t is the final good price, B_t are one-period risk-free bonds with gross return R_t, and T_t are firm profits net of lump-sum taxes. W_t is the real wage, b is the value of non-employment, δ_t is the separation rate, V_t is the number of vacancies, and q_t is the vacancy filling rate. The first-order condition for consumption yields the marginal utility and stochastic discount factor

$$\lambda_t = D_tC_t^{-\tau}, \qquad\qquad Q_{t,t+1} = \beta\frac{\lambda_{t+1}}{\lambda_t}.$$

The saving Euler equation is:

$$1 = \beta\mathbb{E}_t\left[R_t\frac{D_{t+1}}{D_t}\left(\frac{C_{t+1}}{C_t}\right)^{-\tau}\frac{P_t}{P_{t+1}}\right] = \mathbb{E}_t\left[Q_{t,t+1}\frac{R_t}{\Pi_{t+1}}\right],$$

where $\Pi_t = P_t/P_{t-1}$ is inflation. The household surplus from employment is

$$S_t^h = W_t - b + \mathbb{E}_t\left[Q_{t,t+1}(1-\delta_{t+1})S_{t+1}^h\right],$$

representing the present value of a marginal employed household member.

Search and Matching. Aggregate hires are determined by a constant-returns-to-scale matching function $M_t = A_tM(V_t, S_t)$, where A_t is matching efficiency, V_t is the number of vacancies, and S_t is the mass of unmatched workers, given by:

$$S_t = 1 - (1-\delta_t)N_{t-1},$$

which includes workers who were not matched previously and those who separated at the start of the period. Market tightness is defined as $\Theta_t = V_t/S_t$, and the vacancy filling rate:

$$q_t = \frac{M_t}{V_t} = A_tM\left(1, \Theta_t^{-1}\right).$$

Final Goods Producers. Final-good producers aggregate intermediate goods $Y_t(j)$, $j \in [0,1]$, using a CES aggregator: $Y_t = \left(\int_0^1 Y_t(j)^{1-\gamma}dj\right)^{\frac{1}{1-\gamma}}$, taking input and output prices as

given and maximizing profits. Optimality implies the demand for good j:

$$Y_t(j) = \left(\frac{P_t(j)}{P_t}\right)^{-\frac{1}{\gamma}} Y_t.$$

Intermediate Goods Producers. A continuum of identical monopolistic producers $j \in [0,1]$ produces using the linear technology $Y_t(j) = Z_t N_t(j)$, where Z_t is aggregate productivity. Firms face quadratic Rotemberg price adjustment costs,

$$AC_t(j) = \frac{\phi}{2}\left(\frac{P_t(j)}{P_{t-1}(j)} - \Pi\right)^2 Y_t,$$

and hire labor in frictional markets by posting vacancies at cost c. Each firm chooses $N_t(j)$, $V_t(j)$, and $P_t(j)$ to maximize:

$$\begin{aligned}
&\Pi_0(j) = \mathbb{E}_0 \sum_{t=0}^{\infty} Q_{0,t}\left[\left(\frac{P_t(j)}{P_t} Z_t - W_t\right) N_t(j) - cV_t(j) - AC_t(j)\right], \\
\text{s.t.} \quad & Z_t N_t(j) = \left(\frac{P_t(j)}{P_t}\right)^{-\frac{1}{\gamma}} Y_t, \quad [\omega_t] \\
& N_t(j) = (1-\delta_t) N_{t-1}(j) + q_t V_t(j). \quad [\mu_t]
\end{aligned}$$

The multiplier ω_t captures the implicit cost of hiring due to downward movement on the demand curve, while μ_t denotes the marginal value of employment to the firm.

Wage Bargaining. The wage is determined by the Nash sharing rule

$$\eta \mu_t = (1-\eta) S_t^h,$$

where η is the worker's bargaining weight. This implies the real wage is

$$W_t = \eta(1-\omega_t) Z_t + (1-\eta) b,$$

with b the flow value of non-employment and ω_t the implicit marginal hiring cost.

Government. Monetary policy is determined by a Taylor rule for the shadow nominal interest rate:

$$R_t^s = R\left(\frac{\Pi_t}{\Pi}\right)^{\phi_\pi}\left(\frac{Y_t}{Y}\right)^{\phi_y} e^{m_t},$$

where R_t is the systematic component of the policy rule and m_t is an inertial monetary policy shock, $\Pi_t = P_t/P_{t-1}$ is inflation and Y_t is output. The government balances the budget period by period using lump-sum taxes or transfers. Without a ZLB the interest rate is $R_t = R_t^s$, while the presence of a ZLB implies $R_t = \max\{1, R_t^s\}$.

Exogenous States. The discount factor, $d_t = \ln D_t$, productivity, $z_t = \ln Z_t$, and the monetary policy shock, m_t, follow AR(1) processes in logs. The separation rate, $\delta_t = \delta e^{s_t}$, and the matching efficiency, $A_t = Ae^{a_t}$, also include log-linear stochastic components that follow AR(1) processes:

$$z_t = \rho_z z_{t-1} + \sigma_z \varepsilon_t^z, \quad d_t = \rho_d d_{t-1} + \sigma_d \varepsilon_t^d, \quad m_t = \rho_m m_{t-1} + \sigma_m \varepsilon_t^m,$$
$$s_t = \rho_s s_{t-1} + \sigma_s \varepsilon_t^s, \quad a_t = \rho_a a_{t-1} + \sigma_a \varepsilon_t^a.$$

Equilibrium Conditions. We assume that vacancy posting and price adjustment costs are rebated to households in lump sum, so market clearing implies:

$$C_t = Y_t = Z_t N_t.$$

The household Euler equation becomes:

$$1 = \beta \mathbb{E}_t \left[\frac{D_{t+1}}{D_t} \left(\frac{Y_{t+1}}{Y_t} \right)^{-\tau} \frac{R_t}{\Pi_{t+1}} \right].$$

In a symmetric equilibrium, all firms set the same price $P_t = P_t(j)$ and choose the same level of employment $N_t(j) = N_t$ and vacancies $V_t(j) = V_t$, which yields the pricing equation:

$$\frac{\omega_t}{\gamma} = 1 - \phi(\Pi_t - \Pi)\Pi_t + \phi\beta \mathbb{E}_t \left[\frac{D_{t+1}}{D_t} \left(\frac{Y_{t+1}}{Y_t} \right)^{1-\tau} (\Pi_{t+1} - \Pi)\Pi_{t+1} \right].$$

Combining the vacancy posting condition with the Nash wage expression yields the job-creation condition:

$$\frac{c}{q_t} = (1-\eta)\left[(1-\omega_t)Z_t - b\right] + \beta \mathbb{E}_t \left[(1-\delta_{t+1}) \frac{D_{t+1}}{D_t} \left(\frac{Y_{t+1}}{Y_t} \right)^{-\tau} \frac{c}{q_{t+1}} \right].$$

The Taylor rule for the nominal interest rate and the law of motion for employment complete the system. The model admits a minimum state variable representation with state vector $x_t = (N_{t-1}, z_t, d_t, m_t, s_t, a_t)$, and policy functions $\{\Pi(x_t), \omega(x_t), \theta(x_t)\}$.

4.2 State Dependence

The quantitative results will examine the state-dependent effects of monetary policy shocks. As a preliminary step, analyzing the log-linearized system clarifies the model's propagation mechanisms. We study the perfect foresight response to a one-time monetary policy shock, $\varepsilon_0^m = 1$, $\varepsilon_t^m = 0 \ \forall t > 0$, using the log-linearized system. Focusing on two key equations:

$$\text{Phillips curve:} \quad \hat{\pi}_t = \beta \hat{\pi}_{t+1} - \frac{1}{\phi \Pi^2} \hat{\omega}_t$$

$$\text{Job creation:} \quad \hat{q}_t = \frac{\gamma(1 - \beta(1-\delta))}{(1-\gamma-b)} \hat{\omega}_t + (1-\delta)\beta \left[\hat{q}_{t+1} + \hat{r}_t - \hat{\pi}_{t+1}\right]$$

The job-creation condition shows that the sensitivity of the vacancy-filling rate (i.e., market tightness) $\hat{q}_t$ to a cost shock $\hat{\omega}_t$ increases as steady-state flow profits, given by $(1-\gamma-b)$, decline. This mechanism relates to the fundamental surplus in Ljungqvist and Sargent (2017), but in the context of monetary policy. Smaller flow profits amplify the effect of a monetary shock on market tightness, employment, and output. Since $\hat{\pi}_t$ loads on $\hat{\omega}_t$ in the Phillips curve, this channel also indirectly governs the inflation response. The degree of price stickiness, ϕ, determines the strength of this pass-through. In the log-linearized NK-DMP model, elasticities to shocks are constant and pinned down by steady-state values. In contrast, a nonlinear solution allows flow profits to vary with the state, generating state-dependent monetary policy transmission.

4.3 Solution and Estimation

We solve the model locally by expressing states and policies as log-deviations from their steady-state values, e.g., $\theta_t = \ln(\Theta_t/\Theta)$ and $z_t = \ln(Z_t/Z)$. We assume the law of motion for employment n_t, and policy functions for inflation π_t, marginal cost ω_t, and market tightness θ_t are linear in the state vector $x_t = \{n_{t-1}, z_t, d_t, m_t, s_t, a_t\}$. This yields the following functions:

$$\theta_t = b_0^\theta(x) + B^\theta(x) x_t, \qquad \pi_t = b_0^\pi(x) + B^\pi(x) x_t,$$
$$\omega_t = b_0^\omega(x) + B^\omega(x) x_t, \qquad n_t = b_0^n(x) + B^n(x) x_t,$$

where each B^i is a 1×6 vector. Expectations in the equilibrium system can be solved in closed form (see Online Appendix E), and coefficients are recovered using both the TP and DP methods. With an occasionally binding ZLB, the above policies are indexed by the regime, and we use the DP method. In addition, define $b_0^r(x) := \phi_\pi b_0^\pi(x) + \phi_y b_0^n(x)$ and $B^r(x) = \phi_\pi B^\pi(x) + \phi_y B^n(x) + [0 \ \phi^y \ 0 \ 1 \ 0 \ 0]$.

We map the model to the data using seven observables: market tightness θ_t^{obs}, the job finding rate f_t^{obs}, the separation rate s_t^{obs}, employment n_t^{obs}, inflation (GDP deflator) $\pi_t^{1,\text{obs}}$, inflation $\pi_t^{2,\text{obs}}$ (CPI) and the Federal Funds rate r_t^{obs}. Online Appendix Table F1 summarizes the data. The measurement equations incorporate measurement error:

$$\pi_t^{i,\text{obs}} = \pi_{ss} + \pi_t + \sigma_{\pi,i}^{ME}\varepsilon_t^{\pi,i}, \quad i = 1,2 \qquad n_t^{\text{obs}} = n_t + \sigma_n^{ME}\varepsilon_t^n \qquad \theta_t^{\text{obs}} = \theta_t + \sigma_\theta^{ME}\varepsilon_t^\theta,$$
$$r_t^{\text{obs}} = r_{ss} + r_t + \sigma_r^{ME}\varepsilon_t^r \qquad s_t^{\text{obs}} = s_t + \sigma_s^{ME}\varepsilon_t^s \qquad f_t^{\text{obs}} = f_t + \sigma_{\text{jf}}^{ME}\varepsilon_t^f.$$

Letting $y_t^{1,\text{obs}} = (\pi_t^{1,\text{obs}}, \pi_t^{2,\text{obs}}, n_t^{\text{obs}}, \theta_t^{\text{obs}}, r_t^{\text{obs}}, s_t^{\text{obs}}, f_t^{\text{obs}})'$, and solving the model at $x = \hat{x}_{t|t-1}$, the state-space system matrices become:

$$d_t = \begin{pmatrix} \pi_{ss} + b_0^\pi(x) \\ \pi_{ss} + b_0^\pi(x) \\ b_0^n(x) \\ b_0^\theta(x) \\ r_{ss} + b_0^r(x) \\ 0 \\ (1-\alpha)b_0^\theta(x) \end{pmatrix}, \; Z_t = \begin{pmatrix} B^\pi(x) \\ B^\pi(x) \\ B^n(x) \\ B^\theta(x) \\ B^r(x) \\ 0\;\;0\;\;0\;\;0\;\;1\;\;0 \\ (1-\alpha)B^\theta(x) + [0\;0\;0\;0\;0\;1] \end{pmatrix}, \; c_t = \begin{pmatrix} b_0^n(x) \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \; T_t = \begin{pmatrix} B^n(x) \\ 0 & \rho_z & 0 & 0 & 0 & 0 \\ 0 & 0 & \rho_d & 0 & 0 & 0 \\ 0 & 0 & 0 & \rho_m & 0 & 0 \\ 0 & 0 & 0 & 0 & \rho_s & 0 \\ 0 & 0 & 0 & 0 & 0 & \rho_a \end{pmatrix}.$$

4.4 Empirical Framework: State-Dependent Phillips Multiplier

A key concern for policymakers is the unemployment–inflation trade-off, which captures the nominal effects of a monetary policy shock relative to its real effects. We measure the unemployment–inflation trade-off using the Phillips multiplier (PM) of Barnichon and Mesters (2021). The PM measures the average change in inflation associated with a monetary policy shock that changes the unemployment rate by one percentage point over h periods. Its population definition and empirical counterpart are, respectively,

$$T_h = \left.\frac{\partial \bar{\Pi}_{t:t+h}}{\partial R_t}\right|_{\varepsilon_t} \Big/ \left.\frac{\partial \bar{U}_{t:t+h}}{\partial R_t}\right|_{\varepsilon_t} \qquad \text{and} \qquad \mathcal{P}_h = \frac{\mathcal{R}_h^{\bar{\Pi}}}{\mathcal{R}_h^{\bar{U}}}, \qquad h = 0,1,2,\ldots, \tag{40}$$

where $\bar{\Pi}_{t:t+h}$ and $\bar{U}_{t:t+h}$ denote average inflation and unemployment between t and $t+h$, ε_t is the monetary policy shock, and $\mathcal{R}_h^{\bar{\Pi}}$ and $\mathcal{R}_h^{\bar{U}}$ are the corresponding cumulative impulse responses through horizon h.

To estimate a state-dependent Phillips multiplier, we use the state-dependent local-projection approach in Cloyne et al. (2023). For each outcome $j \in \{\pi, u\}$, we estimate

$$y_{t+h}^j = \alpha_h^j + \beta_h^j s_t + \beta_h^{j,u} s_t \left(\frac{u_{t-1} - \bar{u}}{\sigma_u}\right) + \mathbf{w}_t'\boldsymbol{\gamma}_h^j + \varepsilon_{t+h}^j, \qquad h = 0,\ldots,12, \tag{41}$$

where s_t is the federal funds rate and lagged unemployment is the state variable. The control vector $\mathbf{w}_t$ contains four quarterly lags of inflation, unemployment, and the monetary policy shock. We estimate Equation (41) using instrumental variables, instrumenting the federal funds rate by the Romer and Romer (2004) narrative shock series, extended by Wieland and Yang (2020). The interaction with unemployment is instrumented by the interaction between the narrative shock and unemployment.

We obtain the state-dependent Phillips multiplier by plugging the observed standardized lagged unemployment rate into the estimated horizon-specific LP-IV coefficients:

$$\widehat{\mathcal{P}}_{12}(u_{t-1}) = \left(\sum_{h=0}^{12}\left[\widehat{\beta}_h^{\pi} + \widehat{\beta}_h^{\pi,x}\left(\frac{u_{t-1}-\bar{u}}{\sigma_u}\right)\right]\right) \Big/ \left(\sum_{h=0}^{12}\left[\widehat{\beta}_h^{u} + \widehat{\beta}_h^{u,x}\left(\frac{u_{t-1}-\bar{u}}{\sigma_u}\right)\right]\right). \tag{42}$$

We use quarterly U.S. data from Barnichon and Mesters (2021); inflation (annualized quarter-on-quarter PCE), the unemployment rate and the federal funds rate. We use these estimates as an external empirical benchmark for assessing whether the state dependence generated by the model is consistent with the data. A causal interpretation requires the monetary policy instrument and its interaction with unemployment to be exogenous, which is demanding in this setting (Gonçalves et al., 2024).

4.5 Parameter Estimates

Table 5: Calibration for the NKDMP model

Parameter	Value or definition
η: bargaining weight	0.5
τ: risk aversion	1
γ: inverse elasticity in the CES aggregator	0.1
κ: NKPC slope $1/(\phi\Pi_{ss}^2)$	0.007 (Hazell et al., 2022)
Π_{ss}: steady-state gross inflation	1.005
R_{ss}: steady-state gross nominal interest rate	1.01
q_{ss}: steady-state vacancy-filling rate	0.6
θ_{ss}: steady-state market tightness	1
n_{ss}: steady-state employment	0.945

Some parameters are calibrated (Table 5), while the remaining parameters are estimated in a Bayesian framework using a Random-Walk Metropolis–Hastings (RWMH) sampler; priors and posterior estimates are reported in Table 6. All three solution methods—log-linear + KF, Taylor projection + TVKF, and dynamic perturbation + TVKF—are estimated on the common sample 1984Q1–2007Q4. For DP, a single likelihood evaluation takes approximately

0.02 seconds, so 100,000 draws takes about 36 minutes. In addition, we estimate the model with ZLB using Dynamic Perturbation + TVKF on the sample 1984Q1-2019Q4. This takes less than 3.5 hours.

The resulting parameter estimates are quite similar across methods on the pre-ZLB sample. On the full sample that includes the ZLB, what stands out are larger standard deviations and persistence of the discount factor shock and larger standard deviation of the matching efficiency shock. In this model, large and persistent discount factor shocks are needed to reach the ZLB. The larger matching efficiency shocks help rationalize the long period of low employment after 2009.

4.6 Estimated Time-Varying Impulse Responses and Phillips Multiplier

Time-varying policy coefficients. Solving the model locally at each forecasted state allows shock transmission to vary with the state of the economy rather than remaining fixed as in a log-linearization. Figure 3 plots the policy coefficients associated with the monetary policy shock, together with 68% and 90% credible intervals that account for filtering and parameter uncertainty. Since the variables are expressed in log deviations from steady state, these coefficients can be interpreted as elasticities. During recessions, when surplus is low, the elasticities of market tightness and employment increase in absolute value, while the inflation elasticity decreases in absolute value. Online Appendix Figure G1 shows that during the GFC and subsequent ZLB episode, the market tightness and employment elasticities resemble those in other recessions, while the inflation elasticity becomes larger in absolute value.

Figure 3: State-Dependent Policy Coefficients for DP (no ZLB), computed at filtered states

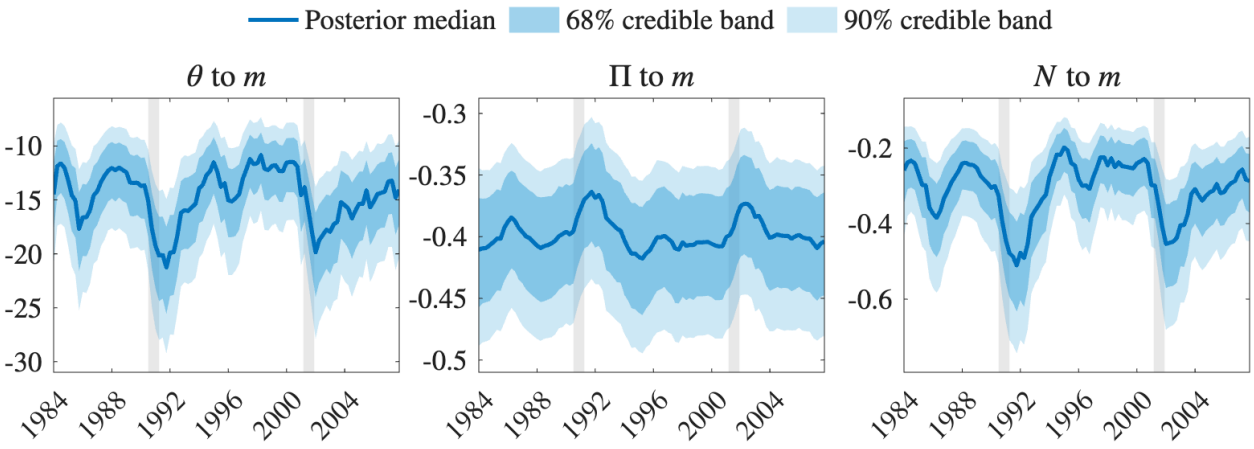


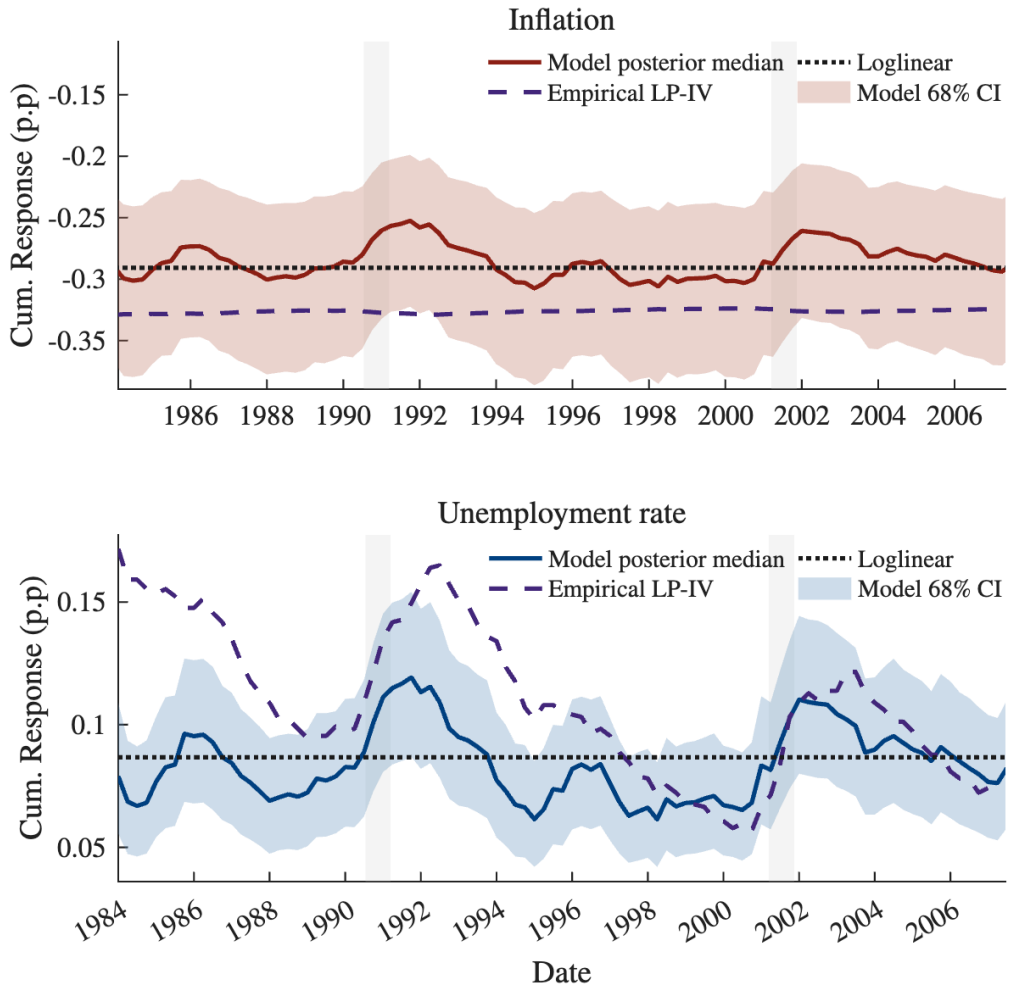
Table 6: Prior distributions (reported by mean and standard deviation) and posterior means with standard deviations across model specifications, solution methods, and samples.

Parameter	Prior (Mean, Std.)	Posterior				
	Model	NKDMP	NKDMP	NKDMP	NKDMP-ZLB	NKDMP-ZLB
	Method	log-linear + KF	TP + TV-KF	DP + TV-KF	DP + TV-KF	DP + TV-KF
	Sample	1984Q1–2007Q4	1984Q1–2007Q4	1984Q1–2007Q4	1984Q1–2007Q4	1984Q1–2019Q4
		Economic parameters				
ϕ_π: inflation Taylor rule	SG(2.00, 0.20)[a]	2.96 (0.24)	3.03 (0.24)	2.97 (0.24)	2.95 (0.23)	3.19 (0.24)
ϕ_y: output Taylor rule	G(0.50, 0.10)	0.19 (0.04)	0.16 (0.03)	0.19 (0.04)	0.20 (0.04)	0.21 (0.04)
ν^*: normalized flow value unemployment	B(0.90, 0.02)	0.96 (0.01)	0.95 (0.01)	0.96 (0.01)	0.97 (0.01)	0.98 (0.00)
α: matching elasticity	B(0.70, 0.10)	0.75 (0.03)	0.75 (0.02)	0.76 (0.03)	0.75 (0.03)	0.68 (0.02)
		Exogenous process parameters				
$100 \times \sigma_z$: std. TFP shocks	IG(1.00, 2.00)	0.40 (0.09)	0.44 (0.10)	0.42 (0.10)	0.37 (0.07)	0.37 (0.06)
$100 \times \sigma_d$: std. DF shocks	IG(1.00, 2.00)	1.13 (0.14)	1.17 (0.18)	1.14 (0.15)	1.17 (0.15)	2.86 (0.47)
$100 \times \sigma_s$: std. separation shocks	IG(1.00, 2.00)	4.96 (0.49)	4.98 (0.39)	5.06 (0.41)	5.02 (0.42)	5.34 (0.39)
$100 \times \sigma_m$: std. MP shocks	IG(1.00, 2.00)	0.35 (0.05)	0.35 (0.05)	0.35 (0.05)	0.34 (0.05)	0.33 (0.04)
$100 \times \sigma_a$: std. matching efficiency shocks	IG(1.00, 2.00)	1.69 (0.34)	1.61 (0.29)	1.79 (0.33)	1.73 (0.31)	2.60 (0.42)
ρ_z: persistence TFP	B(0.50, 0.10)	0.95 (0.01)	0.97 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)
ρ_d: persistence DF	B(0.50, 0.10)	0.87 (0.02)	0.87 (0.02)	0.87 (0.02)	0.87 (0.02)	0.94 (0.01)
ρ_s: persistence separations	B(0.50, 0.10)	0.84 (0.04)	0.83 (0.04)	0.84 (0.04)	0.85 (0.04)	0.84 (0.04)
ρ_m: persistence monetary policy	B(0.50, 0.10)	0.69 (0.05)	0.68 (0.06)	0.69 (0.06)	0.70 (0.06)	0.79 (0.04)
ρ_a: persistence matching efficiency	B(0.50, 0.10)	0.73 (0.07)	0.75 (0.05)	0.76 (0.06)	0.77 (0.05)	0.76 (0.05)
		Measurement error				
$100 \times \sigma_{\pi_1}^{\text{ME}}$: ME GDP deflator inflation	IG(0.50, 5.00)	0.08 (0.01)	0.08 (0.01)	0.08 (0.01)	0.08 (0.01)	0.10 (0.01)
$100 \times \sigma_{\pi_2}^{\text{ME}}$: ME CPI inflation	IG(0.50, 5.00)	0.32 (0.02)	0.33 (0.02)	0.32 (0.02)	0.32 (0.02)	0.38 (0.02)
$100 \times \sigma_N^{\text{ME}}$: ME employment	IG(0.50, 5.00)	0.70 (0.07)	0.71 (0.07)	0.72 (0.08)	0.72 (0.07)	1.13 (0.09)
$100 \times \sigma_{tg}^{\text{ME}}$	IG(0.50, 5.00)	0.59 (0.70)	0.29 (0.12)	0.46 (0.24)	0.80 (0.46)	0.58 (0.47)
$100 \times \sigma_r^{\text{ME}}$: ME FFR	IG(0.50, 5.00)	0.06 (0.01)	0.06 (0.01)	0.06 (0.01)	0.06 (0.01)	0.04 (0.01)
$100 \times \sigma_s^{\text{ME}}$: ME separations	IG(0.50, 5.00)	0.67 (0.66)	0.51 (0.48)	0.51 (0.49)	0.55 (0.54)	0.59 (0.59)
$100 \times \sigma_{jf}^{\text{ME}}$	IG(0.50, 5.00)	5.34 (0.64)	5.41 (0.59)	5.10 (0.62)	5.13 (0.59)	7.07 (0.63)

Notes: [a] *The prior for ϕ_π is a shifted Gamma prior with support on $\phi_\pi > 1$. B denotes Beta, G Gamma, SG shifted Gamma, and IG Inverse Gamma. The reported posterior for ν^* refers to the normalized parameter estimated in the sampler; the model-implied flow value is $b = \nu^*(1 - \gamma - 0.01)$.*

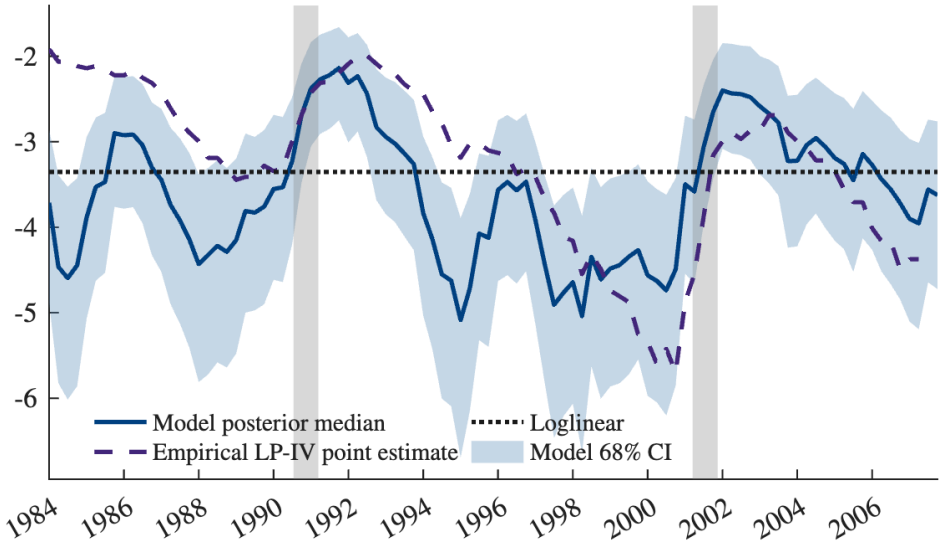
Time-varying impulse response functions. Time variation in the policy functions generates time-varying impulse responses. Online Appendix Figure G2 presents the unemployment and inflation impulse responses over time, while Figure 4 plots their cumulative responses through twelve quarters and compares them with the state-dependent local projections described in Section 4.4. The model and local projection estimates are comparable: both display substantial state dependence in the unemployment response and considerably less variation in the inflation response. This pattern is consistent with the mechanism described in Section 4.2. The cumulative impulse response functions implied by the loglinear model solution and estimates are on average comparable, but lack the time-variation we see in the nonlinear solution and in the data. Online Appendix Figures G3 and G4 present the corresponding results for the model with ZLB.

Figure 4: Model and empirical-based cumulative IRFs over time (no ZLB), over 12 quarters



Time-varying Phillips Multiplier. The cumulative impulse responses in Figure 4 allow us to examine state dependence in the PM given by Equation (42). Figure 5 compares the

Figure 5: Model and empirical-based Phillips Multiplier (no ZLB), over 12 quarters



model- and LP-implied Phillips multipliers over the pre-ZLB sample. The PM ranges from approximately -2 to -5 in the model and from -2 to -5.5 in the local projections. Both estimates are countercyclical, implying a flatter Phillips curve during recessions and a steeper curve during expansions. In contrast, the loglinear solution implies a constant PM around -3.3. Appendix Figure G5 presents the corresponding results for the ZLB model over the longer sample. This figure omits the empirical counterpart because the Romer and Romer (2004) shock series ends in 2007. The longer sample and presence of the ZLB produce a lower (in absolute values) PM on average while preserving the same countercyclical pattern.[8]

5 Extensions

This section considers two extensions that our framework can accommodate.

5.1 Regime Switching

A common extension is the regime-switching DSGE model, e.g., Liu et al. (2011), which underlies frameworks such as the rare disaster model (Barro, 2009). Here, the exogenous process departs from a stationary VAR(1), with dynamics instead governed by a latent discrete state $s_t \in \{0, \ldots, S\}$ evolving via a Markov chain with transition matrix Π:

$$x_{2,t} = A(s_t)x_{2,t-1} + B(s_t)\eta_t, \quad \text{where} \quad s_t \sim \Pi.$$

[8]This is driven by the larger unemployment response in the model with the ZLB, which in turn reflects the larger worker outside option implied by those parameter estimates.

In Appendix H, we show how local methods can accommodate regime switching by assigning regime-specific local linear policy functions, $y_t = g_{s,x}(x_t)$, and transition functions, $x_{t+1} = h_{s,x}(x_t)$ for each $s \in \{0, \ldots, S\}$ where the subscripts indicate the policy rules depend on both regime s and states x. Agents form expectations accounting for regime-dependent shifts in policy rules. The resulting model has a linear state-space representation conditional on regime s_t, allowing likelihood evaluation via the Kalman filter. The latent regime s_t is inferred using the Hamilton (1994) filter, following Kim (1994).

5.2 Heterogeneous agents

Recall the notation for a standard DSGE model in Equation (1):

$$\mathbb{E}_t \left[f(y_{t+1}, y_t, x_{t+1}, x_t) \right] = 0.$$

Heterogeneous agent models can also be written in this form, but the key challenge is that the state vector x_t includes an infinite-dimensional object: a measure μ_t over all idiosyncratic states. For example, in the heterogeneous household model of Krusell and Smith (1998), μ_t is the joint distribution over individual assets a and idiosyncratic efficiency units of labor ε. A common solution is to approximate μ_t using a finite-dimensional object. Examples include histograms (Reiter, 2009), or moment-based polynomial expansions (Winberry, 2018).

For illustration, we focus on a simple heterogeneous household model; a detailed description is provided in Online Appendix I. In brief, the model features households $i \in [0, 1]$ with preferences over consumption c_{it}, represented by

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \log(c_{it}),$$

where β is the discount factor. Households supply labor inelastically with employment status $\varepsilon_{it} \in \{0, 1\}$ (unemployed or employed) varying stochastically, and receive income $w_t[(1 - \tau_t)\varepsilon_{it} + b(1 - \varepsilon_{it})]$, where w_t is the wage, τ_t the labor tax rate, and b denotes the replacement rate of unemployment benefits. Assuming a balanced government budget in every period implies that the labor income tax rate satisfies $\tau_t = b(1 - L_t)/L_t$, where L_t is the aggregate employment rate. Households save in uncontingent assets a_{t+1} earning a return r_t, subject to a borrowing constraint $a_{t+1} \geq 0$. The probability of finding employment $f_t = p(\varepsilon_{i,t+1} = 1 | \varepsilon_{i,t} = 0)$ and separating into unemployment $s_t = p(\varepsilon_{i,t+1} = 0 | \varepsilon_{it} = 1)$ are given by:

$$f_t = \bar{f} e^{\eta_f z_t}, \quad s_t = \bar{s} e^{\eta_s z_t},$$

where η_f and η_s are parameters, and z_t is aggregate productivity (TFP), evolving as an AR(1). This implies aggregate employment L_t follows the law of motion:

$$L_{t+1} = f_t(1 - L_t) + (1 - s_t)L_t.$$

Within this model environment, employment transitions depend on the aggregate state, making recessions periods of heightened unemployment risk (as in Krueger et al., 2016), leading to time-varying precautionary saving motives.

A representative firm produces output using capital and labor $Y_t = e^{z_t} K_t^{\alpha} L_t^{1-\alpha}$, with α the capital share. Capital depreciates at rate δ. The firm maximizes profits so that factors are paid their marginal products r_t and w_t, respectively. The model is closed by market-clearing conditions that equate aggregate capital demand and supply. Since prices depend on the cross-sectional distribution of assets and employment, μ_t enters the equilibrium conditions, requiring households to forecast this distribution. As a result, μ_t is a state variable. Similar to Winberry (2018), we approximate μ_t with the maximum entropy distribution:

$$\mu_{\varepsilon,t}(a) \approx g^0_{\varepsilon,t} \exp\left\{\sum_{i=1}^{n_g} g^i_{\varepsilon,t} a^i\right\}, \text{ for } \varepsilon \in \{0, 1\}, \tag{43}$$

with $\mathbf{g}_t$ coefficients, chosen by matching a finite set of moments $\mathbf{m}_t$.

In addition to approximating the cross-sectional distribution μ_t, to map this model into a standard DSGE model form, we need to approximate the decision rules of the household, which follow from the household Euler equation and the borrowing constraint. We approximate the expected marginal utility from saving using Chebychev polynomials:

$$\mathbb{E}\left[\beta(1 + r_{t+1})c_{t+1}(\varepsilon', a'_t(\varepsilon, a))^{-1}\right] \approx \exp\left\{\sum_{j=1}^{n_\psi} \theta^j_{\varepsilon,t} T_{j-1}(\xi(a))\right\}, \tag{44}$$

with n_ψ the number of Chebychev nodes, T_j denotes the jth-order Chebychev polynomial, $\theta^j_{\varepsilon,t}$ are the coefficients, and $\xi(a) = 2\frac{a-\underline{a}}{\bar{a}-\underline{a}} - 1$. The approximation (44) holds exactly at Chebychev nodes $\{a_j\}_{j=1}^{n_\psi}$.

This results in a minimal state-space representation of the heterogeneous household model with state vector $x_t \equiv (z_t, \mathbf{m}_t, L_t)$ and controls $y_t \equiv (\boldsymbol{\theta}_t, \mathbf{g}_t)$. We use the first and second moment of the capital distribution in our approximation. The equilibrium conditions take the form of Equation (1), with the full system detailed in Online Appendix I. Rather than perturbing this system (Winberry, 2018; Reiter, 2009) or solving it under perfect foresight (Auclert et al., 2021), we use Taylor Projections. This solution method takes into account

the effect of aggregate risk on individual decisions. Integrating this solution approach into filtering algorithm 1 allows researchers to estimate heterogeneous agent models without imposing certainty equivalence. Solution and filtering details are described in Appendix I.

Nonlinearities. In Appendix Figures I1 and I2, we show this model has several interesting nonlinearities that are present in the TP solution, but suppressed by log-linearization. For example, Figure I1 studies state-dependence and shows that TFP shocks in good and bad states imply different output, employment and investment responses. Figure I2 shows the model also has interesting asymmetries, where a positive and negative shock have impulse response functions with different shapes and persistence in the TP solution.

Comparing Filters. We compare our filtering approach to a standard Kalman filter applied to a perturbation-based solution using simulated data ($T = 200$) from the dynamic Taylor Projection solution. Log output is assumed to be observed with measurement error equal to 10% of its standard deviation. As shown in Table 7, TP combined with a time-varying Kalman filter yields substantially lower mean squared errors (MSEs): for TFP z_t, the MSE is less than half that of perturbation; for employment L_t, nearly one-third; and for average unemployed wealth, over 25 times smaller. The final rows show that this gap reflects increasing inaccuracy of the perturbation solution away from steady state: the correlation between z_t^2, which measures the distance from the steady state, and squared filtering error, $e_t^2 = (x_t - \hat{x}_{t|t-1})^2$ for a generic $x_t \in \{z_t, m_u^1, m_u^2, m_e^1, m_e^2, L_t\}$, is approximately 0.38 for TFP and 0.60 for unemployed wealth for the perturbation approach, while close to zero for TP+TVKF.

Table 7: Mean squared error of filtered state versus true value and correlation between squared error and squared TFP. All MSE's are normalized by the unconditional variance of the relevant state.

Statistic	Method	z_t	m_u^1	m_u^2	m_e^1	m_e^2	L_t
MSE	TP + TV-KF	0.026	0.004	0.018	0.016	0.061	0.008
	KF	0.058	0.111	0.140	0.036	0.067	0.024
MSE ratio	KF/TP + TV-KF	2.3	25.3	7.7	2.3	1.1	2.8
$\text{Corr}(z_t^2, e_t^2)$	TP + TV-KF	0.048	0.039	-0.097	-0.028	-0.139	-0.020
	KF	0.377	0.602	0.545	0.146	0.023	0.408

Overall, the results in Figures I1, I2, and Table 7 demonstrate that even a simple heterogeneous household model exhibits nonlinearities that are poorly captured by perturbation-based solutions. This suggests that full-information estimation methods relying on such approximations may yield inaccurate results.

6 Conclusion

This paper presents a computationally efficient framework for full-information estimation of nonlinear DSGE models by integrating the model solution directly into the filtering step. In terms of filtering accuracy and parameter estimates, this approach closely approximates applying the extended Kalman filter to the true global solution, importantly without requiring the global solution itself to be characterized. In three canonical models the method achieves estimation speeds orders of magnitude faster than existing nonlinear approaches, while scaling well to large state spaces. The main application shows that labor market frictions generate meaningful state dependence in monetary policy transmission, with impulse response functions and the Phillips Multiplier varying substantially over the cycle. The framework also extends to models with regime switching and heterogeneous agents, offering a general and tractable approach to estimating nonlinear macroeconomic models.

References

Ahn, H. J. and J. D. Hamilton (2022): "Measuring Labor-Force Participation and the Incidence and Duration of Unemployment," *Review of Economic Dynamics*, 44, 1–32.

Aruoba, S. Borağan, S., P. Cuba-Borda, and F. Schorfheide (2018): "Macroeconomic Dynamics near the ZLB: A Tale of Two Countries," *The Review of Economic Studies*, 85, 87–118.

Auclert, A., B. Bardóczy, M. Rognlie, and L. Straub (2021): "Using the Sequence-Space Jacobian to Solve and Estimate Heterogeneous-agent Models," *Econometrica*, 89, 2375–2408.

Barnichon, R. (2010): "Building a Composite Help-Wanted Index," *Economics Letters*, 109, 175–178.

Barnichon, R., D. Debortoli, and C. Matthes (2022): "Understanding the Size of the Government Spending Multiplier: It's in the Sign," *The Review of Economic Studies*, 89, 87–117.

Barnichon, R. and G. Mesters (2021): "The Phillips Multiplier," *Journal of Monetary Economics*, 117, 689–705.

Barro, R. J. (2009): "Rare Disasters, Asset Prices, and Welfare Costs," *American Economic Review*, 99, 243–264.

Benigno, P. and G. B. Eggertsson (2024): "The Slanted-L Phillips Curve," *AEA Papers and Proceedings*, 114, 84–89.

Bonnabel, S. and J.-J. E. Slotine (2015): "A Contraction Theory-Based Analysis of the Stability of the Deterministic Extended Kalman Filter," *IEEE Transactions on Automatic Control*, 60, 565–569.

Bougerol, P. (1993): "Kalman Filtering with Random Coefficients and Contractions," *SIAM Journal on Control and Optimization*, 31, 942–959.

Cagliarini, A. and M. Kulish (2013): "Solving Linear Rational Expectations Models with Predictable Structural Changes," *The Review of Economics and Statistics*, 95, 328–336.

Childers, D., J. Fernández-Villaverde, J. Perla, C. Rackauckas, and P. Wu (2022): "Differentiable State-Space Models and Hamiltonian Monte Carlo Estimation," Tech. rep., National Bureau of Economic Research.

Christiano, L. J., M. Eichenbaum, and M. Trabandt (2016): "Unemployment and Business Cycles," *Econometrica*, 84, 1523–1569.

Cloyne, J., Ò. Jordà, and A. M. Taylor (2023): "State-Dependent Local Projections: Understanding Impulse Response Heterogeneity," Tech. rep., National Bureau of Economic Research.

Den Haan, W. J., K. L. Judd, and M. Juillard (2011): "Computational Suite of Models with Heterogeneous Agents II: Multi-Country Real Business Cycle Models," *Journal of Economic Dynamics and Control*, 35, 175–177.

Diamond, P. A. (1982): "Wage Determination and Efficiency in Search Equilibrium," *The Review of Economic Studies*, 49, 217–227.

Durbin, J. and S. J. Koopman (2012): *Time Series Analysis by State Space Methods*, vol. 38, OUP Oxford, 2nd ed.

Farmer, L. E. (2021): "The Discretization Filter: A Simple Way to Estimate Nonlinear State Space Models," *Quantitative Economics*, 12, 41–76.

Fernández-Villaverde, J. and O. Levintal (2024): "The Distributional Effects of Asset Returns," Tech. rep., National Bureau of Economic Research.

Fernández-Villaverde, J. and J. F. Rubio-Ramírez (2007): "Estimating Macroeconomic Models: A Likelihood Approach," *The Review of Economic Studies*, 74, 1059–1087.

Galí, J. (2015): *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework and Its Applications*, Princeton University Press, 2 ed.

Galí, J., F. Smets, and R. Wouters (2012): "Unemployment in an Estimated New Keynesian Model," *NBER Macroeconomics Annual*, 26, 329–360.

Gonçalves, S., A. M. Herrera, L. Kilian, and E. Pesavento (2024): "State-Dependent Local Projections," *Journal of Econometrics*, 244, 105702.

Guerrieri, L. and M. Iacoviello (2015): "OccBin: A Toolkit for Solving Dynamic Models with Occasionally Binding Constraints Easily," *Journal of Monetary Economics*, 70, 22–38.

Gust, C., E. Herbst, D. López-Salido, and M. E. Smith (2017): "The Empirical Implications of the Interest-rate Lower Bound," *American Economic Review*, 107, 1971–2006.

Hamilton, J. D. (1994): *Time Series Analysis*, Princeton University Press.

——— (2018): "Why You Should Never Use the Hodrick-Prescott Filter," *Review of Economics and Statistics*, 100, 831–843.

Hazell, J., J. Herreño, E. Nakamura, and J. Steinsson (2022): "The Slope of the Phillips Curve: Evidence from US States," *The Quarterly Journal of Economics*, 137, 1299–1344.

Herbst, E. and F. Schorfheide (2016): *Bayesian Estimation of DSGE Models*, Princeton University Press.

——— (2019): "Tempered Particle Filtering," *Journal of Econometrics*, 210, 26–44.

Juillard, M. and S. Villemot (2011): "Multi-Country Real Business Cycle Models: Accuracy Tests and Test Bench," *Journal of Economic Dynamics and Control*, 35, 178–185.

Kim, C.-J. (1994): "Dynamic Linear Models with Markov-Switching," *Journal of Econometrics*, 60, 1–22.

Kollmann, R., S. Maliar, B. A. Malin, and P. Pichler (2011): "Comparison of Solutions to the Multi-Country Real Business Cycle Model," *Journal of Economic Dynamics and Control*, 35, 186–202.

Krause, M. U., D. Lopez-Salido, and T. A. Lubik (2008): "Inflation Dynamics with Search Frictions: A Structural Econometric Analysis," *Journal of Monetary Economics*, 55, 892–916.

Krueger, D., K. Mitman, and F. Perri (2016): "Macroeconomics and Household Heterogeneity," in *Handbook of Macroeconomics*, Elsevier, vol. 2, 843–921.

Krusell, P., B. Kuruşçu, and A. A. Smith Jr (2002): "Equilibrium Welfare and Government Policy with Quasi-Geometric Discounting," *Journal of Economic Theory*, 105, 42–72.

Krusell, P. and A. A. Smith, Jr (1998): "Income and Wealth Heterogeneity in the macroeconomy," *Journal of Political Economy*, 106, 867–896.

Kulish, M. and A. Pagan (2017): "Estimation and Solution of Models with Expectations and Structural Changes," *Journal of Applied Econometrics*, 32, 255–274.

Le Gland, F. and L. Mevel (2000): "Exponential forgetting and geometric ergodicity in hidden Markov models," *Mathematics of Control, Signals and Systems*, 13, 63–93.

Levintal, O. (2018): "Taylor Projection: A New Solution Method for Dynamic General Equilibrium Models," *International Economic Review*, 59, 1345–1373.

Liu, L. and M. Plagborg-Møller (2023): "Full-information Estimation of Heterogeneous Agent Models Using Macro and Micro Data," *Quantitative Economics*, 14, 1–35.

Liu, Z., D. F. Waggoner, and T. Zha (2011): "Sources of Macroeconomic Fluctuations: A Regime-Switching DSGE Approach," *Quantitative Economics*, 2, 251–301.

Ljungqvist, L. and T. J. Sargent (2017): "The Fundamental Surplus," *American Economic Review*, 107, 2630–2665.

McCrary, S. (2026): "Comparing Local Solution Methods for DSGE Models," SSRN Working Paper.

Mennuni, A., J. F. Rubio-Ramírez, and S. Stepanchuk (2025): "Dynamic Perturbation," *The Review of Economic Studies*, 92, 1157–1192.

Mortensen, D. T. and C. A. Pissarides (1994): "Job Creation and Job Destruction in the Theory of Unemployment," *The Review of Economic Studies*, 61, 397–415.

Ottonello, P. and T. Winberry (2020): "Financial Heterogeneity and the Investment Channel of Monetary Policy," *Econometrica*, 88, 2473–2502.

Petrosky-Nadeau, N. and L. Zhang (2017): "Solving the Diamond–Mortensen–Pissarides Model Accurately," *Quantitative Economics*, 8, 611–650.

Reiter, M. (2009): "Solving Heterogeneous-Agent Models by Projection and Perturbation," *Journal of Economic Dynamics and Control*, 33, 649–665.

Romer, C. D. and D. H. Romer (2004): "A New Measure of Monetary Shocks: Derivation and Implications," *American Economic Review*, 94, 1055–1084.

Schmitt-Grohé, S. and M. Uribe (2004): "Solving Dynamic General Equilibrium Models Using a Second-Order Approximation to the Policy Function," *Journal of Economic Dynamics and Control*, 28, 755–775.

Shimer, R. (2005): "The Cyclical Behavior of Equilibrium Unemployment and Vacancies," *American Economic Review*, 95, 25–49.

Stock, J. H. and M. W. Watson (2020): "Slack and Cyclically Sensitive Inflation," *Journal of Money, Credit and Banking*, 52, 393–428.

Tenreyro, S. and G. Thwaites (2016): "Pushing on a String: US Monetary Policy is Less Powerful in Recessions," *American Economic Journal: Macroeconomics*, 8, 43–74.

Thomas, C. (2011): "Search frictions, real rigidities, and inflation dynamics," *Journal of Money, Credit and Banking*, 43, 1131–1164.

Wieland, J. F. and M.-J. Yang (2020): "Financial Dampening," *Journal of Money, Credit and Banking*, 52, 79–113.

Winberry, T. (2018): "A Method for Solving and Estimating Heterogeneous Agent Macro Models," *Quantitative Economics*, 9, 1123–1151.

Winschel, V. and M. Krätzig (2010): "Solving, Estimating, and Selecting Nonlinear Dynamic Models without the Curse of Dimensionality," *Econometrica*, 78, 803–821.

Online Appendix to "Efficient Estimation of Nonlinear DSGE Models"

Eva F. Janssens Sean McCrary

September, 2026

A Smoothing Algorithm

Algorithm 2 Time-Varying Kalman Smoother with Local Linear Methods

Initialize: Set $t = T$, $r_T = 0$, and $N_T = 0$.
while $t \geq 1$ **do**
Update Step:

$$r_{t-1} = Z_t' F_t^{-1} v_t + L_t' r_t, \tag{A.1}$$
$$N_{t-1} = Z_t' F_t^{-1} Z_t + L_t' N_t L_t, \tag{A.2}$$

where

$$L_t = T_t - K_t Z_t, \qquad K_t = T_t G_t = T_t P_{t|t-1} Z_t' F_t^{-1}. \tag{A.3}$$

The quantities $\widehat{x}_{t|t-1}$, $P_{t|t-1}$, T_t, Z_t, K_t, v_t, and F_t are stored from the forward recursion in Algorithm 1.

Smoothing Step:
Compute the smoothed state estimate:

$$\widehat{x}_{t|T} = \widehat{x}_{t|t-1} + P_{t|t-1} r_{t-1}. \tag{A.4}$$

Compute the smoothed covariance:

$$P_{t|T} = P_{t|t-1} - P_{t|t-1} N_{t-1} P_{t|t-1}. \tag{A.5}$$

Set Back Time: Set $t \leftarrow t - 1$ and repeat.
end while

B Proof of results

B.1 Proof of Theorem 1

Proof. Fix $T \geq 1$, θ, and local method m, and suppress the dependence on θ and m from the notation. Write $S_t^j := S_t^j(\theta),\ e_t := e_{t,m}(\theta),\ d_t := d_{t,m}(\theta)$. Define the one-period filter perturbation $u_t := S_t^m - \widetilde{S}_t^{O|m}$, i.e., using the same expansion point from the local method m, but different state space coefficient matrices. By Equations (25) and (26),

$$u_t = \Psi_{\theta,t}\left(S_{t-1}^m, \Xi^m\left(\hat{x}_{t|t-1}^m; \theta\right)\right) - \Psi_{\theta,t}\left(S_{t-1}^m, \Xi^O\left(\hat{x}_{t|t-1}^m; \theta\right)\right).$$

Write $r_t := r_{t,m}(\theta)$, so that $r_t = \Xi^m\left(\hat{x}_{t|t-1}^m; \theta\right) - \Xi^O\left(\hat{x}_{t|t-1}^m; \theta\right),\ \|r_t\| = e_t$. Parameterize the coefficient segment by $\Xi_t(\lambda) := \Xi^O\left(\hat{x}_{t|t-1}^m; \theta\right) + \lambda r_t,\ \lambda \in [0,1]$. Holding the incoming filter state S_{t-1}^m fixed, the fundamental theorem of calculus and the chain rule give

$$u_t = \Psi_{\theta,t}\left(S_{t-1}^m, \Xi_t(1)\right) - \Psi_{\theta,t}\left(S_{t-1}^m, \Xi_t(0)\right) = \int_0^1 D_\Xi \Psi_{\theta,t}\left(S_{t-1}^m, \Xi_t(\lambda)\right)[r_t]\, d\lambda.$$

For every $\lambda \in [0,1]$, $\Xi_t(\lambda)$ belongs to $\mathcal{L}_{t,m}^{\Xi}(\theta)$. Assumption 2 and the definition of $L_{K,T}$ therefore imply

$$\|u_t\| \leq \int_0^1 \left\|D_\Xi \Psi_{\theta,t}\left(S_{t-1}^m, \Xi_t(\lambda)\right)\right\|_{\text{op}} \|r_t\|\, d\lambda \leq \int_0^1 L_{K,T} e_t\, d\lambda = L_{K,T} e_t. \tag{B.1}$$

Thus, a coefficient error of size e_t at time t changes the output of that period's filter step by at most $L_{K,T} e_t$. We now propagate each of these one-period filter perturbations u_t through the subsequent oracle filter recursion. Fix $t \in \{1, \ldots, T\}$ and define $H_t^{(0)} := S_t^O,\ H_t^{(r)} := \Phi_{t:r,\theta}^O(S_r^m),\ r = 1, \ldots, t$, where $H_t^{(r)}$ uses the local method through time r and then uses the oracle recursion from $r+1$ to t. Hence, by construction, $H_t^{(t)} = \Phi_{t:t,\theta}^0(S_t^m) = S_t^m$. It follows that

$$S_t^m - S_t^O = H_t^{(t)} - H_t^{(0)} = \sum_{r=1}^{t} \left(H_t^{(r)} - H_t^{(r-1)}\right). \tag{B.2}$$

For $r \geq 2$, the composition rule for the oracle propagation maps gives

$$H_t^{(r-1)} = \Phi_{t:r-1,\theta}^O(S_{r-1}^m) = \Phi_{t:r,\theta}^O\left(\Psi_{\theta,r}^O(S_{r-1}^m)\right) = \Phi_{t:r,\theta}^O\left(\widetilde{S}_r^{O|m}\right).$$

Hence, for every $r = 1, \ldots, t$,

$$H_t^{(r)} - H_t^{(r-1)} = \Phi_{t:r,\theta}^O(S_r^m) - \Phi_{t:r,\theta}^O\left(\widetilde{S}_r^{O|m}\right).$$

The two arguments of the common oracle propagation map differ by exactly the one-period perturbation, $S_r^m - \widetilde{S}_r^{O|m} = u_r$. Thus, $H_t^{(r)} - H_t^{(r-1)}$ is the effect at time t of the perturbation introduced at time r, propagated through the oracle filter from r onward. The states S_r^m and $\widetilde{S}_r^{O|m}$ are the endpoints of $\mathcal{L}_{r,m}^S(\theta)$. Assumption 1 therefore gives

$$\left\|H_t^{(r)} - H_t^{(r-1)}\right\| \le C_F \rho^{t-r} \left\|S_r^m - \widetilde{S}_r^{O|m}\right\| = C_F \rho^{t-r} \|u_r\|.$$

Combining this inequality with (B.1) and (B.2) yields

$$d_t = \left\|S_t^m - S_t^O\right\| \le \sum_{r=1}^{t} \left\|H_t^{(r)} - H_t^{(r-1)}\right\| \le C_F L_{K,T} \sum_{r=1}^{t} \rho^{t-r} e_r = C_{\Psi,T} \sum_{r=1}^{t} \rho^{t-r} e_r,$$

which proves (29). Since $\rho \in (0,1)$,

$$d_t \le C_{\Psi,T} \max_{1\le r\le T} e_r \sum_{r=1}^{t} \rho^{t-r} \le \frac{C_{\Psi,T}}{1-\rho} \max_{1\le r\le T} e_r.$$

Together with $d_0 = 0$, taking the maximum over $t = 0, \dots, T$ proves the first part of (30). Moreover, for the second part of (30), after dividing by T:

$$\sum_{t=1}^{T} d_t \le C_{\Psi,T} \sum_{t=1}^{T} \sum_{r=1}^{t} \rho^{t-r} e_r = C_{\Psi,T} \sum_{r=1}^{T} e_r \sum_{t=r}^{T} \rho^{t-r} \le \frac{C_{\Psi,T}}{1-\rho} \sum_{r=1}^{T} e_r.$$

□

B.2 Proof of Theorem 2

Proof. Fix $T \ge 1$, the realized sample $y_{1:T}^{\text{obs}}$, and method m. All constants below are uniform over $\theta \in \mathcal{V}$, but may depend on T, the realized sample, and m.

For each t, the set $\mathcal{G}_{t,m}$ is compact because it is the continuous image of the compact set $\mathcal{V} \times [0,1]$ under the map $(\theta, u) \longmapsto \left((1-u)\zeta_t^O(\theta) + u\zeta_t^m(\theta), \theta\right)$. Assumption 4 therefore implies that

$$B_{v,m,T} := \max_{1\le t\le T} \sup_{(\zeta,\theta)\in\mathcal{G}_{t,m}} \|v_t(\zeta;\theta)\| < \infty, \quad \text{and}$$

$$B_{D,m,T} := \max_{1\le t\le T} \sup_{(\zeta,\theta)\in\mathcal{G}_{t,m}} \left\{\|D_\zeta v_t(\zeta;\theta)\|_{\text{op}} + \|D_\zeta F_t(\zeta;\theta)\|_{\text{op}}\right\} < \infty.$$

Let $n_y := \dim(y_t^{\text{obs}})$ denote the number of observed variables. Thus, $v \in \mathbb{R}^{n_y}$ and $F \in \mathbb{R}^{n_y \times n_y}$. Consider the Gaussian innovation log likelihood, up to its common additive constant,

$$\ell(v, F) = -\frac{1}{2}\left[\log\det F + v' F^{-1} v\right].$$

Fix t and θ, and define $\zeta_0 := \zeta_t^O(\theta),\ \zeta_1 := \zeta_t^m(\theta)$. Write also $F_0 := F_t^O(\theta) = F_t(\zeta_0;\theta),\ F_1 := F_t^m(\theta) = F_t(\zeta_1;\theta), v_0 := v_t^O(\theta) = v_t(\zeta_0;\theta), v_1 := v_t^m(\theta) = v_t(\zeta_1;\theta)$. Assumption 4 gives $F_0, F_1 \succeq \underline{f} I_{n_y}$. Hence, their convex combinations satisfy $F(u) := (1-u)F_0 + uF_1 \succeq \underline{f} I_{n_y},\ u \in [0,1]$. Let $\lambda_1(u), \ldots, \lambda_{n_y}(u)$ denote the eigenvalues of $F(u)$. Each satisfies $\lambda_k(u) \geq \underline{f} > 0$. Consequently,

$$\left\|F(u)^{-1}\right\|_{\text{op}} = \max_{1\leq k\leq n_y} \frac{1}{\lambda_k(u)} \leq \underline{f}^{-1}, \text{ and} \tag{B.3}$$

$$\left\|F(u)^{-1}\right\|_F^2 = \sum_{k=1}^{n_y} \frac{1}{\lambda_k(u)^2} \leq \frac{n_y}{\underline{f}^2} \quad \Rightarrow \quad \left\|F(u)^{-1}\right\|_F \leq \frac{\sqrt{n_y}}{\underline{f}}. \tag{B.4}$$

Let $\Delta F := F_1 - F_0$. By Jacobi's formula and the fundamental theorem of calculus,

$$\log\det F_1 - \log\det F_0 = \int_0^1 \operatorname{tr}\left[F(u)^{-1}\Delta F\right] du.$$

Since $F(u)^{-1}$ is symmetric, the integrand is the Frobenius inner product of $F(u)^{-1}$ and ΔF. Therefore, the Cauchy–Schwarz inequality gives $|\operatorname{tr}[F(u)^{-1}\Delta F]| \leq \|F(u)^{-1}\|_F \|\Delta F\|_F$. Integrating over $u \in [0,1]$ and using (B.4) consequently yields

$$|\log\det F_1 - \log\det F_0| \leq \frac{\sqrt{n_y}}{\underline{f}} \|F_1 - F_0\|_F. \tag{B.5}$$

For the quadratic term,

$$\left|v_1'F_1^{-1}v_1 - v_0'F_0^{-1}v_0\right| \leq \left|v_1'F_1^{-1}v_1 - v_0'F_1^{-1}v_0\right| + \left|v_0'\left(F_1^{-1} - F_0^{-1}\right)v_0\right|.$$

The first term is bounded by $\underline{f}^{-1}(\|v_1\| + \|v_0\|)\|v_1 - v_0\|$, while, using (B.3),

$$F_1^{-1} - F_0^{-1} = F_1^{-1}(F_0 - F_1)F_0^{-1} \quad \Rightarrow \quad \left\|F_1^{-1} - F_0^{-1}\right\|_{\text{op}} \leq \underline{f}^{-2}\|F_1 - F_0\|_F.$$

Using $\|v_0\|, \|v_1\| \leq B_{v,m,T}$, we obtain

$$\left|v_1'F_1^{-1}v_1 - v_0'F_0^{-1}v_0\right| \leq 2B_{v,m,T}\underline{f}^{-1}\|v_1 - v_0\| + B_{v,m,T}^2\underline{f}^{-2}\|F_1 - F_0\|_F. \tag{B.6}$$

Combining, the triangle inequality, (B.5), and (B.6) give

$$|\ell(v_1,F_1) - \ell(v_0,F_0)| \leq \frac{1}{2}\left[\frac{\sqrt{n_y}}{\underline{f}}\|F_1 - F_0\|_F + \frac{2B_{v,m,T}}{\underline{f}}\|v_1 - v_0\| + \frac{B_{v,m,T}^2}{\underline{f}^2}\|F_1 - F_0\|_F\right]. \tag{B.7}$$

It remains to relate the differences in v_t and F_t to the difference between ζ_0 and ζ_1. Define the line segment between these arguments by $\zeta(s) := \zeta_0 + s(\zeta_1 - \zeta_0),\ s \in [0,1]$. By the

fundamental theorem of calculus along this segment, $v_1 - v_0 = \int_0^1 D_\zeta v_t(\zeta(s);\theta)[\zeta_1 - \zeta_0]\, ds$. Taking norms and using the definition of $B_{D,m,T}$ gives

$$\|v_1 - v_0\| \le \int_0^1 \|D_\zeta v_t(\zeta(s);\theta)\|_{\text{op}} \|\zeta_1 - \zeta_0\|\, ds \le B_{D,m,T}\|\zeta_1 - \zeta_0\|. \tag{B.8}$$

Similar steps give

$$\|F_1 - F_0\|_F \le \int_0^1 \|D_\zeta F_t(\zeta(s);\theta)\|_{\text{op}} \|\zeta_1 - \zeta_0\|\, ds \le B_{D,m,T}\|\zeta_1 - \zeta_0\|. \tag{B.9}$$

Combining these inequalities with (B.5) and (B.6), there exists $L_{\ell,m,T} < \infty$, uniform over $t = 1, \ldots, T$ and $\theta \in \mathcal{V}$, such that

$$\left|\ell_t^m(\theta) - \ell_t^O(\theta)\right| \le L_{\ell,m,T} \left\|\zeta_t^m(\theta) - \zeta_t^O(\theta)\right\|, \text{ with } L_{\ell,m,T} = \frac{B_{D,m,T}}{2}\left[\frac{\sqrt{n_y}}{\underline{f}} + \frac{2B_{v,m,T}}{\underline{f}} + \frac{B_{v,m,T}^2}{\underline{f}^2}\right]. \tag{B.10}$$

Write $\bar{x}_t^j := \widehat{x}_{t|t-1}^j = \Pi_x S_{t-1}^j$. By the definition of ζ_t^j and the product norm,

$$\left\|\zeta_t^m - \zeta_t^O\right\| \le d_{t-1,m} + \left\|\Xi^m(\bar{x}_t^m;\theta) - \Xi^O(\bar{x}_t^O;\theta)\right\|.$$

Adding and subtracting $\Xi^O(\bar{x}_t^m;\theta)$ gives $\left\|\Xi^m(\bar{x}_t^m;\theta) - \Xi^O(\bar{x}_t^O;\theta)\right\| \le \left\|\Xi^m(\bar{x}_t^m;\theta) - \Xi^O(\bar{x}_t^m;\theta)\right\| + \left\|\Xi^O(\bar{x}_t^m;\theta) - \Xi^O(\bar{x}_t^O;\theta)\right\|$. The first term is $e_{t,m}(\theta)$. To bound the second term, define $x_t(s) := \bar{x}_t^O + s(\bar{x}_t^m - \bar{x}_t^O)$, $s \in [0,1]$. The fundamental theorem of calculus along this segment gives $\Xi^O(\bar{x}_t^m;\theta) - \Xi^O(\bar{x}_t^O;\theta) = \int_0^1 D_x\Xi^O(x_t(s);\theta)[\bar{x}_t^m - \bar{x}_t^O]\, ds$. Assumption 3 and the definition of $L_{O,T}$ therefore imply

$$\left\|\Xi^O(\bar{x}_t^m;\theta) - \Xi^O(\bar{x}_t^O;\theta)\right\| \le L_{O,T}\left\|\bar{x}_t^m - \bar{x}_t^O\right\|.$$

Moreover, since $\bar{x}_t^j = \Pi_x S_{t-1}^j$ and Π_x is a coordinate projection under the product norm, $\left\|\bar{x}_t^m - \bar{x}_t^O\right\| = \left\|\Pi_x(S_{t-1}^m - S_{t-1}^O)\right\| \le d_{t-1,m}$. Hence, $\left\|\Xi^O(\bar{x}_t^m;\theta) - \Xi^O(\bar{x}_t^O;\theta)\right\| \le L_{O,T} d_{t-1,m}$. Consequently,

$$\left\|\zeta_t^m - \zeta_t^O\right\| \le e_{t,m}(\theta) + (1 + L_{O,T})\, d_{t-1,m}(\theta).$$

Combining with (B.10), and averaging over t gives

$$|Q_{T,m}(\theta) - Q_{T,O}(\theta)| \le L_{\ell,m,T}\frac{1}{T}\sum_{t=1}^T e_{t,m}(\theta) + L_{\ell,m,T}\,(1 + L_{O,T})\,\frac{1}{T}\sum_{t=1}^T d_{t-1,m}(\theta).$$

Because the filters have a common initialization, $d_{0,m} = 0$, and therefore $\sum_{t=1}^{T} d_{t-1,m} \leq \sum_{t=1}^{T} d_{t,m}$. Combining this with the average bound in Theorem 1 it follows that

$$|Q_{T,m}(\theta) - Q_{T,O}(\theta)| \leq C_{\ell,m,T} \frac{1}{T} \sum_{t=1}^{T} e_{t,m}(\theta), \tag{B.11}$$

where one may take $C_{\ell,m,T} = L_{\ell,m,T}\left[1 + (1 + L_{O,T})\,\frac{C_{\Psi,T}}{1-\rho}\right] = L_{\ell,m,T}\left[1 + (1 + L_{O,T})\,\frac{C_F L_{K,T}}{1-\rho}\right]$. Taking the supremum over $\theta \in \mathcal{V}$ and using $\mathcal{E}_{T,m} = \sup_{\theta\in\mathcal{V}}\left\{\frac{1}{T}\sum_{t=1}^{T} e_{t,m}(\theta)\right\}$ proves (36)

$$\sup_{\theta\in\mathcal{V}} |Q_{T,m}(\theta) - Q_{T,O}(\theta)| \leq C_{\ell,m,T}\mathcal{E}_{T,m}.$$

It remains to establish the parameter bound. Define $\eta_{T,m} := \sup_{\theta\in\mathcal{V}} |Q_{T,m}(\theta) - Q_{T,O}(\theta)|$. Optimality of $\widehat{\theta}_{T,m}$ implies $Q_{T,m}(\widehat{\theta}_{T,m}) \geq Q_{T,m}(\widehat{\theta}_{T,O})$. Therefore,

$$\begin{aligned} Q_{T,O}(\widehat{\theta}_{T,O}) - Q_{T,O}(\widehat{\theta}_{T,m}) &\leq \left[Q_{T,O}(\widehat{\theta}_{T,O}) - Q_{T,m}(\widehat{\theta}_{T,O})\right] + \left[Q_{T,m}(\widehat{\theta}_{T,m}) - Q_{T,O}(\widehat{\theta}_{T,m})\right] \\ &\leq 2\eta_{T,m} \leq 2C_{\ell,m,T}\mathcal{E}_{T,m}, \end{aligned}$$

where the final inequality follows from (36).

Suppose first that condition (i) of the theorem holds, and let $\mathcal{C} := \mathcal{V}$. The curvature bound then holds throughout $\mathcal{C}$. Alternatively, suppose that condition (ii) holds. If, toward a contradiction, $\widehat{\theta}_{T,m} \in \mathcal{A}_r$, then the definition of ε_0 gives $Q_{T,O}(\widehat{\theta}_{T,O}) - Q_{T,O}(\widehat{\theta}_{T,m}) \geq 2\varepsilon_0$. On the other hand, $Q_{T,O}(\widehat{\theta}_{T,O}) - Q_{T,O}(\widehat{\theta}_{T,m}) \leq 2C_{\ell,m,T}\mathcal{E}_{T,m} < 2\varepsilon_0$, which is a contradiction. Hence, $\left\|\widehat{\theta}_{T,m} - \widehat{\theta}_{T,O}\right\| < r$. Thus, both maximizers belong to $\mathcal{B}_r$. In this case, let $\mathcal{C} := \mathcal{B}_r$.

Under either condition, $\mathcal{C}$ is convex, contains both $\widehat{\theta}_{T,O}$ and $\widehat{\theta}_{T,m}$, and satisfies $-\nabla^2_\theta Q_{T,O}(\theta) \succeq \mu I,\ \ \theta \in \mathcal{C}$. Consequently, the line segment joining $\widehat{\theta}_{T,O}$ and $\widehat{\theta}_{T,m}$ lies entirely within the region on which the curvature bound holds. Let $d := \widehat{\theta}_{T,m} - \widehat{\theta}_{T,O}$. Since $\widehat{\theta}_{T,O}$ maximizes $Q_{T,O}$ over the convex set $\mathcal{V}$, the constrained first-order condition gives $\nabla_\theta Q_{T,O}\left(\widehat{\theta}_{T,O}\right)' d \leq 0$. The integral form of Taylor's theorem therefore gives

$$Q_{T,O}(\widehat{\theta}_{T,m}) - Q_{T,O}(\widehat{\theta}_{T,O}) = \nabla_\theta Q_{T,O}\left(\widehat{\theta}_{T,O}\right)' d + \int_0^1 (1-s) d' \nabla^2_\theta Q_{T,O}\left(\widehat{\theta}_{T,O} + sd\right) d\, ds \leq -\frac{\mu}{2}\|d\|^2.$$

Consequently, taking square roots of the expression below proves (37):

$$\frac{\mu}{2}\left\|\widehat{\theta}_{T,m} - \widehat{\theta}_{T,O}\right\|^2 \leq Q_{T,O}(\widehat{\theta}_{T,O}) - Q_{T,O}(\widehat{\theta}_{T,m}) \leq 2C_{\ell,m,T}\mathcal{E}_{T,m}.$$

□

C Textbook New-Keynesian Model

Here we provide additional details of the model used in Section 3.2.

C.1 Model

Households. There is an infinitely lived representative household that maximizes:

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t D_t \left\{ \frac{C_t^{1-\tau} - 1}{1-\tau} - \chi \frac{N_t^{1+\eta}}{1+\eta} \right\} \right],$$

where $d_t = \log(D_t)$ is an intertemporal preference shock and follows an AR(1):

$$d_{t+1} = \rho_d d_t + \varepsilon_{t+1}^d, \quad \varepsilon_{t+1}^d \sim N(0, \sigma_d^2),$$

subject to a sequence of budget constraints:

$$P_t C_t + B_t = R_{t-1} B_{t-1} + P_t W_t N_t + T_t,$$

where C_t is consumption of the final good, B_t is an uncontingent nominal bond, R_{t-1} is the nominal interest rate, W_t is the real wage, and T_t contains dividends from firms net of taxes and transfers from the government.

The optimal consumption/saving and labor supply decisions are given by

$$1 = \beta \mathbb{E}_t \left[\frac{D_{t+1}}{D_t} \left(\frac{C_{t+1}}{C_t} \right)^{-\tau} \frac{R_t}{\Pi_{t+1}} \right], \qquad W_t = \frac{\chi N_t^{\eta}}{C_t^{-\tau}},$$

where $\Pi_{t+1} = \frac{P_{t+1}}{P_t}$. The stochastic discount factor is defined as $Q_{t,t+1} = \beta \frac{D_{t+1}}{D_t} \left(\frac{C_{t+1}}{C_t} \right)^{-\tau}$.

Firms. A competitive final good firm combines a continuum of intermediate goods $j \in [0,1]$. Intermediate good j is produced by a monopolist with technology $Y_t(j) = Z_t N_t(j)$ where Z_t is an aggregate technology process common to all firms. These firms choose prices $P_t(j)$ and labor inputs $N_t(j)$ to maximize the present discounted value of profit:

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} Q_{0,t} \left\{ \frac{P_t(j)}{P_t} Z_t N_t(j) - W_t N_t(j) - \frac{\phi}{2} \left(\frac{P_t(j)}{P_{t-1}(j)} - \Pi \right)^2 Y_t \right\} \right],$$

subject to a sequence of constraints:

$$Z_t N_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-1/\gamma} Y_t.$$

Optimal choices of $N_t(j)$ and $P_t(j)$ lead to the following optimal price-setting condition:

$$0 = \frac{Z_t N_t(j)}{P_t} - \phi\left(\frac{P_t(j)}{P_{t-1}(j)} - \Pi\right)\frac{Y_t}{P_{t-1}(j)} - \frac{1}{\gamma}\left(\frac{P_t(j)}{P_t} - \frac{W_t}{Z_t}\right)\left(\frac{P_t(j)}{P_t}\right)^{-1/\gamma-1}\frac{Y_t}{P_t}$$
$$+ \phi\mathbb{E}_t\left[Q_{t,t+1}\left(\frac{P_{t+1}(j)}{P_t(j)} - \Pi\right)\frac{P_{t+1}(j)}{P_t(j)^2}Y_{t+1}\right].$$

In a symmetric equilibrium $P_t(j) = P_t$ and $Y_t(j) = Z_t N_t(j) = Z_t N_t$ for all j, so this simplifies to:

$$\frac{1}{\gamma}\left(1 - \frac{W_t}{Z_t}\right) = 1 - \phi\left(\Pi_t - \Pi\right)\Pi_t + \phi\mathbb{E}_t\left[Q_{t,t+1}\left(\Pi_{t+1} - \Pi\right)\Pi_{t+1}\frac{Y_{t+1}}{Y_t}\right].$$

Government. The government levies lump-sum taxes and runs a balanced budget. The monetary authority follows a Taylor type rule:

$$\frac{R_t}{R} = \left(\frac{\Pi_t}{\Pi}\right)^{\phi_\pi}\left(\frac{Y_t}{Y}\right)^{\phi_y} e^{m_t},$$

where m_t is an exogenous monetary policy shock that follows an AR(1):

$$m_{t+1} = \rho_m m_t + \varepsilon^m_{t+1}, \quad \varepsilon^m_{t+1} \sim N(0, \sigma^2_m).$$

In the model with an occasionally binding zero-lower-bound we define the shadow rate as the typical rule

$$R^s_t = R\left(\frac{\Pi_t}{\Pi}\right)^{\phi_\pi}\left(\frac{Y_t}{Y}\right)^{\phi_y} e^{m_t},$$

and the actual nominal rate as $R_t = \max\{R^s_t, 1\}$.

Steady State. We focus on a non-stochastic steady state with inflation Π equal to the monetary authority target which implies $R = \Pi/\beta$, and the parameter χ normalizes steady state output Y (which equals N) to be unity: $\chi = 1 - \gamma$. Note, the first-order condition for labor supply implies $W = \chi$, so γ is the steady state wage markdown.

Equilibrium. The conditions for market clearing in the labor and goods markets are:

$$Y_t = C_t = N_t Z_t, \qquad W_t = (1-\gamma)N^\eta_t C^\tau_t = (1-\gamma)N^{\eta+\tau}_t Z^\tau_t.$$

A minimal state representation consists of states $x_t = (z_t, d_t, m_t)$ and policies $\{\Pi(x_t), N(x_t)\}$ defined through the equilibrium conditions (substituting market clearing conditions where appropriate). Using the steady state relations and letting lower case letters denote log-

deviations from steady state values e.g. $n_t = \ln N_t - \ln N$, the equilibrium conditions become:[1]

$$\begin{aligned}
\text{Euler/IS}: \quad & 1 = \mathbb{E}_t\left[e^{(\rho_d-1)d_t+\varepsilon^d_{t+1}-\tau\left(n_{t+1}-n_t+(\rho_z-1)z_t+\varepsilon^z_{t+1}\right)+r_t-\pi_{t+1}}\right] \\
\text{NK-Phillips}: \quad & \kappa\left(1-e^{(\eta+\tau)n_t+(\tau-1)z_t}\right) = -\left(e^{\pi_t}-1\right)e^{\pi_t} \\
& \quad + \beta\mathbb{E}_t\left[e^{(\rho_d-1)d_t+\varepsilon^d_{t+1}+(1-\tau)\left(n_{t+1}-n_t+(\rho_z-1)z_t+\varepsilon^z_{t+1}\right)}\left(e^{\pi_{t+1}}-1\right)e^{\pi_{t+1}}\right] \\
\text{Taylor Rule}: \quad & r_t = \phi_\pi\pi_t + \phi_y(n_t+z_t) + m_t.
\end{aligned}$$

C.2 Solution Methods

We solve the equilibrium system above using the local solution methods described in Section 2.5. At a given state x, both Taylor Projection (TP) and Dynamic Perturbation (DP) approximate inflation and employment by affine functions of the state. TP determines the coefficients of these local rules directly at x, using the local rule itself as the continuation policy when evaluating the nonlinear equilibrium conditions. DP instead constructs the continuation policy along an auxiliary equilibrium path and solves backward to recover the local rule at x. In both cases, expectations are evaluated over the distribution of future shocks, so the resulting first-order local policies need not be certainty equivalent.

With the zero lower bound, DP produces separate constrained and unconstrained local policies. The unconstrained nominal-interest-rate policy determines whether the lower bound binds; conditional on this classification, the corresponding regime-specific policy is used. This allows the local solution to account for both the current constraint and expectations of future regime changes. Further details on the construction, implementation, and accuracy of these local solution methods are provided in McCrary (2026).

For the global model solution, we use standard collocation methods with a Chebychev basis. For the model with ZLB, we allow the Chebychev coefficients to vary between regions of the state space where the ZLB binds or not, following Aruoba et al. (2018).

D Multi-Country RBC Filtering Details

State-space representation. All variables are expressed as log deviations from the deterministic steady state. Let $s_t = (k_t', a_t')'$ denote the $2N$-dimensional state vector. The $4N$

[1] We define $\kappa = \frac{1-\gamma}{\phi\Pi^2\gamma}$.

observables are output, investment, consumption, and labor in each country:

$$z_t = (y_t', i_t', c_t', \ell_t')', \qquad y_t = a_t + \alpha k_t + (1-\alpha)\ell_t.$$

Let $\widehat{s}_t \equiv \widehat{s}_{t|t-1}$ denote the forecasted state at date t. A first-order local solution centered at $\widehat{s}_t$ implies

$$\begin{aligned} k_{t+1} &= k_{t+1}^{\circ} + G_t^k(s_t - \widehat{s}_t), \\ q_t &= q_t^{\circ} + G_t^q(s_t - \widehat{s}_t), \qquad q \in \{i, c, \ell\}, \end{aligned}$$

where the superscript $\circ$ denotes the policy evaluated at $\widehat{s}_t$. Partition each policy derivative as $G_t^q = (G_{t,k}^q, G_{t,a}^q)$. The corresponding state-space representation is

$$s_{t+1} = b_t^s + T_t s_t + R\varepsilon_{t+1}, \qquad\qquad z_t = b_t^z + Z_t s_t + \nu_t, \tag{D.1}$$

where $\varepsilon_{t+1} \sim \mathcal{N}(0, I_N)$ and

$$T_t = \begin{bmatrix} G_{t,k}^k & G_{t,a}^k \\ 0 & D_\rho \end{bmatrix}, \qquad R = \begin{bmatrix} 0 \\ L_a \end{bmatrix}, \qquad D_\rho = \operatorname{diag}(\rho_a),$$

$$Z_t = \begin{bmatrix} \alpha I_N + (1-\alpha)G_{t,k}^\ell & I_N + (1-\alpha)G_{t,a}^\ell \\ G_{t,k}^i & G_{t,a}^i \\ G_{t,k}^c & G_{t,a}^c \\ G_{t,k}^\ell & G_{t,a}^\ell \end{bmatrix}.$$

The intercepts ensure that the affine system passes through the local solution:

$$b_t^s = \begin{bmatrix} k_{t+1}^{\circ} \\ D_\rho \widehat{a}_t \end{bmatrix} - T_t \widehat{s}_t, \qquad\qquad b_t^z = z_t^{\circ} - Z_t \widehat{s}_t,$$

where

$$z_t^{\circ} = ((y_t^{\circ})', (i_t^{\circ})', (c_t^{\circ})', (\ell_t^{\circ})')', \qquad y_t^{\circ} = \widehat{a}_t + \alpha\widehat{k}_t + (1-\alpha)\ell_t^{\circ}.$$

Loglinear+KF uses the steady-state loglinear solution, so the transition and measurement matrices are fixed and both intercepts are zero. DP+TVKF resolves the first-order Dynamic Perturbation problem at each forecasted state and reconstructs (b_t^s, T_t, b_t^z, Z_t) before the measurement update.

Measurement errors and initialization. Measurement errors are independent across dates, countries, and observables:

$$\nu_t \sim \mathcal{N}(0, H).$$

For each noise-free simulated series q_t^j, where $q \in \{y, i, c, \ell\}$, let $\widehat{\sigma}_{q,j}$ denote its sample standard deviation. The measurement-error standard deviation is set to

$$\sigma_{\nu,q,j} = 0.05\,\widehat{\sigma}_{q,j},$$

so that

$$H = 0.05^2\,\mathrm{Diag}\left(\widehat{\boldsymbol{\sigma}}_y^{\,2}, \widehat{\boldsymbol{\sigma}}_i^{\,2}, \widehat{\boldsymbol{\sigma}}_c^{\,2}, \widehat{\boldsymbol{\sigma}}_\ell^{\,2}\right).$$

Measurement errors are also independent of the structural innovations. Both filters use initial mean $\widehat{s}_{1|0} = 0$ and initial covariance equal to the sample covariance matrix of the simulated state path. This initialization abstracts from differences arising solely from the treatment of the initial state.

Calibration and simulation. The timing experiment considers $N \in \{2, 4, 8, 16, 32, 50\}$. The common parameters are $\alpha = 0.36, \beta = 0.96^{1/4}, \delta = 0.025, \phi = 1$. Negishi weights and risk aversion equal one in every country. The labor-disutility shifters normalize steady-state labor to one. Country-specific Frisch elasticities and productivity persistence parameters are

$$\eta_j = 0.3 + 0.4\frac{j-1}{N-1}, \qquad \rho_{a,j} = 0.85 + 0.10\frac{j-1}{N-1}.$$

Productivity innovations have standard deviation 0.02 and pairwise correlation 0.10:

$$L_a L_a' = 0.02^2\left(0.90 I_N + 0.10\iota_N\iota_N'\right).$$

For each N, one sample of length $T = 200$ is generated from the first-order DP solution. Measurement errors are added after simulation of the noise-free series. The same observed sample, measurement-error covariance, and initialization are used by both filters.

Timing protocol. Each reported time is the average of 100 evaluations of the same simulated sample. Every timed evaluation recomputes the loglinear QZ solution and then evaluates the complete likelihood. Under DP+TVKF, the QZ solution supplies the terminal and auxiliary policy, while the local DP problem is resolved at each of the $T = 200$ forecasted states. The projected times multiply the average evaluation time by 100,000, corresponding to a representative number of likelihood evaluations in Bayesian estimation. Further

details on the reduced equilibrium system, the DP backward recursion, and the closed-form conditional expectations are provided in McCrary (2026).

E NK-DMP Model

We first write the equilibrium conditions for a given continuation policy. We then evaluate the expectations for affine continuation policies and specialize the construction to TP, DP, and DP with a zero lower bound (DP-ZLB).

E.1 Normalization and equilibrium conditions

As in the preceding model description, productivity and the preference shifter have unit deterministic steady states. We use Q_t for the vacancy-filling rate and Ω_t for the pricing multiplier, denoted by q_t and ω_t in the main model description. Their log deviations are distinguished below. The non-employment payoff is denoted by b, as in the main text.

At the deterministic steady state,

$$R = \Pi/\beta, \qquad \Omega = \gamma, \qquad A = Q\Theta^{\alpha}, \qquad \delta = \frac{Q\Theta(1-N)}{N(1-Q\Theta)}.$$

The calibration $\Theta = 1$, $Q = 0.6$, and $N = 0.945$ therefore implies $A = 0.6$ and $\delta \simeq 0.0873$. The steady-state job-creation condition is

$$\frac{c}{Q}\Big[1 - \beta(1-\delta)\Big] = (1-\eta)(1-\gamma-b).$$

Define

$$\begin{aligned} n_t &= \log(N_t/N), & \theta_t &= \log(\Theta_t/\Theta), & \pi_t &= \log(\Pi_t/\Pi), \\ \omega_t &= \log(\Omega_t/\gamma), & q_t &= \log(Q_t/Q), & r_t &= \log(R_t/R). \end{aligned}$$

Let $y_t = n_t + z_t$ be log output relative to its steady state and let

$$x_t = (n_{t-1}, z_t, d_t, m_t, s_t, a_t)', \qquad u_t = (\theta_t, \pi_t, \omega_t, n_t)'.$$

Thus current employment n_t becomes the first component of x_{t+1}. Conditional on x_t and u_t, the next state is

$$x_{t+1} = \bar{x}_{t+1|t} + J\xi_{t+1}, \qquad \bar{x}_{t+1|t} = (n_t, \rho_z z_t, \rho_d d_t, \rho_m m_t, \rho_s s_t, \rho_a a_t)', \qquad J = \begin{pmatrix} 0_{1\times 5} \\ I_5 \end{pmatrix}. \quad \text{(E.1)}$$

Here $\xi_{t+1} \sim \mathcal{N}(0, \Sigma)$ is independent of time-t information, with $\Sigma = \text{diag}(\sigma_z^2, \sigma_d^2, \sigma_m^2, \sigma_s^2, \sigma_a^2)$. In terms of the standardized innovations in the main text, $\xi_{t+1}^i = \sigma_i \varepsilon_{t+1}^i$.

An arbitrary continuation policy. Fix a continuation rule $g^+ = (g_\theta^+, g_\pi^+, g_n^+)$ on the next-period state space and set

$$(\theta_{t+1}, \pi_{t+1}, n_{t+1}) = g^+(x_{t+1}), \qquad q_{t+1} = a_{t+1} - \alpha\theta_{t+1}, \qquad \Delta y_{t+1} = n_{t+1} + z_{t+1} - n_t - z_t.$$

The superscript + identifies a continuation rule; it does not require that this rule equal the current policy. No continuation rule for ω_{t+1} is needed in the expectations below. For weights $k = (k_y, k_\pi, k_q, k_s)$, define the single expectation

$$\mathcal{E}_t(k; g^+) = \mathbb{E}_t[\exp\{d_{t+1} - d_t + k_y \Delta y_{t+1} + k_\pi \pi_{t+1} + k_q q_{t+1} + k_s s_{t+1}\}]\,. \tag{E.2}$$

The expectation is conditional on the current state and current choices, with x_{t+1} given by (E.1); the required moments are assumed finite. Suppressing g^+ in its arguments, the Euler, price-setting, job-creation, and employment equations are

$$1 = e^{r_t}\mathcal{E}_t(-\tau, -1, 0, 0), \tag{E.3}$$

$$\frac{e^{\omega_t} - 1}{\phi\Pi^2} = -\Big(e^{\pi_t} - 1\Big)e^{\pi_t} + \beta\Big[\mathcal{E}_t(1-\tau, 2, 0, 0) - \mathcal{E}_t(1-\tau, 1, 0, 0)\Big], \tag{E.4}$$

$$e^{-q_t} = \frac{Q(1-\eta)}{c}\Big[(1 - \gamma e^{\omega_t})e^{z_t} - b\Big] + \beta\Big[\mathcal{E}_t(-\tau, 0, -1, 0) - \delta\mathcal{E}_t(-\tau, 0, -1, 1)\Big], \tag{E.5}$$

$$e^{n_t - n_{t-1}} = 1 - \delta e^{s_t} + Q\Theta e^{a_t + (1-\alpha)\theta_t}\left[\frac{e^{-n_{t-1}}}{N} - 1 + \delta e^{s_t}\right]. \tag{E.6}$$

In these equations $q_t = a_t - \alpha\theta_t$. The factor β drops out of (E.3) because $\beta R/\Pi = 1$. The shadow rate and the rate entering the Euler equation satisfy

$$r_t^s = \phi_\pi \pi_t + \phi_y(n_t + z_t) + m_t, \qquad r_t = \begin{cases} r_t^s, & \text{without a ZLB,} \\ \max\{\underline{r}, r_t^s\}, & \text{with a ZLB,} \end{cases} \qquad \underline{r} = -\log R. \tag{E.7}$$

Thus a binding ZLB means $R_t = 1$, or $r_t = \underline{r}$.

E.2 Affine continuation policies and local coefficients

For a fixed affine continuation rule, write

$$g_v^+(x') = b_0^{v,+} + B^{v,+}x', \qquad v \in \{\theta, \pi, n\}, \qquad B^{v,+} \in \mathbb{R}^{1\times 6}. \tag{E.8}$$

Let ι_i be the unit vector selecting innovation i from ξ_{t+1}. Substitution of (E.1) gives

$$v_{t+1} = \bar{v}_t + \ell_v' \xi_{t+1}, \qquad \bar{v}_t = b_0^{v,+} + B^{v,+} \bar{x}_{t+1|t}, \qquad \ell_v = J'(B^{v,+})'.$$

The bars and loadings in this subsection always refer to the fixed continuation rule. In particular,

$$\begin{aligned} \Delta y_{t+1} &= \overline{\Delta y}_t + \ell_\Delta' \xi_{t+1}, & \overline{\Delta y}_t &= \bar{n}_t - n_t + (\rho_z - 1) z_t, & \ell_\Delta &= \ell_n + \iota_z, \\ q_{t+1} &= \bar{q}_t + \ell_q' \xi_{t+1}, & \bar{q}_t &= \rho_a a_t - \alpha \bar{\theta}_t, & \ell_q &= \iota_a - \alpha \ell_\theta. \end{aligned}$$

Consequently, every expectation in (E.3)–(E.5) follows from

$$\ell(k) = \iota_d + k_y \ell_\Delta + k_\pi \ell_\pi + k_q \ell_q + k_s \iota_s, \tag{E.9}$$

$$\mathcal{E}_t(k; g^+) = \exp\Big\{ (\rho_d - 1) d_t + k_y \overline{\Delta y}_t + k_\pi \bar{\pi}_t + k_q \bar{q}_t + k_s \rho_s s_t \Big\} \mathcal{M}\Big(\ell(k)\Big), \tag{E.10}$$

$$\mathcal{M}(w) = \mathbb{E}[e^{w' \xi_{t+1}}] = \exp\Big(\tfrac{1}{2} w' \Sigma w\Big).$$

For example, the Euler equation uses $\ell(-\tau, -1, 0, 0) = \iota_d - \tau(\ell_n + \iota_z) - \ell_\pi$. The two pricing terms follow from $(e^{\pi_{t+1}} - 1)e^{\pi_{t+1}} = e^{2\pi_{t+1}} - e^{\pi_{t+1}}$; the two job-creation terms follow from $1 - \delta e^{s_{t+1}}$. Thus productivity risk and the covariance with separation risk enter through the same moment formula.

Recovering the current local rule. At an expansion point $\bar{x}$, approximate the four current choices by $u(x) = b_0 + Bx$, with $B \in \mathbb{R}^{4 \times 6}$ and rows ordered as (θ, π, ω, n). Let $F(x, u; g^+)$ collect the four residuals in (E.3)–(E.6), after substituting the applicable rate rule. The value and slope conditions are

$$F(\bar{x}, \bar{u}; g^+) = 0, \qquad F_x(\bar{x}, \bar{u}; g^+) + F_u(\bar{x}, \bar{u}; g^+) B = 0, \qquad b_0 = \bar{u} - B\bar{x}. \tag{E.11}$$

Derivatives hold the continuation coefficients fixed but include the dependence of x_{t+1} on current employment. For a given g^+, $B = -F_u^{-1} F_x$ when the expected residual is continuously differentiable and F_u is nonsingular. These conditions are imposed within each smooth current regime when a ZLB is present.

Taylor Projection. TP uses the current affine rule as its own continuation:

$$b_0^{v,+} = b_0^v, \qquad B^{v,+} = B^v, \qquad v \in \{\theta, \pi, n\}.$$

Under this restriction, (E.11) is a joint system for the four current intercepts and their 24 slopes. The coefficients are held fixed when differentiating with respect to the state, but the continuation rule is still differentiated through its state argument.

Dynamic Perturbation. Given an auxiliary path $\{x^{[h]}\}_{h=0}^{H}$, with $x^{[0]} = \bar{x}$, and a terminal continuation rule at $h = H$, DP works backward. For $h = H-1, \ldots, 0$, at $x^{[h]}$ set

$$b_0^{v,+} = b_{0,h+1}^{v}, \qquad B^{v,+} = B_{h+1}^{v}, \qquad v \in \{\theta, \pi, n\},$$

and use (E.11) to recover $(b_{0,h}, B_h)$. The rule at $h = 0$ is the desired local solution. The index h labels the auxiliary path, not an observed calendar period. All intercepts use the common, uncentered coordinates in (E.8); a rule stored as $\bar{v}_h + B_h^v(x - x^{[h]})$ must first be written with $b_{0,h}^v = \bar{v}_h - B_h^v x^{[h]}$.

E.3 DP with a zero lower bound

DP-ZLB carries two affine continuation rules $g^{+,U}$ and $g^{+,C}$, for the unconstrained and constrained regimes. Define $\bar{v}_t^j$, ℓ_v^j, $\overline{\Delta y}_t^j$, and $\bar{q}_t^j$ for each $j \in \{U, C\}$ exactly as above. Here $\bar{v}_t^j$ is the regime specific policy rule evaluated at $\bar{x}_{t+1|t}$; it is not a mean conditional on regime j. The regime label describes the constraint status at the date of the policy; both current regimes incorporate expectations of subsequent switching.

Future regime selection. The approximation classifies the future regime using the unconstrained continuation shadow rate. Its distance from the bound is

$$\begin{aligned} r_{t+1}^{s,U} - \underline{r} &= \bar{h}_t + \ell_h' \xi_{t+1}, \\ \bar{h}_t &= \phi_\pi \bar{\pi}_t^U + \phi_y(\bar{n}_t^U + \rho_z z_t) + \rho_m m_t - \underline{r}, \\ \ell_h &= \phi_\pi \ell_\pi^U + \phi_y(\ell_n^U + \iota_z) + \iota_m. \end{aligned} \tag{E.12}$$

Set $\mathcal{R}_U = \{\xi : \bar{h}_t + \ell_h'\xi > 0\}$ and $\mathcal{R}_C = \{\xi : \bar{h}_t + \ell_h'\xi \le 0\}$, and use $g^{+,j}$ on $\mathcal{R}_j$. The expectation in (E.2) then becomes

$$\mathcal{E}_t(k; g^+) = \sum_{j \in \{U,C\}} \exp\Big\{(\rho_d - 1)d_t + k_y \overline{\Delta y}_t^j + k_\pi \bar{\pi}_t^j + k_q \bar{q}_t^j + k_s \rho_s s_t\Big\} \mathcal{M}_{j,t}\Big(\ell^j(k)\Big), \tag{E.13}$$

where $\ell^j(k)$ is (E.9) for regime j and

$$\mathcal{M}_{j,t}(w) = \mathbb{E}_t\Big[e^{w'\xi_{t+1}} \mathbf{1}\{\xi_{t+1} \in \mathcal{R}_j\}\Big].$$

Let $\varsigma_t^2 = \ell_h' \Sigma \ell_h$. For $\varsigma_t > 0$, completing the square in the Gaussian density gives

$$\mathcal{M}_{U,t}(w) = e^{\frac{1}{2} w' \Sigma w} \Phi\left(\frac{\bar{h}_t + \ell_h' \Sigma w}{\varsigma_t} \right), \quad \mathcal{M}_{C,t}(w) = e^{\frac{1}{2} w' \Sigma w} \Phi\left(-\frac{\bar{h}_t + \ell_h' \Sigma w}{\varsigma_t} \right), \tag{E.14}$$

where Φ is the standard normal cumulative distribution function.

Current regimes and the backward recursion. At each auxiliary point, use the two next-point continuation rules in (E.13). Solve (E.11) separately with $r_t = r_t^{s,U}$ for the current unconstrained policy and with $r_t = \underline{r}$ for the current constrained policy. Current employment, and hence $\bar{x}_{t+1|t}$ and $\bar{h}_t$, is recomputed within each of these solves. The constrained rate has zero slopes and intercept $\underline{r}$. When obtaining policy slopes, we differentiate the full expression (E.13), including the Gaussian probability terms and the dependence of the boundary on current employment.

The four equilibrium equations therefore remain (E.3)–(E.6): only the continuation expectation and the current rate rule change. A single continuation policy with $\mathcal{R}_U = \mathbb{R}^5$ recovers ordinary DP. The moment formulas are exact for the stated affine or piecewise affine continuation approximation, not for an arbitrary nonlinear equilibrium policy.

F Data

Table F1: Data details: estimation and filtering

Variable	Details	Transformation	Source
Employment n_t^{obs}, one minus unrate	Percent, Quarterly, Seasonally adjusted	log and Hamilton (2018) filtered	UNRATE (FRED)
Market tightness θ_t^{obs}	Vacancies over unrate, Quarterly	log and Hamilton (2018) filtered	Barnichon (2010), JTSJOL, UNRATE (FRED)
Nominal rate r_t^{obs}	3-Month Treasury Bill Secondary Market Rate, Discount Basis	Quarterly*, log	DTB3 (FRED)
Inflation $\pi_{1,t}^{\text{obs}}$	Gross domestic product (implicit price deflator)	Quarter-to-quarter P_t/P_{t-1}, log	GDPDEF (FRED)
Inflation $\pi_{2,t}^{\text{obs}}$	Consumer Price Index	Quarter-to-quarter P_t/P_{t-1}, log	CPIAUCSL (FRED)
Separation rate s_t^{obs}	Shimer (2005)	Quarterly average, log and Hamilton (2018) filtered	UEMPLT5, UNRATE (FRED)
Job finding rate f_t^{obs}	Shimer (2005)	Quarterly average, log and Hamilton (2018) filtered	UEMPLT5, UNRATE (FRED)

Notes: (*) 3-month treasury bill (R3M) converted to a quarterly interest rate $(1 + R3M/100)^{(1/4)}$.

G Supplemental Figures for NKDMP Application

Figure G1: State-Dependent Policy Coefficients for DP (with ZLB), computed at filtered states

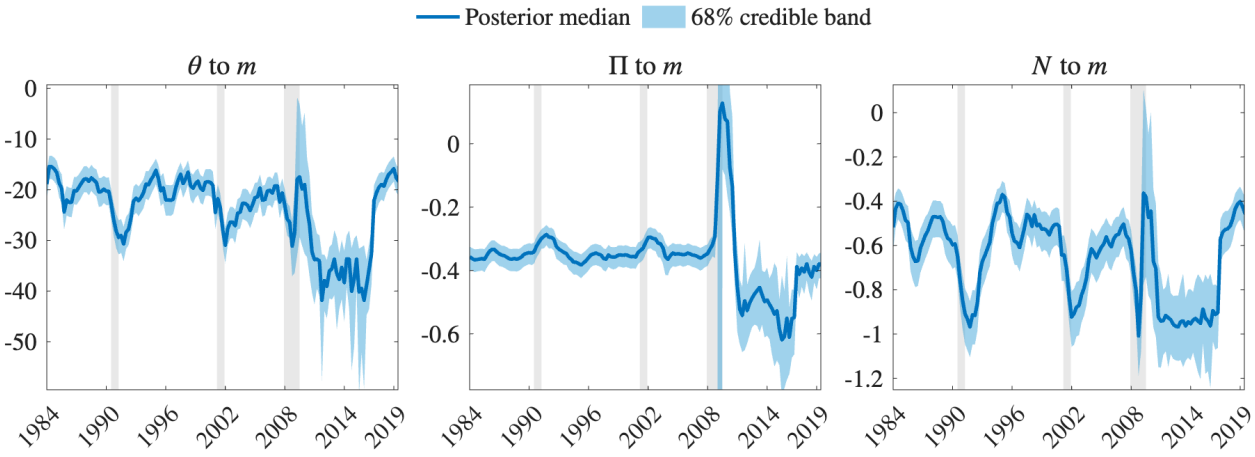


Figure G2: Model-based impulse response functions over time (no ZLB)

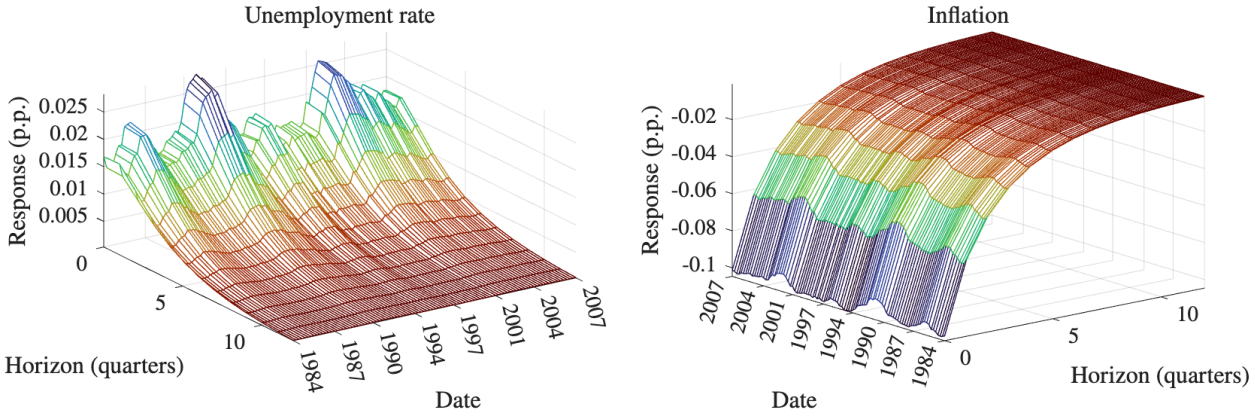


Figure G3: Model-based impulse response functions over time (model with ZLB)

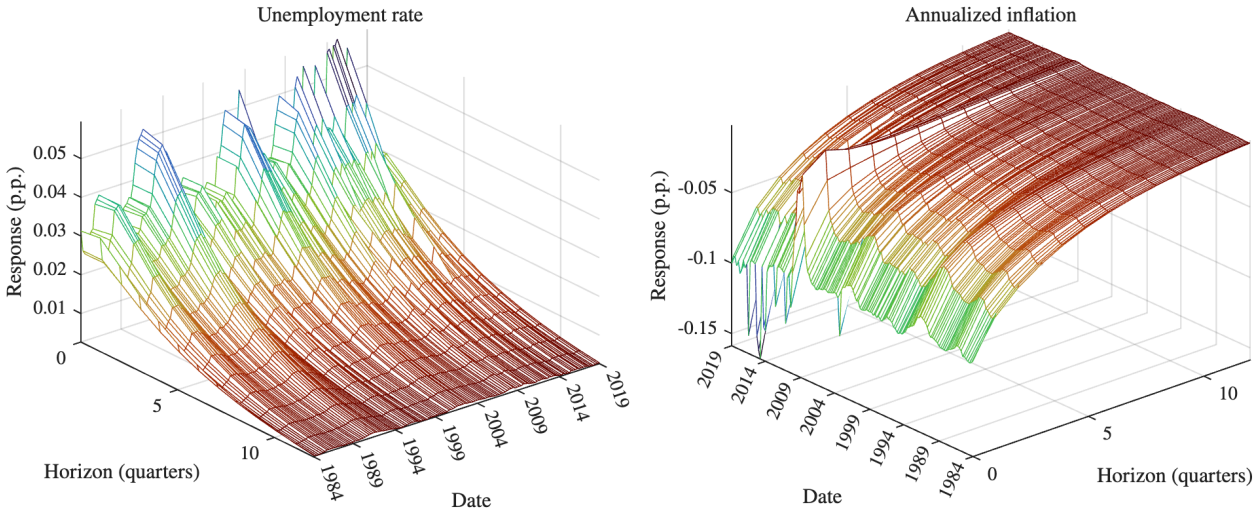


Figure G4: Cumulative (over 12 quarters) impulse response functions over time (with ZLB)

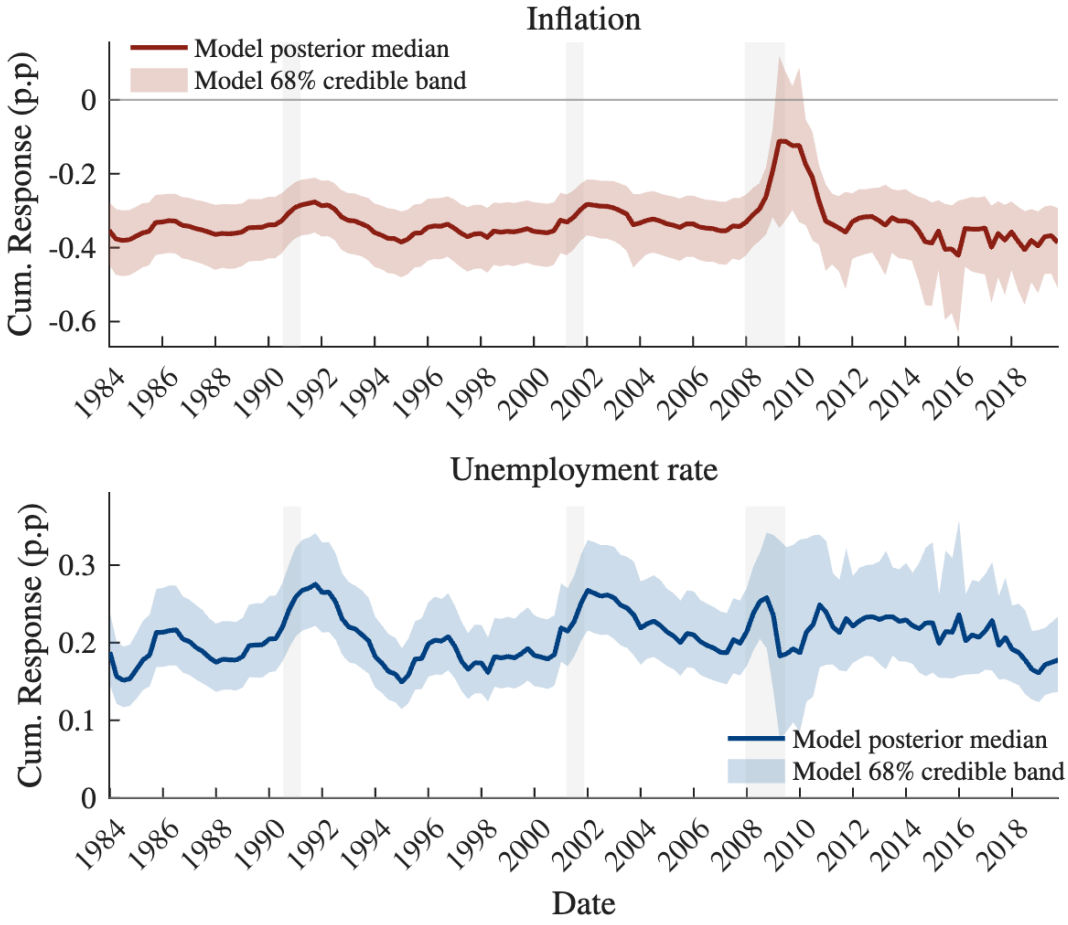
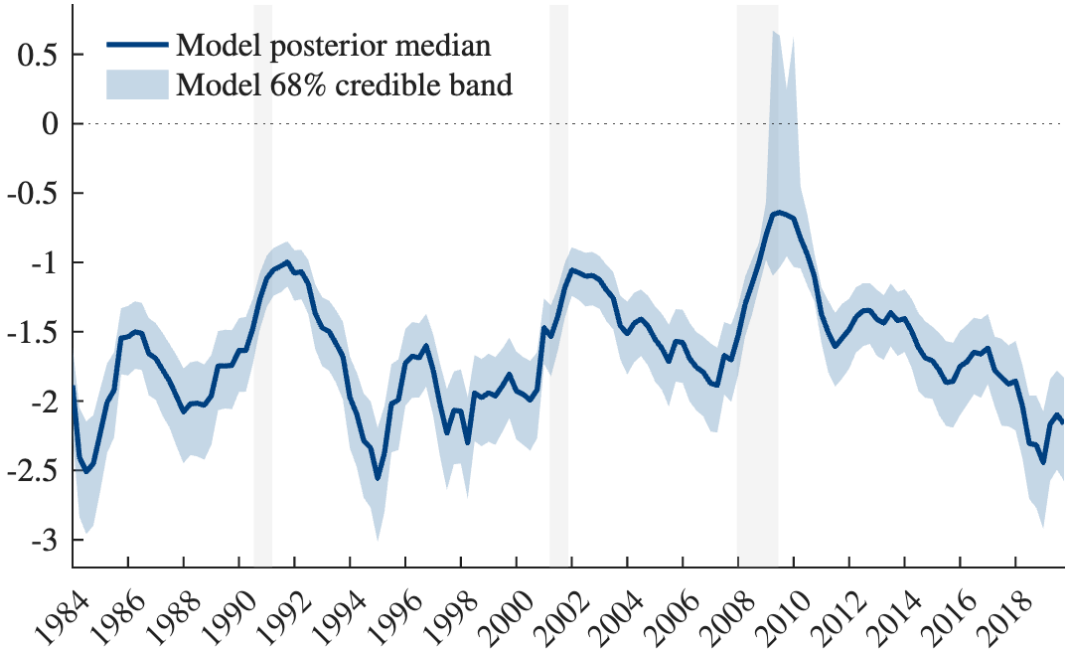


Figure G5: Model-based Phillips Multiplier (with ZLB)



H Stochastic Growth Model with Alternative Stochastic Processes

We illustrate how to accommodate an alternative stochastic process within our methodology: regime-switching.

H.1 Basic stochastic growth model

To simplify exposition, we work through a small example: the stochastic growth model. For any exogenous driving process for total factor productivity, the equilibrium for this model can be summarized as:

$$C_t^{-\tau} = \beta \mathbb{E}_t \left[C_{t+1}^{-\tau} \left(\alpha e^{z_{t+1}} K_{t+1}^{\alpha-1} + 1 - \delta \right) \right]$$
$$K_{t+1} = e^{z_t} K_t^{\alpha} + (1-\delta) K_t - C_t.$$

In a deterministic steady state, $z = 0$, and the solution is:

$$K = \left[\frac{1}{\alpha} \left(\frac{1}{\beta} - 1 + \delta \right) \right]^{\frac{1}{\alpha - 1}}, \quad C = K^{\alpha} - \delta K.$$

In what follows we consider log deviations from steady state value $c_t = \log(C_t/C)$ and $k_t = \log(K_t/K)$, so that $C_t = Ce^{c_t}$ and $K_t = Ke^{k_t}$. Substitution of this change of variables

into the equilibrium gives:

$$e^{-\tau c_t} = \mathbb{E}_t\left[e^{-\tau c_{t+1}}\left\{(1-\beta(1-\delta))\, e^{z_{t+1}+(\alpha-1)k_{t+1}} + \beta(1-\delta)\right\}\right]$$
$$e^{k_{t+1}} = \frac{1}{\alpha}\left(\frac{1}{\beta} - (1-\delta)\right) e^{z_t+\alpha k_t} + (1-\delta)e^{k_t} - \left(\frac{1}{\alpha}\left(\frac{1}{\beta} - (1-\delta)\right) - \delta\right) e^{c_t}.$$

H.2 Regime Switching

A common extension in macroeconomic models allows exogenous state dynamics to follow a regime-switching process. This class includes models with rare disasters and other forms of time-varying volatility. Specifically, let $s_t \in \{0, 1\}$ denote a latent regime indicator that evolves according to the transition matrix

$$s_t \sim \Pi, \qquad \Pi = \begin{bmatrix} p_0 & 1-p_0 \\ 1-p_1 & p_1 \end{bmatrix},$$

where $p(s_{t+1} = 1 \mid s_t = 1) = p_1$ and $p(s_{t+1} = 0 \mid s_t = 0) = p_0$. Although the framework extends naturally to multiple regimes, we focus on the two-state case for exposition. The latent state s_t governs the evolution of productivity z_t:

$$z_{t+1} = \begin{cases} \rho_0 z_t + \sigma_0 \eta_{t+1} & \text{if } s_t = 0, \\ \rho_1 z_t + \sigma_1 \eta_{t+1} & \text{if } s_t = 1, \end{cases} \qquad \eta_t \sim N(0,1). \tag{H.1}$$

As before, let $y_t = z_t + \alpha k_t$, and define the state vector as $x_t = (k_t, z_t)$. The policy function for consumption and the law of motion for capital are regime-dependent as denoted by:

$$c_t = a_{j0} + a_{j1}k_t + a_{j2}z_t,$$
$$k_{t+1} = b_{j0} + b_{j1}k_t + b_{j2}z_t$$

so the coefficient vectors $\mathbf{a}_j$ and $\mathbf{b}_j$ are assumed to vary with the regime $j = s_t$.

Expectations The regime-switching policy rules imply that the realization of consumption at time $t+1$ depends on the state sequence s_t, s_{t+1}:

$$c_{t+1} = \Big\{a_{j0} + a_{j1}b_{i0} + a_{j1}b_{i1}k_t + (a_{j1}b_{i2} + a_{j2}\rho_i)z_t + a_{j2}\sigma_i\eta_{t+1}, \text{ if } \ s_t = i, s_{t+1} = j$$

Using iterated expectations, the expectation terms in the Markov model can be written as:

$$\mathbb{E}_t[\cdot] = \mathbb{E}_{s_{t+1}}[\mathbb{E}_{z_{t+1}}[\cdot|s_{t+1}, s_t, z_t]|s_t].$$

Applying this to the first expectation term in the Euler equation yields:

$$\mathbb{E}_{z_{t+1}}\left[e^{-\tau c_{t+1}+z_{t+1}+(\alpha-1)k_{t+1}}|s_{t+1},s_t,z_t\right]=$$
$$\exp\left\{-\tau(a_{j0}+a_{j1}b_{i0}+a_{j1}b_{i1}k_t+(a_{j1}b_{i2}+a_{j2}\rho_i)z_t)+\rho_i z_t+(\alpha-1)k_{t+1}+\frac{(1-\tau a_{j2})^2\sigma_i^2}{2}\right\},$$

if $s_t=i$ and $s_{t+1}=j$. Here, k_{t+1} is determined by s_t and other known variables at time t. Similarly,

$$\mathbb{E}_{z_{t+1}}\left[e^{-\tau c_{t+1}}\right]=\exp\left\{-\tau(a_{j0}+a_{j1}b_{i0}+a_{j1}b_{i1}k_t+(a_{j1}b_{i2}+a_{j2}\rho_i)z_t)+\frac{(\tau a_{j2})^2\sigma_i^2}{2}\right\}.$$

These expressions show that the regime-switching model also allows expectations to be evaluated analytically under the Taylor projection approach, without numerical integration.

State Space Representation Conditional on s_t, this is simply a linear state space system, with, for $s_t=j$, a measurement equation

$$\begin{bmatrix}y_t\\c_t\end{bmatrix}=\begin{bmatrix}0\\a_{j0}\end{bmatrix}+\begin{bmatrix}\alpha & 1\\a_{j1} & a_{j2}\end{bmatrix}\begin{bmatrix}k_t\\z_t\end{bmatrix}+\varepsilon_t,\quad \varepsilon_t\sim N(0,H).$$

and, again for $s_t=j$, the state transition equation is given by

$$\begin{bmatrix}k_{t+1}\\z_{t+1}\end{bmatrix}=\begin{bmatrix}b_{j0}\\0\end{bmatrix}+\begin{bmatrix}b_{j1} & b_{j2}\\0 & \rho_j\end{bmatrix}\begin{bmatrix}k_t\\z_t\end{bmatrix}+\begin{bmatrix}0\\\sigma_j\end{bmatrix}\eta_t,\quad \eta_t\sim N(0,1).$$

Filter To initialize the filter, set $x_{1|0}$, $P_{1|0}$, and $\hat{\xi}_{1|0}$, where $\hat{\xi}_{1|0}$ contains the regime probabilities $\mathbb{P}(s_1=j)$ prior to observing any data. These can, for example, be initialized using the stationary distribution implied by Π. Solve the model at $x_{1|0}$ using Taylor projection. Then, for each regime $j\in\{0,1\}$, perform the update and prediction steps of Algorithm 1.

For each regime, compute the regime-specific likelihood contribution $\ell_t^{(j)}$ by evaluating the forecast error v_t under a normal distribution with mean zero and variance F_t. The regime probabilities are then updated as:

$$\hat{\xi}_{t|t}^{(j)}:=\mathbb{P}(s_t=j\mid y_{1:t}^{\text{obs}})=\frac{\ell_t^{(j)}\cdot\hat{\xi}_{t|t-1}^{(j)}}{\sum_{j'=0}^{1}\ell_t^{(j')}\cdot\hat{\xi}_{t|t-1}^{(j')}},\quad j\in\{0,1\}.$$

Following Hamilton (1994) and Kim (1994), the predicted regime probabilities are:

$$\hat{\xi}_{t|t-1} = \Pi' \cdot \hat{\xi}_{t-1|t-1},$$

where $\hat{\xi}^{(i)}_{t-1|t-1} = \mathbb{P}(s_{t-1} = i \mid y^{\text{obs}}_{1:t-1})$ is the filtered probability from the previous period, and $\hat{\xi}^{(i)}_{t|t-1}$ is the forecasted probability of regime i at time t.

Latent variables are then filtered and forecasted via regime-weighted averages. For instance:

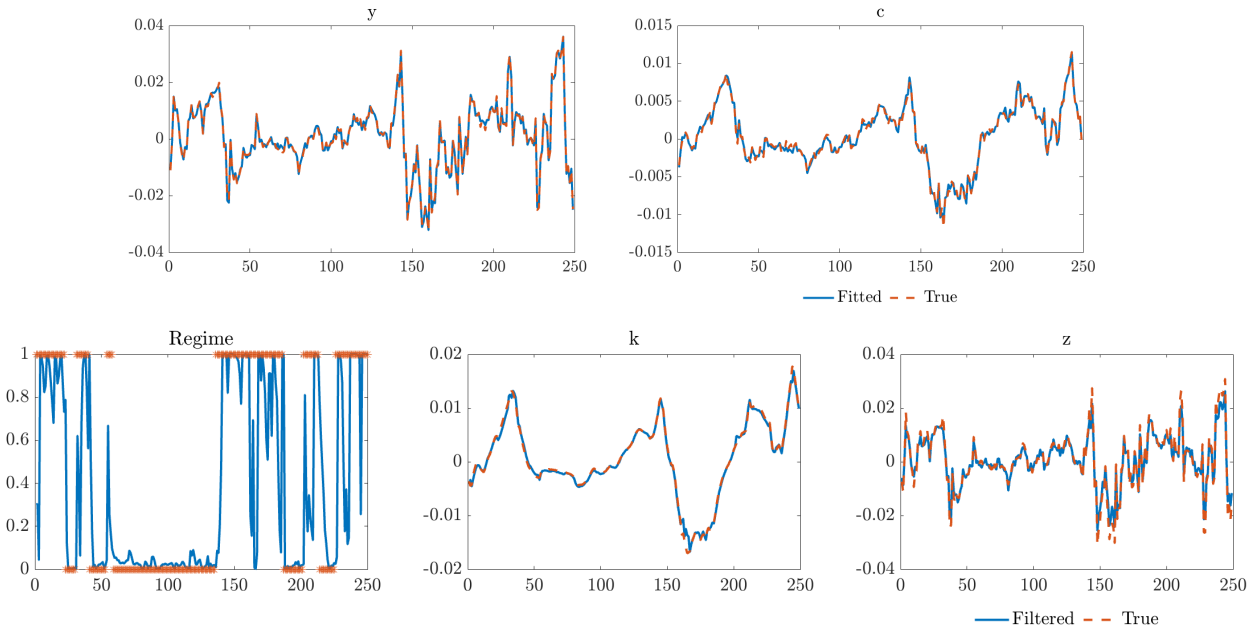
$$\hat{z}_{t|t} = \sum_{j=0}^{1} \hat{\xi}^{(j)}_{t|t} \cdot \hat{z}^{(j)}_{t|t}, \quad \hat{z}_{t+1|t} = \sum_{j=0}^{1} \hat{\xi}^{(j)}_{t+1|t} \cdot \hat{z}^{(j)}_{t+1|t},$$

where $\hat{z}^{(j)}_{t|t}$ denotes the filtered state conditional on regime j. The model is then re-solved using Taylor projection at $\hat{x}_{t+1|t}$, and the Kalman update and prediction steps are repeated.

Parameters We use the following parameterization: $p_0 = 0.97$, $\rho_0 = 0.8$, $\sigma_0 = 0.01$, $p_1 = 0.95$, $\rho_1 = 0.95$, $\sigma_1 = 0.002$, risk aversion $\tau = 2$, $\beta = 1.04^{-1/4}$, $\alpha = 0.36$ and $\delta = 0.025$.

Demonstration To illustrate the performance of the filter described above, we simulate a dataset of length $T = 250$ from the dynamic Taylor Projection model solution. We assume measurement error is 10% of the standard deviation of the observed data in the simulation below. We then recover the latent states using the filter. Results are visualized in Figure H1. As we can see in this figure, the filtered states track the true values used in the simulation closely. It takes about 1.3 seconds to run the filter 100 times on a dataset of length $T = 250$.

Figure H1: Fitted values (top) and filtered and true states values (bottom)



I Heterogeneous Agent Models

Equilibrium. The aggregate states of the model are $x_t = (z_t, \mu_t)$, with μ_t the distribution of households over employment and assets, i.e., (ε, a). The equilibrium unknowns are next period's assets, $a'(\varepsilon, a; z, \mu)$, interest rate $r(z, \mu)$, wages $w(z, \mu)$ and distribution $\mu'(z, \mu)$, which have to satisfy three sets of conditions:

1. For given prices and law of motion of the distribution, household optimization satisfies: $c(\varepsilon, a; z, \mu)^{-1} \geq \beta \mathbb{E}\left[(1 + r(z', \mu'(z, \mu)))c(\varepsilon', a'(\varepsilon, a; z, \mu); z', \mu'(z, \mu))^{-1} | \varepsilon, z, \mu)\right]$ with equality if $a'(\varepsilon, a; z, \mu) > 0$. Here $c(\varepsilon, a; z, \mu) = w(z, \mu)((1-\tau)\varepsilon + b(1-\varepsilon)) + (1+r(z, \mu))a - a'(\varepsilon, a; z, \mu)$ is optimal consumption.

2. Interest rate and wages satisfy:

$$r(z, \mu) = \alpha e^z K^{\alpha-1} L^{1-\alpha} - \delta$$
$$w(z, \mu) = (1 - \alpha) e^z K^{\alpha} L^{-\alpha}$$

   where $K = \sum_{\varepsilon} \int a d\mu(\varepsilon, a)$.

3. Law of motion for the distribution: for all measurable sets Λ_a,

$$\mu'(z, \mu)(\varepsilon, \Lambda_a) = \sum_{\tilde{\varepsilon}} \pi(\varepsilon | \tilde{\varepsilon}, z) \int 1\{a'(\tilde{\varepsilon}, a; z, \mu) \in \Lambda_a\} \mu(\tilde{\varepsilon}, da)$$

Computation. In what follows, when states and policies are expressed in log-deviations from steady state, they are denoted by $\hat{\cdot}$, and $\bar{\cdot}$ denotes a steady state value. In the Taylor Projection approach, we use two moments to approximate the cross-sectional distribution $\mu_{\varepsilon,t}(a)$, so the local linear policy functions take the form:

$$\mathbf{g}_t = \bar{\mathbf{g}} + b_0^g + B^g \begin{bmatrix} \hat{z}_t & m_{0,t}^1 & m_{1,t}^1 & m_{0,t}^2 & m_{1,t}^2 & L_t \end{bmatrix}'$$
$$\boldsymbol{\theta}_t = \bar{\boldsymbol{\theta}} + b_0^{\theta} + B^{\theta} \begin{bmatrix} \hat{z}_t & m_{0,t}^1 & m_{1,t}^1 & m_{0,t}^2 & m_{1,t}^2 & L_t \end{bmatrix}',$$

where $\mathbf{g}_t$ and $\boldsymbol{\theta}_t$ are vectors, so B^g and B^{θ} are matrices. Note, these equations are of the form $y_t = \bar{y} + B \times (1, x_t)'$, such that the Taylor projection is centered at the steady state value when $B = 0$. Thus, the unknown policies are linear deviations from the steady-state value. In addition to the local linear policy rules, the Taylor Projection solution yields linearized laws of motions for endogenous states, i.e., moments $i = 1, 2$, $\varepsilon \in \{0, 1\}$, and aggregate

employment:

$$m^i_{\varepsilon,t+1} = \bar{m}^i_\varepsilon + b^{m,\varepsilon,i}_0 + B^{m,\varepsilon,i}\begin{bmatrix}\hat{z}_t & m^1_{0,t} & m^1_{1,t} & m^2_{0,t} & m^2_{1,t} & L_t\end{bmatrix}', \tag{I.1}$$

$$L_{t+1} = \bar{L} + b^L_0 + B^L\begin{bmatrix}\hat{z}_t & m^1_{0,t} & m^1_{1,t} & m^2_{0,t} & m^2_{1,t} & L_t\end{bmatrix}'. \tag{I.2}$$

Under dynamic Taylor Projection, solving the model locally at different points in the state space yields different policy coefficients, allowing the solution to capture state dependence with respect to the current aggregate state x_t. Such state dependence has been shown to be empirically relevant. For instance, in heterogeneous firm models, Ottonello and Winberry (2020) argue that monetary policy transmission varies with the distribution of firms' distance to default. Dynamic Taylor Projections can accommodate this type of state-dependent shock transmission.

After dimension-reduction, the model has six states: $x_t = (z_t, m^1_{0t}, m^2_{0t}, m^1_{1t}, m^2_{1t}, L_t)'$. These include aggregate productivity z_t; the first and second moments of the wealth distribution for unemployed households, (m^1_{0t}, m^2_{0t}); the first and second moments for employed households, (m^1_{1t}, m^2_{1t}); and aggregate employment L_t. The model also includes 24 control variables:

$$y_t = \left(\theta^1_{0,t}, \ldots, \theta^9_{0,t}, \theta^1_{1,t}, \ldots, \theta^9_{1,t}, g^0_{0,t}, g^1_{0,t}, g^2_{0,t}, g^0_{1,t}, g^1_{1,t}, g^2_{1,t}\right)'.$$

Here, $\theta^j_{\varepsilon,t}$ for $j = 1, \ldots, 9$ are the Chebyshev coefficients used to approximate the continuation value, and $g^k_{\varepsilon,t}$ for $k = 0, 1, 2$ are the coefficients that parametrize the density function of wealth for each employment status.

We approximate the moments using $n_\ell = 15$ Gauss–Legendre nodes and weights $\{a_\ell, \omega_\ell\}_{\ell=1}^{n_\ell}$, where the implied density at each node is given by:

$$h^\ell_{\varepsilon,t} = g^0_{\varepsilon,t} \exp\left(g^1_{\varepsilon,t} a_\ell + g^2_{\varepsilon,t} a^2_\ell\right) \omega_\ell.$$

We also use $n_\psi = 9$ Chebyshev nodes $\{a_i\}_{i=1}^{n_\psi}$ to approximate the unconstrained expected marginal utility. We can write the approximate equilibrium conditions compactly as:

$$\mathbb{E}_t\left[f\left(y_t, y_{t+1}, x_t, x_{t+1}\right)\right] =$$

$$\mathbb{E}_t \begin{bmatrix} z_{t+1} - \rho_z z_t - \sigma_e \eta_{t+1} \\ m^k_{\varepsilon,t+1} - \frac{p(\varepsilon|0,z_t)(1-L_t)\sum_\ell \{a_{t+1}(0,a_\ell)\}^k h^\ell_{0,t} + p(\varepsilon|1,z_t)L_t \sum_\ell \{a_{t+1}(1,a_\ell)\}^k h^\ell_{1,t}}{\varepsilon L_{t+1} + (1-\varepsilon)(1-L_{t+1})} \\ \exp\left\{\sum_{j=1}^{n_\psi} \theta^j_{\varepsilon,t} T_{j-1}(\xi(a_i))\right\} - \beta(1+r_{t+1}) \sum_{\varepsilon_{t+1}} p(\varepsilon_{t+1}|\varepsilon, z_t) c_{t+1}\left(\varepsilon_{t+1}, a_{t+1}\left(\varepsilon, a_i\right)\right)^{-1} \\ 1 - \sum_\ell h^\ell_{\varepsilon,t} \\ m^k_{\varepsilon,t} - \sum_\ell \{a_\ell\}^k h^\ell_{\varepsilon,t} \\ L_{t+1} - f_t(1-L_t) - (1-s_t)L_t \end{bmatrix},$$

for $k = 1, 2$ moments, $\varepsilon = 0, 1$ employment states, and $i = 1, \ldots, 9$ asset grid points, leading to $30 = \dim(x_t) + \dim(y_t)$ total equilibrium conditions.

State-space system. The state transition equation follows from Equation (I.1)-(I.2) and $\hat{z}_t$ being an AR(1). For the measurement equation, assume, for example, that we observe deviations of aggregate output. Then, our measurement equation is given by

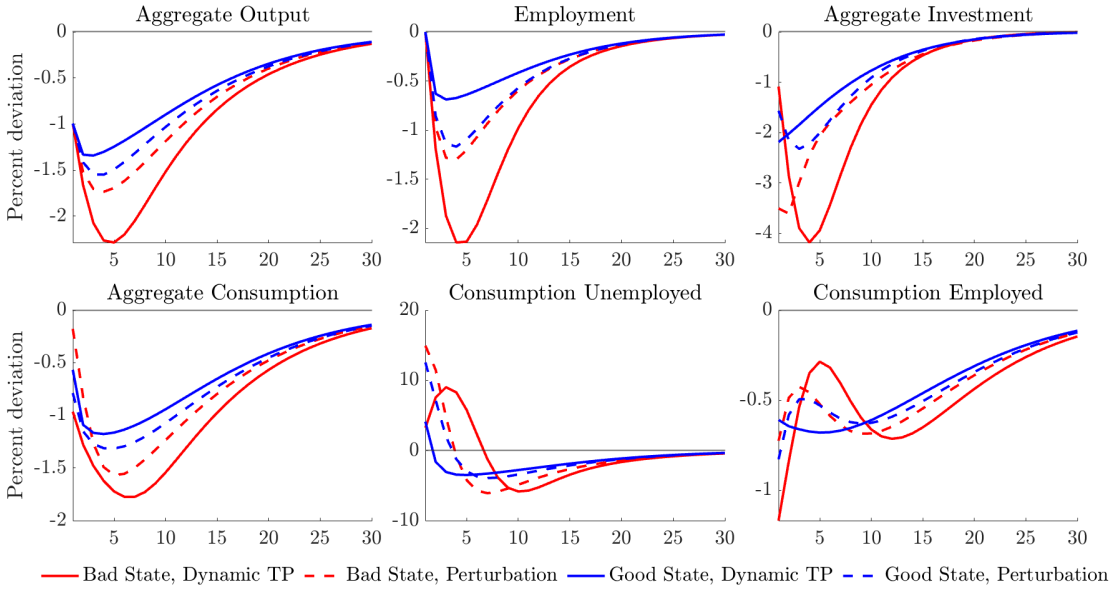
$$y_t^{\text{obs}} = b_0^y + B^y \begin{bmatrix} \hat{z}_t & m^1_{0,t} & m^1_{1,t} & m^2_{0,t} & m^2_{1,t} & L_t \end{bmatrix}' + \varepsilon_t^{\text{ME}}, \quad \varepsilon_t^{\text{ME}} \sim N(0, H).$$

Alternatively, one could incorporate micro-level data, such as repeated cross-sections, into the estimation procedure, following the approach in Liu and Plagborg-Møller (2023). This involves evaluating the cross-sectional data points using the density in Equation (43) evaluated at many draws from the model-implied distribution of the smoothed states $\mathbf{m}_t$ and $\mathbf{g}_t$. The resulting micro log-likelihood can then be combined with the macro log-likelihood derived from the filtered states.

Parameters. The discount factor is set to $\beta = 0.96$. Unemployment benefits are set to $b = 0.15$. The steady-state separation and job-finding probabilities are set to $\bar{s} = 0.038$ and $\bar{f} = 0.5$, respectively. The elasticities of the job-finding and separation rates with respect to aggregate productivity z_t are $\eta_f = 15$ and $\eta_s = -9$. The production function features a capital share of $\alpha = 0.36$ and a depreciation rate of $\delta = 0.10$. TFP follows an AR(1) process with persistence $\rho_z = 0.859$ and innovation standard deviation $\sigma_z = 0.01$.

State Dependence. We use the dynamic Taylor Projection (TP) method to show that even a relatively simple heterogeneous household model exhibits nonlinearities that standard perturbation methods fail to capture. We compare impulse response functions from the TP and perturbation-based solutions,[2] evaluating responses to a one-standard-deviation negative TFP shock under two initial conditions: (i) a *good* state (TFP one standard deviation above steady state for five periods) and (ii) a *bad* state (TFP one standard deviation below). As shown in Figure I1, perturbation significantly understates state dependence. Impulse response functions differ not only in magnitude but also in shape and timing. For example, TP produces a hump-shaped investment response only in the bad state, while perturbation predicts a similar response in both. Similar discrepancies arise in the consumption paths of employed and unemployed households.

Figure I1: State-Dependence in a Heterogeneous Household Model: Perturbation versus Dynamic Taylor Projection



Asymmetry. Figure I2 plots the responses of aggregate variables to a ± 1 standard deviation productivity shock at the steady state $x_t = \bar{x}$, expressed as log deviations from steady-state values. Because the perturbation solution is linear in levels, the log transformation introduces mild asymmetries. In contrast, the dynamic Taylor projection solution produces more pronounced asymmetries across all aggregates. To aid visual comparison, the response to a positive shock is multiplied by minus one. For variables such as output, employment, and consumption, asymmetries show up through magnitude. For other variables,

[2]Following Levintal (2018), first-order TP converges to a perturbation solution as $\sigma_z \to 0$.

such as investment, they affect the shape: investment exhibits a hump-shaped response to negative shocks that is absent following a positive shock. This notable asymmetry is not captured by the perturbation solution.

Figure I2: Asymmetry in a Heterogeneous Household Model: Perturbation versus Dynamic Taylor Projection, response to a one-standard deviation change in z_t.

